CONSTRAINTS ON THE EMERGENCE AND INTERPRETABILITY OF LINGUISTIC CONVENTIONS

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Abstract

Human communication takes place across many circumstances, and humans are adept at flexibly using language to achieve goals even in constrained conditions. The formation of linguistic conventions is often studied in iterated reference games, where over repeated reference to the same targets, a describer–matcher pair establishes partner-specific shorthand names for targets. In this dissertation, we expand the range of empirical evidence about when conventionalization occurs and what types of conventions are formed, all within the framework of iterated reference games. In Chapter 2, we ask which properties of the group's interaction structure facilitate successful communication, using a repeated reference game paradigm where we manipulated several key constraints on the group's interaction, including group size, the amount of feedback that matchers could give to directors, and the availability of peer interaction between matchers. In Chapter 3, we explore the developing pragmatic and referential communicative abilities of young children using a simplified, child-friendly paradigm. In Chapter 4, we consider the nature of conventions in iterated reference games, in particular to what extent the partner-specificity of these linguistic conventions causes them to be opaque to outsiders. Overall, we find that some signatures of convention formation can occur across a wider variety of circumstances than previously known, and that the descriptions used in iterated reference games are relatively comprehensible to outsiders.

Dedication

To be written later

Acknowledgments

To be written later

Contents

List of Tables

List of Figures

Chapter 1

Introduction: Computational psycholingistic approaches applied to understanding convention formation in iterated reference games

1.1 Motivations and definitions

The use of language is one candidate for what distinguishes humans from other species. Language enables coordination between individuals, enforcing social norms via gossip, and communicating of complex and abstract ideas which allows for the cultural accumulation of technologies beyond what any one mind could come up with. Especially with the technological innovation of writing, language allows for communicating across long distances of space and time.

Humans and other social species can communicate in a variety of modalities, but the structure of language makes it more expressive compared to other systems. We learn native languages in the early years of life and this gives a shared set of symbol meaning mappings and combinatorial rules that can be used to communicate. Even young children realize that language is a very powerful system for transmitting ideas from mind to mind. Language is ubiquitous to (at least my) daily life, but it also feels like the more one considers language, the more there is to explain and the more impressive our linguistic capabilities seem.

The word "language" can refer to a suite of related concepts, all of which can be the targets of study. There's the abstract idea of a specific language, for instance, English as a set of grammatical rules and word meanings. This is a bit idealized as the difficulty with specifying formal rules that cleanly generate the accepted utterances of a language demonstrate. From a more statistical perspective, there's also language as it is used by a certain community, not a static entity but something that evolves slowly over time. This too is a bit of an idealized abstraction as the grammars in the mind of individuals don't exactly align with each other, so the meaning is probabilistic or distributional. The idea of an individual's grammar still abstracts over the neural architecture that produces and processes language. Whatever the structure of linguistic representations, they also have to be constructed out of all the individual exposures to speech acts (and written language).

We can also study language based on the artifacts of what gets uttered or written in a language, abstracting away from the humans with intentions who produce and comprehend them. There's language as a capacity for learning a system for expressing ideas and being able to express and understand ideas in a system. From a different formalism, we can abstract away from the details of language and view it as a particular communication channel, subject to the same bounds as any other signal-transmission method. But language also can have substantial social and cultural relevance. Generalizing across purposes, we could say that language is a tool that we use to communicate with each other, possibly twisting or bending constructions and meanings to accomplish goals of various types.

These different levels of meaning of the word language both make it a fascinating thing to study, but also lead to disagreements around what is studied in linguistics or the "language sciences". Here I will tend to focus on language as a tool for communication, but artifacts of language, language as a shared set of expectations, and language as a thing the mind processes are also lenses that will occur.

In studying language use, what counts as language? Some might want to focus only on conversational face to face language use for communicative purposes, and exclude derivatives like wordplay or phonetic codes or language models as things that use the artifacts of language and some of their properties but maybe aren't language. However, beyond the focus on language as a particularly interesting communication modality, I do not assign primacy to any particular parts of language. We may find that some types of language use are easier to process than others, or that some uses require inferences that are domain general and external to the language faculties, but I will not assume a priori that certain parts of language are more privileged or central than others.

Though not to the extent of the word language, the word pragmatics also means different things to different people, so I will clarify here that I mean pragmatics expansively as something along the lines of meaning in context or context as a component to inferring meaning (Bohn et al., 2019).

I start with this definitional digression because I think it's important to scope the topic. This dissertation will focus on how we use language to communicate around the edges of expressivity – how we extend our lexica to accommodate communicative needs that are on the border and how we are able to coordinate with each other in these atypical uses of language. These linguistic stretches can be ad hoc and temporary, but these individual atypical usages are also what can spread and build up into the adoption of new words and meanings and language change.

Why care about this? I think questions around language use *in context* is in a middle ground between being tractable enough that experiments can usefully inform models and theories and being analogous enough to explain real-world behaviors. Broadly, context contributes to the level of complication as it is another source of variance, but language use is usually deeply embedded in context. What we do about the context is an interesting question both at more fine-grained levels of language specific processing and at more general levels of language as a communication tool.

In the rest of the introduction, I will first explain the motivation for using iterated reference games as a model system. I will then cover three main approaches from psycholinguistics that I believe have useful applications to convention formation in iterated reference games: efficiency, Bayesian pragmatics, and information theory. Then I will review findings and theories related to the conversational literature and consider how these findings could be subsumed under the larger psycholinguistic frameworks. Finally, I will suggest what I see as some gaps in the literature and close with a preview of a few gaps in particular that I will address in the later chapters of this dissertation.

1.2 Why iterated reference games

Having covered in the first section some of what I think makes language an interesting domain to study, I now introduce the paradigm that will serve as a focus for this dissertation.

Here we will focus on a particular paradigm for studying adaptive language use: the iterated reference game. A reference game consist of a set of people communicating with each other about a set of potential referents (often images). One person, the describer, has a specific designated target (usually pre-identified as the target by the experimenter) that they seek to communicate to the matcher (or matchers). This act of reference requires coordination between describer and matcher on a symbol – meaning alignment. Iterated reference games involve repeated reference to the same targets within the same group of people, so that a certain conversational history builds up that could shape the construction and understanding of later symbols. Crucially, this task uses referents that are somehow non-trivial to identify, usually by some combination of low-codeability or high-similarity to each other.

As described, this task is agnostic to communication modality, and iterated reference games can be done using gestures or drawing (Bohn & Frank, 2019; Garrod et al., 2007; Hawkins, Sano, et al., 2023). However, I am interested in how this task interacts with the rich baseline of symbol – meaning mappings afforded by natural language. That said, I believe that some of what we learn about language also transfers to other modalities of communication and vice versa.

One observation from real-life language use is that people use words in creative ways to express meanings sometimes. For instance, even if I don't know the name for something (or if it is known but evading memory as in tip-of-the-tongue events), I can still communicate about it in a more circuitous manner. And if I have the communicative need that lead to repeated reference to the same object or concept, the ad-hoc referring expression can calcify into a settled name for the object.

Communicative acts serve a variety of purposes, but a simple, yet central one is reference: connecting a symbol with a meaning to pick out a target. This is a subgoal needed for more complex acts – it's hard to tell a story or debate someone without agreement on what different words or phrases refer to.

Combining these observations – that language can be adapted to fill communicative needs, that repeated need leads to changes in referring form, and that reference is a easy to elicit communicative act that is still important – suggests the utility of iterated reference games. This is an ahistorical account – but I suspect that considerations like these – of striking the right balance between simiplicity, naturalness, and interesting elicited behavior – are some of the motivations behind the use of iterated reference games in the literature.

This task is commonly used. The earliest studies using it that I'm aware of are from the 1960s (Krauss & Bricker, 1967; Krauss & Weinheimer, 1964, 1966) , and it seems to have become more widely used in the 1980s (Clark & Wilkes-Gibbs, 1986). Over time it has been used in a wide variety of populations to study a wide variety of communicative and linguistic phenomena. The primary targets of study are coordination and the formation of conventions or conceptual pacts [Dahan (2025); Ghaleb et al. (2024); Eliav et al. (2023); Hawkins, Frank, et al. (2020); Metzing & Brennan (2003); Brennan & Clark (1996); among others]. Some studies focus particularly on iterated reference games as a way of studying collaboration in natural conversations (Clark & Wilkes-Gibbs, 1986; Dahan, 2024; Schober & Clark, 1989; Wilkes-Gibbs & Clark, 1992). Different studies focus on different aspects of the coordination, including the negotiation of referring expressions (Bangerter et al., 2020; Knutsen & Le Bigot, 2018) and the role of feedback (Fox Tree & Clark, 2013; Krauss & Glucksberg, 1977; Krauss & Weinheimer, 1966). These same types of games are used to study how speakers may design their utterances for their specific partners (Horton & Gerrig, 2002a, 2005; Turner & Knutsen, 2021; S. O. Yoon & Brown-Schmidt, 2014, 2019a, 2019b) and how speakers might generalize patterns of social conventions across interlocuters (Hawkins, Goodman, et al., 2020; Hawkins et al., 2021). Iterated reference games are used as the dialogue condition in studies on the differences between monologue and dialogue (H. P. Branigan et al., 2011; Fox Tree, 1999; Murfitt & McAllister, 2001; Tolins et al., 2017). Iterated reference games are used to study conversation and memory (McKinley et al., 2017; Vanlangendonck et al., 2018). There are also studies using this paradigm to model cultural conflicts between merging teams (Weber & Camerer, 2003), look at speech-gesture tradeoffs (de Ruiter et al., 2012) and study the effects of alcohol on social dynamics (Garrison et al., 2024).

In addition to studies on typical populations, iterated reference games are also used to study the abilities of special populations. Iterated reference games are used with children and teenagers (H. P. Branigan et al., 2016; Leung et al., 2024; Turner & Knutsen, 2021) and with older adults (Horton & Spieler, 2007; Hupet et al., 1993; Lysander & Horton, 2012) to study cognitive and linguistic abilities across the lifespan. Iterated reference games are used to study types of memory in people with amnesia (Duff et al., 2005; Duff, Gupta, et al., 2011; Duff, Warren, et al., 2011; S. O. Yoon et al., 2017). Iterated reference games have also been used to study various other populations with neurological and psychological conditions Chantraine et al. (1998).

1.3 Frameworks

One question is whether the phenomena related to convention formation and iterated reference games are best thought of as an isolated, special occurrence or whether the results can be subsumed within a more general and broader coverage theory of language use. I review three broad approaches that are common within computational psycholinguistics: efficiency, Bayesian pragmatics, and information theory.

1.3.1 Efficiency

One unifying framework gaining traction in psycholinguistics is efficiency, the idea that language and language use is under pressure to support efficient communication by maximizing the ratio of relevant information transmitted to effort. Efficiency is a high level framework that can only be tested against data when it is fleshed out with linking assumptions; however, comparisons between language use that is attested versus counterfactual languages can bound what types of assumptions are needed for this framework to be parsimonious.

Efficiency is thought to arise from trade-offs between communicative expressivity and some combination of learnability and ease of production (Kirby et al., 2015; Piantadosi et al., 2012). The state of language is thus in a dynamic equilibrium between the usage pressure to be able to communicate many ideas (pushing complexity of language) and some constraints pushing for simplicity of language, which could be either the requirement that languages be learnable or pressures for ease of production.

Evidence for efficiency comes from the argument that features of language are distributed much more closely to the Pareto frontier than would be expected by chance. A historically well-known example is that word frequencies follow a power-law distribution, which Zipf (1949) explains in terms of a "principle of least effort", where words that are more frequent are also shorter. It is worth noting that this type of power-law distribution is common across domains and generated by a variety of processes, so while well known, it is not the strongest evidence for efficiency as a driver of linguistic distributions (Piantadosi, 2014). Stronger evidence comes from the lexical partitioning of subdomains such as color, number, and kinship terms, where the distribution of systems falls on the Pareto frontier between complexity (number of terms) and informativity (how many bits each term provides) (Gibson et al., 2019; Kemp et al., 2018; Zaslavsky et al., 2018). Syntactic features of language such as harmonic word order and dependency length also appear to be optimized for increased expressivity with minimized processing effort (Gibson et al., 2019; J. Hawkins, 1995).

Efficiency arguments are based on the language artifacts of grammars and transcripts, but efficiency pressures act on language use as a process, not language as a static code (Gibson et al., 2019). Thus efficiency can be seen as imposing a joint constraint on the entire communicative process: minimizing the total time and effort involved in going from an idea in one person's head to a sufficiently close idea in another person's head. A corollary of this framing is that while shorter utterances are usually considered more efficient, utterance length in syllables or clock-time is not actually what is under pressure, they would not be efficient if they took longer to produce or parse than the more verbose alternatives.

As with other high level frameworks, the parsimony of the efficiency framework is entirely based on its auxiliary linking theories (Gershman, 2018). Thus, the question becomes what sort of auxiliary assumptions are needed to connect a given phenomena under the efficiency umbrella, and how reasonable are those assumptions.

As an illustration, I look at the issues of ambiguity and redundancy in referring expressions.

Redundancy and over-informative referring expressions

How do we reconcile the idea that language is efficient with violations of efficiency in language use ("look at the yellow banana") and the ubiquity of ambiguous utterances?

Terms such as "redundant" and "over-informative" are commonly used to describe situations where people produce modifiers that do not restrict the extension of a noun phrase, e.g. "blue cup" when only one cup is salient. In this example, we might think that the substring "cup" is a alternative utterance that could uniquely identify the target (Rubio-Fernandez et al., 2021). People do produce these non-restricting modifiers in reference tasks, especially for color, but this behavior seems to run counter to the idea of efficient language use.

Formalizing claims of redundancy or over-informativity requires a definition of what would be minimally informative, which in turn depends on a commitment to a fully specified semantic-pragmatic system, at least for the range of utterances to be considered. In the more general case, this quickly gets quite thorny: for instance, if specificity implicatures are within the option space, are those calculated before or after informativeness is measured (Bergen et al., 2016)? One could sidestep the theory issue by empirically measuring the information content of different utterances by how they shift the entropy of the distribution of inferred meanings (Degen et al., 2020), but this does not scale up well in part due to issues with compositionality.

The flip side of "redundancy" is ambiguity: many, many utterances are ambiguous. In general, strong contextual factors render the ambiguity a non-issue (Piantadosi et al., 2012), but this means we can't judge language outside of the physical and social context it is used in. Determining what is efficient language use in context requires not just analyzing phrases and their alternatives, but also how long utterances take to generate and comprehend, which may be highy contingent on contextual factors and conversational history.

The idea that utterances should have "just enough" information (Grice, 1989) has inspired a wealth of empirical research into what utterances people produce and what utterances people comprehend. By comparing these two halves of language use, we can determine how calibrated the utterances people produce are to what other people will understand. If calibration is reasonable, that would support an efficiency framework with minimal linking assumptions; if the calibration is not, we would need to invoke other assumptions (for instance, bottlenecks on production, goals such as politeness rather than just informativity) and then consider whether they are still parsimonious.

1.3.2 Bayesian pragmatics

Another broad coverage framework in cognitive science is modeling humans as Bayesian agents making resource rational choices. Within language, this tends to show up as a Bayesian pragmatic models, where (boundedly) rational Bayesian agents reason about each other to determine the meanings of utterances.

A prominent model within Bayesian pragmatics is the Rational Speech Acts (RSA) framework, an information-theoretic, computational framework for making quantitative predictions about pragmatic inferences in contexts (Frank & Goodman, 2012; Goodman & Frank, 2016). The basic idea of the RSA family of models is to picture two interlocuters recursively reasoning about how the other would produce or interpret utterances, grounding out in a listener (or speaker) who behaves in a pre-specified “literal” way.

Computational frameworks such as RSA provide a way to factor together different trade-offs and determine their relative weights in a model (Goodman & Frank, 2016). A softmax is taken over the scores of the options to produce a distribution of interpretations and utterances, with some parameters such as the degree of optimality fit based on data. This framework is usually run with one or two levels of recursion, where it tends to produce a reasonable fit to human experimental judgments, consistent with work finding that most people reason pragmatically at a low recursion depth (Franke & Degen, 2016).

RSA models have been used to predict some instances of ad hoc pragmatics as well as conventionalized pragmatic implicatures (Bergen et al., 2016; Degen et al., 2020; Goodman & Stuhlmüller, 2013). Models generally include a utility or informativity term that relates to how well an utterance resolves uncertainty in favor of the target referent. It is common to also include factors such as the prior likelihood of referring to each target (salience prior) and some cost on utterances where longer or more complex utterances are penalized (Goodman & Frank, 2016). Some models also go beyond informativity, incorporating options to infer the question-under-discussion (Kao, 2014; Qing et al., 2016) or balance informativity with politeness (E. J. Yoon et al., 2020). RSA-type models have also been used as a model of word learning in children (Bohn et al., 2022; Ohmer et al., 2022).

A full RSA model would incorporate all of these components and infer their weights. However, for tractability, usually only those features that are considered relevant to the domain of interest are included. Because of the flexible framework, it is possible to model many sources and levels of uncertainty, and then integrate out that uncertainty to make predictions, but also update on the sources of uncertainty in response to input.

The Challenge of Semantics

Perhaps the largest challenge to RSA models is the question of how to ground out the models in a "literal" listener or speaker. For the most part, RSA is tested in toy domains where the set of possible utterances is small and it is possible to enumerate a set of meanings (Bergen et al., 2016; Frank & Goodman, 2012; Goodman & Stuhlmüller, 2013). In some domains, a soft or continuous semantics is used to represent that some dimensions of meaning might be more strongly informative than others (Degen et al., 2020). This semantics supports the prediction of patterns of "redundant" color adjective use in referring expressions.

Soft semantics can run into conflict with compositionality. One can have soft semantics without explicit compositionality, if there is a way to determine match between expression and meaning. One could use a pre-trained neural network system that embeds utterances into semantic space and then taking distances to the target meaning, although this requires making assumptions about what "literal" meaning is. The system might have implicit compositionality, but there are no guarantees about the relationships between the embeddings of "red", "apple" and "red apple" in these systems. This may be more severe for expressions like "green apple on the red box" and "red apple on the green box" where syntactic relations are key to meaning which sentence-level embeddings may not fully preserve (Erk, 2022). Depending on the domain, embedding systems may or may not have satisfactory match to human semantic judgments.

Alternatively, one could have compositionality, but this requires a process for mapping the meanings of pieces into the meanings of the whole. The process could be deterministic rules, or an algorithm or neural model implementation, but one still needs a way to break down the units of meaning and build them up. In formal semantics, this is often door with a system of logical relations and lambda calculus which can assign that "red apple" is the intersection of "red" and "apple". However, this would then predict that "apple" is equally true for members of "red apple" and "blue apple", which from Degen et al. (2020) doesn't match people's judgments due to typicality effects.

One reason to want both compositionality and soft semantics is to be able to learn and update the prior on the basis of interactions. Given that an expression – meaning pairing was used successfully (or unsuccessfully), one may want to locally update the lexicon to reflect this additional information.

In less toy domains, there is not a satisfactory answer: some situations can be handled by empirically measuring likelihoods in an exhaustive ways, but this holistic approach is not compatible with incremental RSA (Cohn-Gordon et al., 2018) or larger sets of utterances that require compositionality to be defined. Some work has used grounded neural language models for alternative generation and likelihood measures, which has scaling potential in domains that have the right grounded datasets (Cohn-Gordon et al., 2018; Monroe & Potts, 2015; White et al., 2020). In order to extend RSA towards more realistic and open-ended scenarios, an important question to grapple with is what form of meaning (even at a computational level) will appropriately support pragmatic reasoning.

1.3.3 Information Theory

Information theory is a formalism that is used with both Bayesian pragmatics and efficiency as a link to connect abstract ideas to concrete measures and predictions. Information theoretic concepts such as entropy, surprisal, and rate distortion theory have been key to quantitative theories in computational work on language.

The ideas of information theory provides a link from functionalist accounts of language usage to the artifacts of language systems, by focusing on what the constraints are that would render the language systems we observe an optimal solution to the functional question of communication (Futrell & Hahn, 2022). One advantage of information theory is that it provides ways to calculate measures of complexity in a mostly theory neutral way (Futrell & Hahn, 2022). Information theory based theoretical proofs and toy models have been applied to domains such as language evolution (Plotkin & Nowak, 2000).

Measures derived from information theory such as surprisal, conditional entropy, and mututal information have been very productive in language science both for unifying aspects of language typology (ex. dependency locality) (Futrell & Levy, 2017) and for explaining patterns in language processing (ex. surprisal theory and entropy reduction) (Hale, 2001, 2016; Levy, 2008). The idea of rate distortion and a limited channel compressing a meaning through a message has been key to testing efficiency claims related to syntax, word order, and grammatical markers (Futrell & Levy, 2017; Mollica et al., 2021).

There is also a connection between information theory and pragmatics, as RSA formulations are closely related to solutions to information-theoretic joint optimization between utility and effort (Zaslavsky et al., 2020).

One difficulty with applying information theoretic measures is that it requires a probability distribution to work with; thus, similarly to other frameworks, there are questions of what distributional language models or datasets to use to capture the baseline language.

1.4 Phenomena

The communication and conversation literature, especially with regard to referring expressions, has provided useful descriptive characterizations of how people communicate in naturalistic and experimental settings. For iterated reference games in particular, there is an accumulation of fairly consistent and reliable findings of reduction, convention, and partner-specificity. While there are consistently observable trends in the conversation literature broadly and in iterated reference games specifically, there has not been strong identification of necessary and sufficient conditions or dose-response relationships between circumstances (or experimental settings) and the size of the effects. Thus, there are not quantitative accounts for the phenomena. Instead, the effects tend to be characterized with verbal theories that posit causes.

It is difficult to directly test these theories as they often fail to make explicit predictions about what would happen under conditions other than those tested or what experiments would adjudicate between theories. Many of these theories present certain results as "special" but our question as we go will be to what extent these phenomena could be explained by the broader coverage theories of efficiency and rational communication.

As I go through different theories and phenomena, I will consider the available evidence, problematize the verbal theories, and attempt to see how the above approaches from computational psycholinguistics might be used to more formally model or subsume the empirical results.

1.4.1 Mentalizing and non-mentalizing approaches

One historically important distinction that recurs with conversation, and interactions between agents more generally, is whether and how agents track other agents' internal states of knowledge and how this factors into their interaction. While an intuitively appealing distinction, it's not clear how to test this.

One could build, for instance, model-based and model-free algorithms and then compare which better captures an observed phenomenon, but this seems to harken back to associationist versus rule-based accounts in linguistics, which don't seem as relevant now. These aren't opposites so much as points on a spectrum and appearances may also be deceiving – what looks rule based could be instantiated in a non-symbolic, associationist way. On the other hand, a more sophisticated approach could look more associationist under performance load.

While is it not clear how to test these theories, it is still worth discussing the frameworks for historical context.

Mentalizing tradition and "common ground"

The "mentalizing" tradition treats humans as representing other humans as agents with internal states that include knowledge and goals. Within this broad school, there is variation in how these representations are implemented, how information gets added or modified, what exactly is tracked, and when representations (versus heuristics) are used.

Within this tradition, many use the term "common ground" to refer to knowledge that two agents share. In some cases, it is used in a pre-theoretic way to mean roughly "things you think another person will understand and won't be surprised if you reference" (Garrison et al., 2024; Leung et al., 2024). For instance, Hanna et al. (2003) defines common ground as the "mutual knowledge, beliefs, and assumptions" held by the interlocuters. This meaning is roughly comparably to "givenness" in other domains (Fay et al., 2010).

However, "common ground" is also sometimes used in a theoretically loaded way, originating from the privileged versus mutual versus common knowledge framework (Clark & Wilkes-Gibbs, 1986). Under this usage, "common ground" is defined via infinite recursion in knowing that the other person knows that the first person knows that …; this is the usage that comes up in formal semantics where many things must be introduced to common ground via accommodation, which could itself be seen as a kludge to explain how people get away with making statements whose presuppositions aren't known to be shared (Horton & Keysar, 1996; Pickering & Garrod, 2004). In practice, humans don't tend to do more than a couple layers of recursion in their pragmatic reasoning (Franke & Degen, 2016). Thus, it is generally not important to distinguish knowledge types at deeper recursion levels than mutual knowledge that both people know to be mutual.

Even if we accept that common ground has grown away from its formal semantic roots to just be a convenient shorthand for "mutual knowledge", there is still a question of how agents know (or decide) what is mutually known. Many approaches have tried to characterize when something is mutually known (Brown-Schmidt, 2012; Clark & Wilkes-Gibbs, 1986; Horton & Keysar, 1996). This has predominately taken a deterministic approach with rules such as "if it's in shared visual presence, it's mutually known". Visual presence may be a good heuristic, but it seems like whether something is mutually known may be better represented in a more graded way.

These terms seem to be easily subsumed within a Bayesian inference and Bayesian pragmatics framework. While the "mentalizing" approach places special emphasis on "common ground", it's unclear whether this form of modeling another's knowledge should be treated differently from cases where one models another's knowledge that one does not personally have. For instance, in deciding who to learn from, someone might determine that another agent knows what's behind a door and then use this information to for instance, interpret a quantifier like "some" (Goodman & Stuhlmüller, 2013).

Another implication from the Bayesian approach is that we do not need to deal with deterministic rules and can instead reason under uncertainty. This may be useful for accounting for hedging behavior where there seems to be uncertainty over what an interlocuter will understand. Joint updating can also help for times when something was initially thought to be likely in the shared prior, but then new information decreases this likelihood. The framework of flexible inference and probabilistic knowledge states is a powerful tool.

Non-mentalizing approach: Interactive alignment theory

In contrast to mentalizing approaches, "interactive alignment theory" attempts to explain how people can successfully collaborate on reference tasks without reasoning about each others' mental states (Gandolfi et al., 2022; Pickering & Garrod, 2004). The motivation for non-mentalizing accounts is the apparent difficulty of mentalizing, coming from accounts such as naive egocentrism (which will be discussed more later). Given that humans reason socially about each other readily and from a young age (Rakoczy, 2022), it's not clear how well the motivation for a non-mentalizing approach holds up.

Regardless, at the time, there was desire for an approach to explain communication and coordination behavior without positing explicit reasoning about other minds, because it was that was too computationally expensive and slow. Under this framework, Pickering & Garrod (2004) tried to account for observed alignment between interlocuters by extending ideas from structural priming.

Structural priming occurs when processing a grammatical structure (for instance, a double object construction) increases the likelihood of expressing an idea using that structure rather than an alternative (producing a double object rather than a prepositional object construction) (Tooley & Traxler, 2010). In production, this occurs even when there is not lexical overlap (ex. different verbs), but is increased if there is lexical overlap. A similar effect occurs in comprehension, but it seems to only happen when there is the same lexical head item (Tooley & Traxler, 2010). Many potential mechanisms behind structural priming have been suggested, but there is not a consensus. While typically observed in controlled settings, the tendency to reuse constructions to express ideas is also observed in dialogue (Pickering & Ferreira, 2008). Some other forms of alignment in dialogue can be observed in natural corpora. Interlocuters on Twitter exhibit style alignment although the direction of accommodation is moderated by social factors which suggests it may not be fully automatic (Doyle et al., 2016). Analyzes of Twitter and Switchboard corpora reveal lexical repetition, but not category alignment, between interlocutors for non-content word categories (Doyle & Frank, 2016).

Pickering & Garrod (2004) claims that the alignment occurs via "priming" and is "resource-free and automatic", without providing a further explanation of what this means or how this works in terms of memory and processing. Unfortunately, as expressed, interactive alignment theory doesn't provide an algorithmic or mechanistic level theory. There are a few issues with this theory. For one, it is unclear that structural priming would predict this cross-level trickle-up or trickle-down effect that is key to interactive alignment; the claim that people tend to say "the ball that's red" after their partner says "the block that's blue" is quite different from a claim that this also aligns their "situation models". Another issue is that it's really hard to test for intent (or lack thereof); for instance, repetition could be due to some sort of "priming", or explicit reasoning, or production-side ease-of-use constraints (MacDonald, 2013).

Perhaps the largest problem with interactive alignment theory is that it does not account for one of the key features of iterated reference games: that people converge to conventionalized referring expressions that evolve and shorten after both people understand each others expressions. Thus, one requires an explanation for change, not just repetition or mimicking. To my knowledge, interactive alignment does not have an explanation for this, in part perhaps because their studies tend to be of a different sort. The "common ground" tradition generally using asymmetric director/matcher designs (dating back to at least Krauss & Weinheimer (1966)) and the "interactive alignment" traditions using symmetric designs where people both have partial information about a maze that they navigate together. Thus it is possible the two approaches are built around trying to explain differing sets of experimental results.

Figuring out exactly what alignment happens in conversation may well be worthwhile, but it does not seem that useful for the phenomena in iterated reference games in particular. That said, it is reasonable to be concerned that full inference is not necessarily cognitively feasible. One way to explore this would be with approximations of full Bayesian inference in the resource-rationality tradition or with information theoretic limits on memory capacity that don't allow for full partner representation.

1.4.2 Partner specificity

One key phenomenon from iterated reference games is that describers change their behavior when the set of matchers changes, usually shifting back to longer descriptions when a new matcher is introduced. This change in behavior is described as "partner-specificity", with the idea being that the conventional names developed with one partner as specific to that partnership (Brennan & Clark, 1996; Hawkins et al., 2021; Metzing & Brennan, 2003). The idea of partner-specificity is also referenced with regard to how different pairs diverge to different names for the targets (Hawkins, Frank, et al., 2020). Partner specificity is part of the mentalizing tradition and assumes that partners (and their background and knowledge states) are being represented in the minds of speakers. The form of the representation is not explicitly stated, so these models could be compatible with heuristic or distributed representations, but include explicit thinking about the audience and are incompatible with the non-mentalizing "priming" account.

Empirical evidence from experiments where one director talks with multiple partners suggests that people do "partial pooling" over their partners (Hawkins, Goodman, et al., 2020; S. O. Yoon & Brown-Schmidt, 2019a). That is, a speaker A will show some variation in their expressions when talking to partner B versus partner C, but there will be some generalization between partners as well, so that A talking with

B is more like A talking with C than D talking to E. When coupled with a tendency for descriptions to shorten within a pair, this leads to a jagged pattern of reference length: when switching to a new partner, speakers use longer utterances, but not as long as their initial utterance with their first partner (S. O. Yoon & Brown-Schmidt, 2019a).

This pattern could be efficient if the context from the initial utterances is requisite to understand the later parts. The jagged pattern in particular is suggestive of a mix of speaker production effort and shaping the matcher's lexicon. If we consider the elaborated and shortened forms of the concepts (ex "arms up, like a goalie diving for a ball" versus "goalie") to be synonyms (in the same way chimp and chimpanzee are synonyms), then the switch to short form after establishment and the temporary switch back with a new listener could be explained under a uniform information density model, provided that the longer form occurs in a context when it is more surprising and informative (Mahowald et al., 2013).

Partial pooling and partner specificity can be instantiated in a hierarchical model of Bayesian pragmatics, such as those introduced in CHAI or lexical uncertainty models (Bergen et al., 2016; Hawkins, Franke, et al., 2023; Potts et al., 2015; Schuster & Degen, 2020). Testing out these model frameworks with natural language data is not yet tractable due to the need to figure out computational and semantic issues, but the models seem to provide coverage at least in theory.

How can we think of partner specificity from an information-theroretic lens? One approach might be to think about what the mutual information is between the conversation history, the current utterance, and the visual context of targets. Within a partnership, the conversation history can serve as a sort of encoder that allows later conversational elements to use a more efficient coding to map between language and referent. Under partner specificity, one would say that conditioned on conversation history, the target and convention predict each other, but that conditioned on no history or a different conversation history they would not predict each other. This abstracts over having a model of how conversation history unfolds, but it's at least an operationalization that could be testable using machine learning models.

1.4.3 Audience design & effort sharing

One set of theories in communication that are often tested and explored with iterated reference games have to do with how two (or more) people split the communication load.

One relevant concept is "audience design", the idea that speakers seem to be sensitive to the knowledge state of their listener and say things that are easy for the listeners to comprehend. This seems to occur sometimes in everyday life where we speak differently to a 5 year old than to an adult, or explain things differently to a colleague than to a lay person. However, despite the introspection that we design utterances for the audience in some cases, it is difficult to identify if it is happening intentionally in referring games.

Confusingly, the phrase "audience design" sometimes implies intention on the part of the speaker (Horton & Gerrig, 2002b, 2005), and sometimes is used when utterances are constructed based on what's easy for the speaker, and listener ease is a side effect (Horton & Keysar, 1996; MacDonald, 2013; Rogers et al., 2013). For instance, audience design could both occur in times when the speaker gives an elaborated description to a naive listener (inferred to be intentional if contrastive with their description to a non-naive listener), but speakers may also tend to start descriptions with given material which is both more accessible to the speaker and convenient for the listener.

Audience design is sometimes used very broadly (V. S. Ferreira, 2019) when production choices make it easy for listeners to comprehend, even if this is not customized to the listener. In a very broad sense, efficiency pressures for things like uniform information density could be construed as audience design. Beyond iterated reference games, speakers do behave differently depending on the listener; speakers tell a story differently when retelling it to the same listener versus when telling the story again to a new listener (Galati & Brennan, 2010).

Intention versus side-effect are difficult to distinguish between because speakers and listeners often share recent context, find the same things salient, and linguistically what is easier to produce is often easier to process. Thus, disentangling speaker and listener ease may require careful experimental designs where ease of production and ease of comprehension are separated (F. Ferreira, 2004).

Questions around audience design are related to larger issues of how interlocuters split the communicative burden with one another. Depending on the task and the communication modality, there may be many options for how to balance the communicative load (Clark & Wilkes-Gibbs, 1986; Fay et al., 2010; Fox Tree & Clark, 2013). For instance, a listener could describe what options they see or otherwise prompt the speaker. We might expect the load splitting to vary based on the capacities of the interlocuters (ex. a speaker might craft their utterances more when talking to a child versus an adult) and the capacities of the channels (ex. speakers may use different approaches if listeners can interrupt).

While much of the work on iterated reference games and communication more generally has focused on interactions that are dyadic at any given time, much conversation in the real world takes place in larger groups. Multi-way conversations complicate the theories of audience design and partner specificity by introducing a larger audience of more partners. Two main questions are whether speakers "aim low" or "aim high" in balancing the needs of the listeners and whether speakers track individual listeners or an aggregate (S. O. Yoon & Brown-Schmidt, 2017, 2019a).

Empirical results indicate that speakers are sensitive to the knowledge states of listeners in a gradient way. Describers seem to track the knowledge states of different matchers independently when they alternate between addressing matchers (S. O. Yoon & Brown-Schmidt, 2017). When addressing a group of matchers with disparate knowledge levels all at once, speakers generally give descriptions that are fairly similar to what they give only naive listeners (S. O. Yoon & Brown-Schmidt, 2017). However, there is also some gradient, where descriptions vary based on the proportion of naive listeners in groups of up to 4 addressees (S. O. Yoon & Brown-Schmidt, 2019a). With pairs of listeners who differ both in their background knowledge and in what context they are seeing the target in, speakers produce descriptions that are sensitive to the interaction between background knowledge and distinctiveness of the target in context (S. O. Yoon & Brown-Schmidt, 2019b).

Speakers also use strategies in group contexts that don't occur as often in dyadic contexts, such as referring to a target with both the name that one person will understand and a elaborated description that will help another person get on the same page (S. O. Yoon & Brown-Schmidt, 2019a). The ability to track partner's knowledge states presumably would degrade as groups got bigger, but this paradigm has not been used for groups large enough for this to happen.

One angle on larger groups is network structures, where (non-linguistic) convention formation has been studied on networks of up to 50 people (Guilbeault et al., 2021). In this communication game with targets varying over continuous space, large groups tend to end up with fairly consistent category boundaries across independent networks, while small networks can support more idiosyncratic category boundaries. This work is suggestive that there may be group-size and group/network-structure dependencies on what sort of conventions arise. There has been some work on group and network designs in traditional iterated reference games, but only on relatively small groups.

One question related to partner specificity and audience design is whether the observed behavior is optimal. People are generally fairly pragmatic in their communication, so we might think that using different descriptions based on the composition of listeners is adaptive and calibrated. Speakers behave as if the descriptions that are appropriate to give vary based on the listeners knowledge state and their shared conversation history.

One source of evidence about the potential optimality is whether other people (naïve matchers) can understand the descriptions, and whether this varies by circumstance. This has not been addressed from the efficiency and optimality angle, but the role of dialogue and conversational history has been studied. Naïve matchers tend to do better the more their observation history resembles that of the original conversation—when hearing descriptions in order instead of in reverse order (Murfitt & McAllister, 2001), when listening to the entire game instead of starting in the third round (Schober & Clark, 1989), and when seeing yoked trials from a single game rather than trials sampled across 10 games (Hawkins, Sano, et al., 2023). Descriptions produced in dialogue are easier to understand than those produced in monologue (Murfitt & McAllister, 2001). The gradient suggests that some of the development in descriptions may be adaptive, but there has not been work grappling with whether the far above chance performance of naïve matchers poses problems for audience design theories.

It certainly suggests that from a purely audience design listener-focused perspective, speakers are not efficient, in that to the extent that new people can understand later utterances without context, a counterfactual more rapid adoption of short forms could achieve a certain level of accuracy just as well as the observed pattern. This is an area that could benefit from empirical work around what the optimal pattern could be.

There are also some alternative auxiliary hypotheses that could save the efficiency idea before we are forced to conclude that humans are miscalibrated. One possibility is that the communicative bottleneck is firmly on the production side – speakers are doing the best they can to rapidly produce descriptions, and producing more succinct descriptions earlier would be too costly in terms of effort or time. Another possibility is that social norms and task dynamics dissuade the use of short form descriptions even if they are likely to be understood: perhaps the initial descriptions are more polite, or perhaps people are (overly) cautious in their efficiency/accuracy trade-offs.

Bayesian pragmatic models could be extended to cover situations discussed here. Rather than just communicate pragmatically with one person, balancing multiple people's needs would require balancing different goals. Again the difficulty of specifying base interpretations and compositionality and reasoning may pose some difficulties, but a toy model of the situation could work. Similarly, a lexical uncertainty Bayesian pragmatics model would be an approach to capturing the new listener behavior in eavesdropper scenarios. In terms of effort splitting, that could also at least in theory be represented by changing RSA parameters such as depth of reasoning, starting lexical semantics, or cost functions.

1.4.4 Convention formation

One of the key phenomena of interest with iterated reference games is that over repetitions with the same partner, dyads tend to form shared "conventions" (also called "conversational pacts") about how to refer to the initially ambiguous targets. These conventions tend to be partner- and context-specific: changes in the speaker, audience members, or changes in the context can all license the use of a new description (Ibarra & Tanenhaus, 2016; Metzing & Brennan, 2003; S. O. Yoon & Brown-Schmidt, 2014). Absent these changes, pairs tend to stick near the semantic space of successful descriptions.

What exactly does convention formation refer to? There is ambiguity about what level of specificity convention formation and conceptual pacts refer to. It could be on the lexical level, such as calling a figure “ballerina”. It could be conceptualizing the figure as a ballet dancer with a tutu (manifesting in descriptions with semantic association, but not lexical overlap, such as “ballerina” and “dancing in a tutu”). It could also be a general paradigm for how to describe figures, such as in terms as humans in different postures. Horton & Gerrig (2002b) distinguish between “lexical entrainment” when the same words are reused, and “conceptual similarity” when there is broader similarity that does not repeat the same words. These levels often co-occur, but in order to computationally quantify the phenomenon of convention formation, we have to make choices about how to operationalize the similarity: is it word overlap, lemma overlap, distance in semantic space, or something else.

In addition to convention indexing the stickiness of some properties of the description, there is also a pattern of change from typically a description of a set of features to a more holistic label. Typically, in convention formation, holistic or analogic descriptions are the ones that stick (Clark & Wilkes-Gibbs, 1986), but this isn’t absolute. Groups can successfully coordinate on reference using many different types of names, including ones that pick up on low-level or meta-level features. The range of successful options makes explaining convention formation harder, and suggests suggests a strong path dependency for how reduced utterances evolve, potentially influenced by factors such as relationships and humor value. How to quantify this transition or model why it occurs are open problems.

The semantic meaning of a description is not a priori related to its length, but these two features empirically tend to correlate in iterated reference games (Hawkins, Frank, et al., 2020). Thus “reduction” or the shortening of utterances is sometimes used as a shorthand and measurement proxy for the semantic changes (Clark & Wilkes-Gibbs, 1986; Garrison et al., 2024; Hawkins, Franke, et al., 2023), and Bovet et al. (2023) considers it a measure of common ground. Convention and reduction are sometimes conflated with partner-specificity, as these phenomena often co-occur with different pairs forming different conventions and changes in group composition leading to (temporarily) longer descriptions (Clark & Wilkes-Gibbs, 1986; Wilkes-Gibbs & Clark, 1992). Better tools for measuring semantic distance separate from length may help address the theoretical issues of how entangled these concepts are. For instance, when a describer gives a more elaborated description based around the same conceptualization for a new addressee, where does that fall in terms of conventionalization?

It remains an empirical question whether the shortening of utterances, convention formation, and partner-specificity of descriptions are inseparable or merely occur together in the experimental paradigms considered in the literature.

From an information-theoretic perspective, we can think of convention formation as convergence towards a modified encoding that is more efficient at representing the specific set of target meanings than the baseline language. These are not meanings that are not usually commonly needed and thus don't have initially high codeability, but over local usage, a (temporary) warp of the meanings can be achieved that is locally more optimized to express the target meanings.

CHAI models

Scaling up RSA to handle reference games requires solving at least two problems. One is specifying a semantics system that can handle the abstract, metaphoric, and parts-based descriptions that are used – this could be seen as the problem of accounting for initial reference.

Secondly, one must explain how to go from one successful utterance to a different (often shorter) utterance. One RSA-style model that attempts to explain why multi-part descriptions are produced initially, but shorter descriptions are produced later is CHAI, a framework to bridge different levels of convention formation (Hawkins, Franke, et al., 2023).

CHAI incorporates parameterized hierarchical uncertainty over lexica that allow for variability in how words will be interpreted that can be integrated over. Listeners' lexica are not fully known which accounts for background differences that may be shared by a sub-population and differences from communication history that may be individual-specific. The results of inference update these priors, and propagate the new information to both the individual and group lexical representations. CHAI can qualitatively account for various convention-formation phenomena in toy systems (Hawkins, Franke, et al., 2023).

CHAI's explanation for reduction is consistent with the ambiguity arguments of Piantadosi et al. (2012): initially the context is not sufficiently constraining to resolve the ambiguity of a short utterance, so multiple descriptive pieces are needed to triangulate the meaning, but as conversational history accrues, the context is sufficient to disambiguate the shorter description.

In toy models of interlocuters playing a reference game with soft semantics, initial utterances use multiple properties to collectively increase the degree of certainty in the target. This successful reference then shapes the prior over word meanings, until the degree of certainty afforded by only one word is sufficient to pick out a referent.

The same approach CHAI uses to generalize from individuals to groups and populations could also be used to generalize across stimuli, if one could get the baseline semantics worked out. Then, a successful reference (like "man with arms up") would not only update the meaning of that phrase or components, but could also lend weight to the over-hypothesis about using human and body part based descriptions, and thus update the speaker's beliefs about the listener's semantics for the words "head" and "leg".

1.4.5 Developmental work

One angle to understanding linguistic phenomena is to look at their developmental trajectory to understand when the phenomena arises and how it changes across development.

There have been a limited number of direct studies of iterated references games with children, although there is a much wider body of work on young children's communicative and pragmatic abilities.

One influential early study suggested that 4-5-year-old preschoolers struggle with child-child referential communication (Glucksberg et al., 1966). In their paradigm, one child was given a set of 6 blocks in a specific order. Their task was to describe the image on each block so their partner could pick out their corresponding block. As children described and selected blocks, they stacked them on pegs. While 4-5-year-old children succeeded on practice trials with familiar shapes and visual access to each other's blocks, children failed on critical trials where the blocks had abstract drawings and there was no visual access. Even after multiple rounds with the same images, children were not able to correctly order the blocks. Glucksberg et al. (1966) attributed children's communicative failures to their production of ego-centric descriptions that did not account for the other child's perspective. Similar experiments with older children indicated a gradual improvement through adolescence both for initial accuracy and for the increase in accuracy across repetitions. Still, even the 9th grade sample was noticeably worse than the adult college student sample (Glucksberg & Krauss, 1967; Krauss & Glucksberg, 1969).

Since Glucksberg et al. (1966), few studies have revisited the question of whether children can successfully communicate in an iterated reference game. In one study, 8-10-year-olds exhibited adult-like patterns of increasing accuracy, increasing speed, and shorter descriptions across repetitions, but children's accuracy was still far below adult performance and highly variable between dyads (H. P. Branigan et al., 2016). 4-6-year-olds successfully used conventionalized gestures with their partners in an iterated reference game where children could only use gestures to communicate (Bohn & Frank, 2019). 4-8-year-old children succeeded at playing an iterated reference game with a parent using a simple, child-friendly tablet-based task (Leung et al., 2024). In this task, even 4-year-olds had an initial accuracy above 80%, which rose to above 90% in later repetitions (Leung et al., 2024).

One issue that arises is developmental work is assessing whether the paradigms are measuring the abilities of interest or whether task demands or cognitive limitations are masking nascent abilities. The overall trend in developmental work has been discovering evidence of abilities emerging earlier than previously thought, due to better methods and designs that require fewer additional cognitive resources. Thus, we might be skeptical of whether the negative results of Glucksberg et al. (1966) are an accurate reflection of children's communicative abilities. The perspective taking abilities of 4 to 5 year olds do seem to be greater than previously thought (San Juan et al., 2025).

What could psycholinguistic frameworks add to our understanding of developmental pragmatics? If we had a computational model of adult performance, the question would be what aspects to change or restrict to approximate child behavior. For an

RSA type model, we could consider a lower optimization parameter, or reducing the amount of computation (sampling fewer options). Children might also do a similar process, but with a different size and baseline semantics in the lexicon. Would their updates in a repeated model be smaller or bigger? Even if we can't fit the model, the modeling formalism suggests ways of thinking about the differences between children and adults in a dimensional way.

If we take efficiency to mean an optimal tradeoff between expressivity and effort, then children might have different weightings between the two, ending up on a different frontier or a different part of the frontier. For instance, cognitive capacity constraints, or things like working memory or executive control could mean that certain types of utterances are particularly effortful, and thus shift the optimal curve to include less expressive utterances. For child – adult dyads, the shift in speaker / listener capacity may be substantially different than for adult - adult dyads, rendering different approaches more efficient. This may be difficult to test since it would require figuring how where the bottlenecks are empirically, or having good enough models to compare fits to the data.

From in information theory lens, we could conceive of communication channels involving children as being extra noisy or having a more limited channel capacity or both.

1.4.6 Incremental approaches to processing and production

Another angle from which to view reference is at the level of fine grained time course. As a referring expression is incrementally processed, what is the listener doing? How early do effects related to speaker identity or pragmatics occur?

The typical approach to understanding the unfolding of comprehension in real time is via visual world eye tracking where looks are interpreted as meaning to potential targets. Alternative methods to get at incremental interpretation and processing difficulty can also be used.

One idea that visual world approaches to reference try to address is whether there is an initial stage of interpretation that differs from the final resolution. For instance, certain influences like the literal meaning of the utterance might be available sooner than pragmatic or more contextualized interpretations. This is sometimes characterized as “top-down” versus “bottom-up” but there may also be different sources such as linguistic context or social context or visual access.

The relative balance and time course of top-down and bottom-up influences has been a big question in psycholinguistics and neurolinguistics for a long time (Gwilliams & Davis, 2022; Horton & Gerrig, 2005; Horton & Keysar, 1996; Tanenhaus et al., 1995). One debate in particular, in the (non-iterated) reference game literature specifically concerned the role of egocentrism in the early stages of linguistic processing. One side of the debate was the Egocentric Anchoring and Adjustment theory, where Keysar et al. (2000) argued for an initially egocentric perspective on the basis that people often initially look at objects that are good matches to a description even if the object is not mutually visible to the speaker Keysar et al. (2003). In particular, this theory claims a two-stage model where there is an initial (completely) egocentric perspective that can later be overcome, in contrast to the (much less controversial) idea that there is a general egocentric bias where one of the sources of information is egocentric but it can be in parallel to theory of mind information (Rubio-Fernández, 2008). The ego-centric first claim rests on a couple dubious linking hypotheses: that if people consider an interlocuter’s perspective, their prior should be that the interlocuter only refers to mutually known things; and separately, that looking at an object is a sign that it is considered a potential referent (eye-tracking data is widely interpreted this way, but there is evidence that this proxy is only approximate, Degen et al., 2021). Their studies also always had the egocentric-only lure be extremely salient, by making it by far the best match to the description.

The counterpoint to initial ego-centrism presented by Hanna et al. (2003) is a constraint-based theory where many factors can play into comprehension. Many factors may influence language production and comprehension, to varying extents, and on differing time courses. Their studies use egocentric lures that are equivalently good to the shared visibility target and find early perspective taking, and confirm this with other experiments in different paradigms (Hanna et al., 2003; Nadig & Sedivy, 2002).

The larger picture of language processing and neuroscience strongly suggests that context information (at least from linguistic context) is used to predict upcoming linguistic material, and thus that "top-down" information can affect even early stages of linguistic processing such as phoneme identification and lexical processing (Gwilliams et al., 2023; Gwilliams & Davis, 2022). Predictive coding is a theory with good coverage of neural evidence in many domains, including language, with language processing regions seemingly sensitive to both lexical and structural information in predicting upcoming words (Shain et al., 2020). However, we might still wonder if earlier behavioral readouts are more influenced by perceptual information and less pragmatic, or if cognitive load also sways the balance of influence.

1.4.7 Production constraints

Compared to the theoretic landscape around psycholinguistic processing, production is much less studied because it's a lot harder to study experimentally.

There is agreement that production requires conceptualization, formulation, and articulation, but that is about all that is agreed upon (Bürki, 2018). From interference studies, we can get some sense of when certain elements in sentences are planned (Momma et al., 2016). On a word level, the phonological realization of words that are more frequent or usually appear in more predictable contexts is reduced relative to words that usually convey more information (Bürki, 2018). This could be consistent with either a production facilitation account or an audience design account – speaker and listener language statistics are similar so it is difficult ot disentangle. For production beyond the scale of single words, that are just more questions, including what the planning looks like, how the time courses of different stages of production may overlap, and what size and types of units are stored in memory (Bürki, 2018). There's a lot we don't know about the mechanics of production even in situations with less context-sensitivity than iterated reference games.

As with behavioral measures, neuroscience measures are also used more for processing than for production, and articulatory processes are better studied than the higher level planning processes. fMRI studies indicate that language production tends to activate the same regions as language processing (unsurprisingly), but these regions are especially active during phrase structure building, suggesting that production may have higher costs than comprehension (Hu et al., 2022). However, fMRI does not have precise time course resolution, so it can't resolve questions of how information flows in the brain.

The increased load from production is consistent with accounts that suggest that utterance design (and language evolution) are substantially shaped by biases to ease production. MacDonald (2013) points out three production biases: easy first, plan reuse, and reduce interference that could explain some patterns of utterance production, including in reference games.

One approach to identifying speaker-side difficulties in iterated reference games is to look for disfluencies (S. O. Yoon & Brown-Schmidt, 2014). Intuitively, producing descriptions of abstract figures is difficult and there may be load based on task difficulty that leads to non-optimal productions (ex using "square head" as part of a description when it is not diagnostic) that are reminiscent of children's describe/identify difficulties.

There are approaches one could take to try to understand more what production-side pressures are, including time-course locked measures such as eye-tracking and disfluency analysis. Carefully controlled studies and comparisons to monologue conditions may also help. Varying time pressure or pre-articulation planning time could also be used to help sort out what's going on (Horton & Keysar, 1996). But as the larger questions around production generally show, this is hard to study and may require very careful and clever experiment designs.

What do the broader coverage frameworks have to say about production? Production-side constraints are one of the factors posited to drive linguistic efficiency, but there aren't fleshed out predictions about speakers would or wouldn't say. In general RSA operates on a level of abstraction that only considers entire utterances and so only considers production cost in terms of the utterance sampling procedure (if not all options are considered) and the costs assigned to each utterance. However, given that most iterated reference games allow the speaker to keep talking, one could (again, in theory) adapt incremental RSA (Cohn-Gordon et al., 2018) to explain potential order effects within what parts of descriptions are produced first.

One information-theoretic computational-level model of incremental language production comes from Futrell (2023). This model applies the Rate Distortion Theory of Control to production, suggesting a trade-off between incremental steps that are good for the communicative goal (ex. being informative) with what is easy to produce, that is what is more automatic and predictable. As with other information-theoretic models, it has good explanations in toy or qualitative domains for explaining production ordering preferences, the use of filled pauses, and good-enough word production.

1.5 Where does this leave us?

A core theoretical question is whether the observed patterns of reduction, convention, and partner-specificity are due to specific mechanisms that only operate in a limited scope of iterated-reference-game-like situations or whether these results are explainable by broader coverage theories of pragmatic language use.

The phenomena observed in iterated reference games do not appear under other circumstances where people use pragmatic reasoning (including regular reference games). One possibility is that there are mechanisms specific to certain circumstances, and these mechanisms lead to the observed phenomena. For instance, some theories may focus only on conversational coordination with common ground and a backchannel. The exact scopes are often ambiguous, but they are not trying to account for all language use or even all pragmatic referential language productions. In some cases, the implicit scope may be smaller than the range of circumstances where the phenomena is observed, as is the case for theories that focus on dyadic mechanisms.

Alternatively, it could be that the same mechanisms that account for pragmatics broadly can also account for the phenomena in iterated reference games. The particular phenomena of interest in iterated reference games arise from general mechanisms playing out in certain circumstances such as when targets have low codeability and reference is repeated to the same target with the same partner. These circumstances are incorporated into the model as a parameters (what the targets are, what the contextualized lexicon is, what the communication channel is) and certain combinations of the parameters lead to the emergence of things like partner-specific convention formation.

Most of the iterated reference game literature has been focused on descriptive results and developing small-scope theories. Here I've been examining what sorts of parameterizations and auxiliary hypotheses would be needed for broad-scope theories to account for the results of iterated reference games. I believe our default hypothesis should be that convention formation emerges from the same mechanisms that account for all the rest of language use. As we attempt to figure out what set of mechanisms account for language use, iterated reference games are an interesting model for highly contextual language production and comprehension.

Having surveyed the phenomena and small-scope theories of iterated reference games, I now return to the three frameworks and synthesize how they might apply and what I think the interesting open questions are.

1.5.1 Efficiency

Sometimes, judging whether something is efficient or not may be clear cut, if all reasonable sets of assumptions return the same result. It is perhaps easier to judge something to be inefficient if there is a shorter alternative that is both easier to produce and easier to comprehend. In the general case, however, efficiency is very hard to cache out in specific predictions. What's efficient for an utterance in isolation may not be efficient when considered over an entire life of language use.

The formation of conversational pacts is often characterized as efficient, but this claim has not been cached out in formal models (Clark & Wilkes-Gibbs, 1986; Hawkins, Frank, et al., 2020). "Efficiency" is an informal description of these pacts that captures that shorter expressions are *more* efficient, but a given shorted expression could still be mis-calibrated to a particular situation. It could be too short, leading to misunderstanding, or still longer than it needs to be for effective communication. Mis-calibration of this type could contradict claims about partner specificity of reduction, but would not have to. Communicators could intend to be optimally calibrated but run into production difficulties, masking their competence. Either way, current claims of efficiency in conversation do not connect directly with the formal claims of optimal efficiency in the "language design" literature, and it is unknown whether speakers and listeners are calibrated or not.

Even with a specified task goal (communicate the target to a partner), what we tell people and what their actual utility function is are not the same. It's still a conversation between people so communication norms may also play a role, and that's difficult to control while retaining naturalism.

There are some open questions that are testable, mostly focused on figuring out what sorts of linking hypotheses would be required for the behavior to be (mostly) efficient.

One is figuring out how understandable utterances are from different points in the conversation, to basically test the idea that speakers are reducing "as fast as possible" for listeners; that is, both that extra words in an utterance are adding utility (adding information, adding certainty), and that the context of earlier utterances (and their repetition) is necessary for understanding later utterances (i.e. reducing faster would negatively impact accuracy). These are addressable questions, both with models and information theory applied to utterances and with testing new matchers on manipulated samples or orders of utterances.

The other issue is trying to get a handle on production-side demands. If speakers have the hard task, then perhaps what they say is mostly driven by ease of planning. This is harder to test, but one could do monologue studies and dialog studies with eye-tracking and time course measures to try to get at it in more fine grained detail.

Generally efficiency is usually applied to language the culturally evolved system based out of pressures that apply at each step. But it doesn't actually require that the individual steps be fully at the frontier, merely that there be enough pressure towards expressivity and ease overall that the system shifts in this direction. So, even if it turns out that people really don't seem efficient at this task, it doesn't hurt the efficiency framework that much because this task may be substantially out of distribution and thus, people's behavior on it is not shaping language use on a large scale.

1.5.2 Bayesian pragmatics

Bayesian pragmatics is a very promising large-scale framework for explaining a lot of linguistic behavior in context and incorporating history and non-linguistic context into a model of utterance production and comprehension. The main challenge is figuring out how to scale up to handle the natural language that comes up in reference games.

There are also some modeling considerations for how to extend models to deal with different types of data – how to deal with multiple listeners simultaneously, how to abstract across items to learn conceptual paradigms (i.e. abstracting from items to an approach of describing images in terms of what the person is doing). One question that comes up with Bayesian pragmatic models is how many free parameters they have and whether one is actually getting a parsimonious model with broad coverage, or just a set of related models very tuned to individual datasets. This is a broad issue in modeling and the closer we can get to a generative or cognitive model (even if fairly abstracted) the better.

So one way of making progress and seeing how broadly Bayesian models can be applied would be extending the existing models for a wider variety of designs. Group communication, either in the form of one speaker - multiple listeners (S. O. Yoon & Brown-Schmidt, 2014, 2019a, 2019b) or network structures (Hawkins, Goodman, et al., 2020), poses a challenge as there is not yet a non-dyadic extension of RSA that can do partial pooling across listeners and jointly optimize for the responses of multiple interlocuters. It should be possible to design extensions to CHAI to accommodate different group structures.

However, it would still lead to questions of whether the small-scale models can predict the patterns see in natural language, full scale experiments. And that requires solving problems related to the open vocabulary. How do you set baseline (literal) meanings for any utterance someone might produce, and how do you allow those to update in a sensible way? With toy models, one can do composition semantics and update based on the compositionality. While we could ground semantics in some neural model (ex. in some sort of VLM) that doesn't give composition units that can be individually updated. Thus, I think some of the challenge of getting a robust test of Bayesian pragmatics models requires a lot of NLP and ML engineering.

CHAI, while not a full explanation for the full patterns of real-world data, is a big step forward in at least providing a framework that actually explains why reduction would be optimal. It remains to be seen whether RSA-style models can predict the slope of reduction and the content of words that stay versus drop, but I believe the attempt would push forward our understanding of pragmatics and communication. Even without a cached out model, this approaches can inform how we think about experiments and results.

1.5.3 Information theory

Unlike efficiency and Bayesian pragmatics which are frameworks, information theory is more of a tool. It's robust to see that in iterated reference games, the utterances tend to get shorter, but what's actually going on with the language? In order to quantify that, it's useful to have some sort of metrics to use. One could theorize that words that are more informative are more likely to stick around and become the conventions. To test this, one would need a measure of "information" to apply, which may now be possible to test using computational models of language and stimuli.

Information theory also provides measures for trying to understand how context supports understanding. This could be to formalize ideas of common ground by considering the joint information between the conversation history (as the common ground) and the current utterance. Again, to measure this at scale we would need models to approximate the information, but there are now models that could be used.

The idea of task relevant information also makes this an interesting place to apply linguistic measures of information theory. For normal psycholinguistic texts, information theoretic predictors like surprisal or entropy reduction have substantial predictive value on processing difficulty as indexed by measures of reading time (or neural signals) (Smith & Levy, 2013; Wilcox et al., 2021). While surprisal is a measure of information, we might think that it's not the same as how many useful bits of information are conveyed. (Gibberish is quite high surprisal, but low information.)

One way that information theory might usefully frame what happens in iterated reference games is that the language is adapting to the specific task of optimally coding the target options. An optimal code would only need 3.5 bits to express each of the 12 options (a common number of options in iterated reference games). Many more bits than this of natural language are used in early rounds – suggesting that initially standard natural language is not a good fit for this task. Some of this language is conveying task-relevant information, just in a wordy way, while other parts may be not-task relevant at all (ex: describing features that do not vary across targets). Over time, people probably both drop some of the filler and identify the task-relevant components, but they also develop a more efficient code for those features. This formulation makes some testable predictions about the entropy of distributions over utterances at different points during the game.

1.5.4 Summary

From this survey of approaches as well as phenomena and theories, I think there are a few takeaways. I've highlighted some places where further work is needed to answer the question of how well broad-coverage psycholinguistic approaches explain iterated reference game results. Some of this is completely outside the reference domain – we could do with a better understanding of (general) language planning and production, and we could benefit from natural language processing models that do compositionality that could be plugged into Bayesian frameworks to interface with real data. There are also some computational and modeling directions, highlighted above.

One other direction is better characterizing the **explanandum**: what exactly are the phenomena we are hoping to characterize? While there have been many studies of iterated reference games across some variety of conditions, some theoretically relevant aspects of the space are under-explored. Some of the reference game specific theories focus on narrow parts of iterated reference game space, while all the broad coverage approaches expect a mostly continuous relationship between circumstantial knobs and the outcome measures. From a modeling perspective, having more data and a wider range of data also reduces some concerns that we are over-fitting to a very narrow circumstance.

Thus, I believe one necessity to substantial progress in the iterated reference game domain is a better empirical and descriptive understanding of the phenomena of interest. How do circumstantial knobs (group size, communication modality, stimuli, etc.) push around the extend to which we observe reduction, convention formation, and partner specificity? We don't have to finely map the entire landscape, but some sense of the geography of experimental space is necessary (Almaatouq et al., 2022).

More variable experimental circumstances could also address the question of how closely related the different phenomena of interest are. At least where it's been studied, reduction, semantic alignment within a group, and partner specificity have patterned together (to the point where some metrics are used as proxies for each other). How closely are they actually linked? Does reduction only occur via convention formation? A greater range of experimental situations could tell how well these measures separate versus correlate in different situations, which will in turn clarify what theory should explain.

1.6 Outline of the dissertation

In this dissertation, we expand the empirical foundation of iterated reference games in three directions, with the aim of addressing some theoretical assumptions and questions and providing some data for computational modeling in the future. We also apply some computational models to survey coarse-grained patterns of changes in semantic similarities.

In Chapter 2, we address the generalization patterns of the key patterns of results in iterated reference games, by examining the trajectories of convention formation in games of varying small group sizes and varying communication channel capacities. This expands the empirical foundation to mimic more of the different circumstances in which communication occurs in the real world, and allows for a test of how entangled the different reduction phenomena really are.

In Chapter 3, we turn to the developmental timeline of the communication and coordination skills embedded in iterated reference tasks. Here we re-visit historical claims that children are too egocentric to produce successful ad-hoc referential expressions, which is fairly incompatible with more recent work showing nascent pragmatic language use and communication skills in preschoolers. This different context again provides a test of how entangled different observed reduction phenomena really are and provides insights into how language use fits into broader cognitive and linguistic development.

Finally, in Chapter 4, we measure the level of opacity in the referential expressions produced in iterated reference games (using the corpus of reference games from Chapter 2). Taking inspiration from lexical uncertainty models, we use both naive human matchers and computational proxies to determine how far from "baseline semantics" referential expressions are at different stages of convention formation. These patterns are a test of at least the strongest versions of efficiency and audience design claims and give us a better understanding of what sort of partner-specificity is at play.

In the last chapter, I conclude with a discussion of where this leaves us and what directions seem most promising from here.

Chapter 2

Interaction structure constrains the emergence of conventions in group communication

2.1 Introduction

Much of human social life revolves around communication in groups. At school, teachers address large classrooms of children (Cazden, 1988); at home, we chat with groups of friends and family members over dinner (Tannen, 2005); and at work, we attend meetings with colleagues and managers (Caplow, 1957; Zack, 1993). Such settings present considerable challenges that do not arise in the purely two-party (dyadic) settings typically studied in psychology (H. Branigan, 2006; Ginzburg & Fernandez, 2005; Traum, 2004). For example, producers need to account for the fact that different comprehenders in the group may have different mental states or levels of background understanding (Fox Tree & Clark, 2013; Horton & Gerrig, 2002a, 2005; Weber & Camerer, 2003; S. O. Yoon & Brown-Schmidt, 2014, 2019a), while comprehenders must account for the fact that utterances are not necessarily tailored to them (Carletta et al., 1998; Cohn-Gordon et al., 2019; Fay et al., 2000; Metzing & Brennan, 2003; Rogers et al., 2013; Tolins & Fox Tree, 2015; S. O. Yoon & Brown-Schmidt, 2014). What enables producers and comprehenders to nevertheless overcome these challenges and navigate multi-party settings with relative ease?

One promising set of hypotheses centers on the group's *interaction structure*, the set of constraints placed on the group's shared communication channel. Many different aspects of interaction structure have been implicated in the effectiveness of dyadic communication, including the availability and quality of concurrent feedback (Krauss & Bricker, 1967; Krauss & Weinheimer, 1966; Kraut et al., 1982), the bandwidth of the communication modality (Dewhirst, 1971; Krauss & Glucksberg, 1977), and the group's access to a shared workspace (Clark & Krych, 2004; Garrod et al., 2007). Yet larger groups introduce qualitatively different dimensions of interaction structure, leading to a large but often inconsistent body of findings even for these well-understood factors (Hiltz et al., 1986; Swaab et al., 2011). While communication is generally expected to deteriorate as groups get larger (MacMillan et al., 2004; Seaman & Basili, 1997), the structural "thickness" of the feedback channel may slow such deterioration (Ahern, 1994; Parisi & Brungart, 2005).

In this paper, we develop an experimental paradigm for evaluating the relative contribution of these factors: a *multi-party repeated reference game.* The ability to distinguish one particular entity from other possible entities, known as *reference*, is one of the most primitive and ubiquitous functions of communication. Reference games (Lewis, 1969; Wittgenstein, 1953) have been widely used to study dyadic communication under controlled conditions in the lab. They provide a clear metric of communicative effectiveness: how many words are required before a matcher successfully chooses a target image from a context of distractors? *Repeated* reference games, where the same target images appear multiple times in succession, were introduced to examine how interlocutors establish shared reference in the absence of conventional labels (Clark & Wilkes-Gibbs, 1986; Krauss & Weinheimer, 1964). At the beginning of the game, long and costly descriptions are typically required to succeed. A key finding, however, is that dyads become increasingly efficient over the course of interaction. Fewer words are required to achieve the same accuracy, but referring expressions also become more impenetrable to outsiders (Schober & Clark, 1989; Wilkes-Gibbs & Clark, 1992). The evolution of referring expressions over repetitions shows the characteristic dynamics of conventions: *stability*, or convergence on labels within a group, and *arbitrariness*, or divergence to different across groups, suggesting that dyads leverage their shared communication history to coordinate on expectations about how to label the target images (Hawkins, Frank, et al., 2020).

In principle, repeated reference games provide a strong operationalization of communicative effectiveness for the problem of multi-party communication: describers must simultaneously achieve shared reference with multiple matchers. However, empirically studying multi-party communication raises a number of difficulties in practice. A much larger pool of participants must be recruited to achieve sufficient power at the relevant unit of analysis – the group – spanning a very high-dimensional space of possible parameter settings (Almaatouq et al., 2022). We address this problem by drawing on recent technical advances that have made it newly possible to achieve such samples using interactive web-based platforms (Almaatouq et al., 2021; Haber et al., 2019; Hawkins, Frank, et al., 2020). Repeated reference games in web-based platforms have previously replicated earlier results from face-to-face studies, although people produce fewer words in text modalities than oral modalities (Hawkins, Frank, et al., 2020). The text-based chat modalities arguably more closely resemble the interfaces used by modern teams who increasingly communicate through group text threads or popular platforms like Slack or Discord.

We leverage our platform to explore effects of group size and interaction channel thickness in a series of three experiments. While we find that small groups reliably converge on group-specific "shorthand" regardless of the interaction structure, larger groups require thicker channels – richer conversational feedback among members – to achieve the same degree of coherence. Thus, increasing group size alone does not impede communication; rather, larger groups may require stronger social and linguistic cues to establish common ground among all members. More broadly, our work suggests that studying communication in larger groups is necessary to unveil critical aspects of interaction structure that have not been evident in typical dyadic settings.

2.2 Results

We recruited 1319 participants through Prolific, an online crowd-sourcing platform. Participants were organized into 313 groups of size two to six for a communication game (Figure 2.2A). On each trial, everyone in the group was shown a gallery of 12

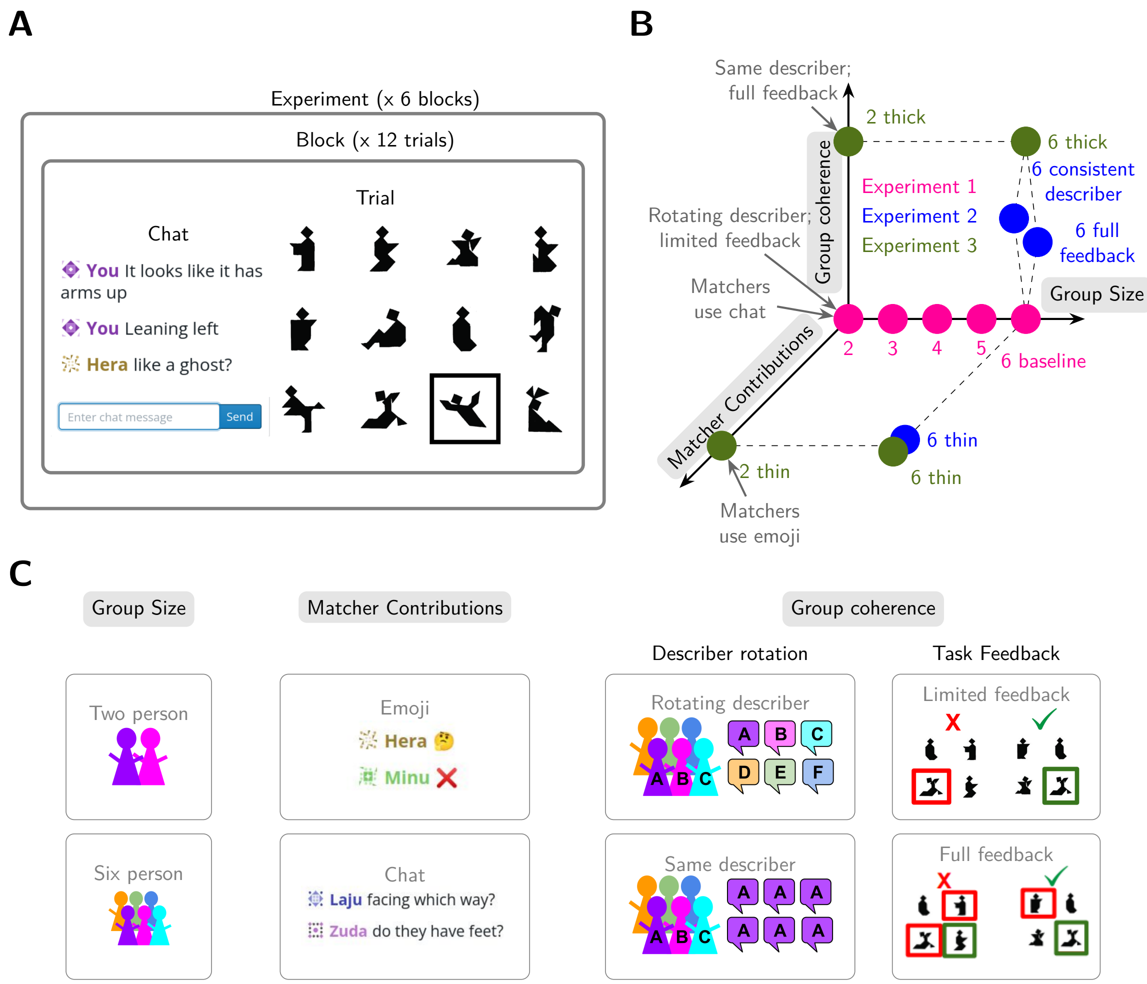


Figure 2.1: (A) Participants played a repeated reference game in groups of size 2 to 6. On each trial, a describer described the target image to the group of matchers. Each image appeared once per block for six blocks. (B) Experiments varied along 3 dimensions: Group size, group coherence, and matcher contributions. (C) Experiment 1 (pink) varied group size from 2 to 6 players while holding group coherence and matcher contributions constant. Experiment 2 (blue) held group size constant at 6 and manipulated the other dimensions. Experiment 3 (green) tested 4 corners of the space, crossing group size (2 vs. 6 players) with the thickness of interaction structure (high vs. low coherence and matcher contributions).

tangram images (Clark & Wilkes-Gibbs, 1986; Hawkins, Frank, et al., 2020; Ji et al., 2022). One player was designated the *describer* and the others were designated the *matchers*. The describer was asked to use a chat box interface to describe a privately indicated *target* image. After all matchers guessed which of the 12 images was the target, they received task feedback and proceeded to the next trial. The game consisted of 72 trials structured into 6 repetition blocks, where each image appeared as the target exactly once per block.

We manipulated the interaction structure of this game across 11 distinct conditions in 3 distinct pre-registered experiments (Figure 2.2B). We systematically sampled points along four dimensions parameterizing different aspects of the interaction space. We manipulated *group size* (ranging from two to six), *role stability* (whether or not participants took turns in the describer role), richness of *task feedback* (whether or not matchers were able to see each other's responses), and richness of the *matcher contributions* (whether matchers were able to freely respond through a chatbox or could only use emojis; Figure 2.2C). Other factors, such as the set of stimuli and background knowledge about one's partners, were held constant across games.

2.2.1 Overview of experiments

Experiment 1 began by investigating how performance scaled with group size. Based on prior qualitative work, we predicted that larger groups face a more challenging coordination problem. We continuously varied the number of players from 2 to 6 while keeping other factors constant. For these conditions, the describer role rotated after each block, so that all players had at least one turn as describer. Matchers had access to an unrestricted chat box, but only received binary task feedback about whether their individual selection was correct without revealing others' selections or the intended target.

Experiment 2 focused on the most challenging 6-player groups and explored the role of interaction structure. Each condition in Experiment 2 varied one aspect of the experiment relative to the Experiment 1 6-player baseline. We tried two variants that we expected to increase group coherence and improve performance, and a third variant we expected to interfere with the ability to establish mutual understanding and thus impede performance. In the first variant, we maintained the same describer throughout rather than a rotating describer, such that the same individual has the opportunity to aggregate feedback across trials and track which matchers are struggling with which targets. In the second variant, we gave the group of matchers full feedback about what every other member of the group had selected, and we showed the intended target. In the third variant, we changed how matchers could make contributions to the group. In contrast to prior experiments, where matchers could contribute freely to the chat; here, we limited matchers to sending four discrete emojis (green check, thinking face, red x, and laughing-crying face) that could convey simple valence and level of comprehension, but not any referential content.

Experiment 3 crossed the extremes of group size from experiment 1 (2 vs. 6 people) with the extremes of group interactions from Experiment 2 (*thick* vs. *thin* interaction structure). In the *thick* condition, we maintained a consistent describer, gave all matchers full task feedback, and allowed them to freely use a chat box. In the *thin* condition, we forced the describer to rotate on each block, restricted feedback to their own binary correctness, and restricted matcher contributions to the four emojis. Note that the 2-player thick game most closely resembles the design of classic repeated reference games (Clark & Wilkes-Gibbs, 1986).

2.2.2 Smaller and higher-coherence groups are more accurate

Our first set of hypotheses focused on group performance: how accurately and efficiently groups were able to perform the referential task. We characterize group performance along two complementary metrics: (1) matcher accuracy and (2) describer efficiency. Matcher accuracy is given by the percent of matchers on each trial who successfully selected the target referent. Describer efficiency is given by the number of words produced by the describer to achieve that degree of matcher accuracy in the group. The degree to which describers are able to communicate more efficiently without negatively impacting matcher accuracy is indicative of convergence on a more effective shared communication protocol within the group.

We begin by examining matcher accuracy, the extent to which the intended target was reliably transmitted to all matchers. We constructed a series of 5 logistic mixed-effects regression models predicting accuracy as a function of condition and repetition block (separate models were run for experiment 1, each condition in experiment 2, and experiment 3). For this and other effects, there was substantial variation at the

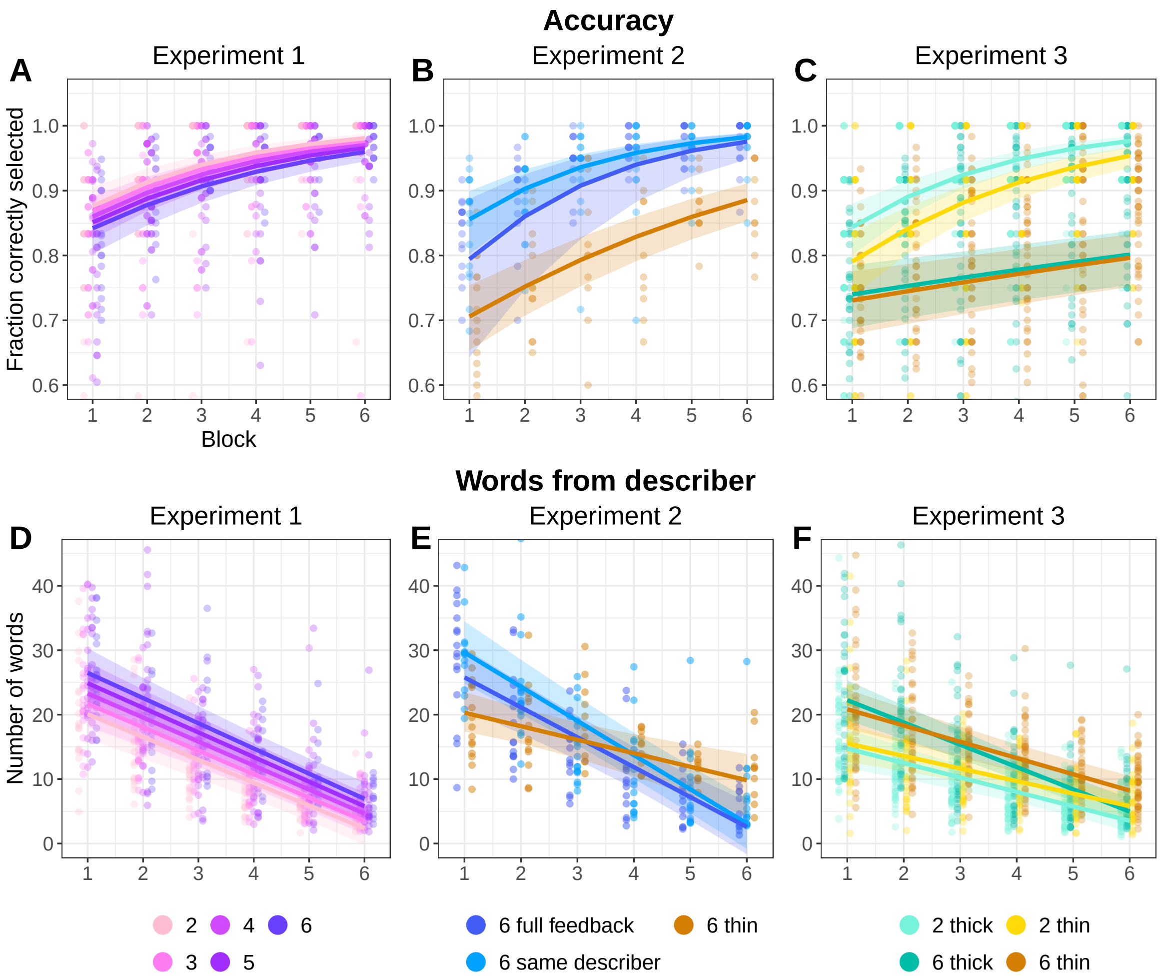


Figure 2.2: Behavioral results across all three experiments. (A-C). Matcher accuracy at selecting the target image. (D-F). Number of words produced by the describer each trial. For all, small dots are per game, per block means, and smooth lines are predictions from model fixed effects with 95% credible intervals. Y-axes are truncated, and a few outliers points are not visible.

tangram and game levels, with some tangrams being markedly easier than others and some groups performing differently than others. This wide variation made it difficult to precisely estimate population-level main effects, leading to wide credible intervals. See Appendix Figure A11 for a visualization of the relative magnitudes of population effects and game and tangram level variations.

Across all conditions, we observed strong positive effects of repetition block, indicating improved performance over time (Figure 2.2A-C, Appendix Tables A4-A8). In Experiment 3, larger games began with lower initial accuracy ($\beta = -0.64$, 95% CrI = $[-1.05, -0.25]$) and improved more slowly ($\beta = -0.34$, 95% CrI = $[-0.43, -0.25]$) than smaller games, although group size differences were not reliable in Experiment 1 (Table A4), and these experiment 3 differences were not robust in a sensitivity analysis (Appendix Figure A4C and Table A58). Among large groups in Experiment 2, accuracy was higher in the thicker conditions than in the condition with thin interaction structure (Tables A5-A7), although effects of game thickness were not reliable in Experiment 3 (Table A8).

Because each experiment only explored a slice of the full parameter space, we also considered an exploratory analysis that pooled data across experiments, aiming to mitigate the loss in power from running entirely separate regression models. Specifically, we aggregated data from all experiments into a post-hoc mega-analytic model predicting accuracy as a function of repetition block, game thickness (thin v. not-thin) and game size. Overall, we found evidence that accuracy increased over time ($\beta = 0.46$, 95% CrI = $[0.4, 0.52]$) but the rate of increase was reduced for thin games ($\beta = -0.12$, 95% CrI = $[-0.21, -0.02]$) and larger games ($\beta = -0.07$, 95% CrI = $[-0.09, -0.05]$) compared to smaller or thicker games. That is, smaller groups and groups with higher coherence tended to be more accurate, though the magnitude and reliability of these effects varied across individual experiments.

2.2.3 Smaller and higher-coherence groups are more efficient

After establishing that groups were able to communicate accurately, we turned to the challenges faced by describers when deciding how much information to provide. Specifically, we predicted that larger and more heterogeneous groups may initially require more information, but that thicker interaction structure may similarly allow describers to communicate more effectively over time. We tested these predictions using linear mixed-effects models predicting the number of words a describer produced on each trial as a function of condition and block. These models counted all words the describer produced, including after matcher contributions (similar effects were found in models predicting the length of describer's utterances before any matcher contributions, see Tables A21-A24).

First, as predicted, describers in larger groups produced longer descriptions at the outset than describers in smaller groups (Figure 2.2D-F). This effect held for the continuous manipulation of group size for Experiment 1 ($\beta = 1.6$, 95% CrI = $[0.62, 2.6]$) as well as the 2-person versus 6-person manipulation in Experiment 3 ($\beta = 7.51, 95\%\text{CrI} = [3.63, 11.3]$). Smaller groups also continued to use shorter descriptions than larger groups over the course of the game. In Experiment 1, the rate at which efficiency increased was similar across different size groups ($\beta = -0.09$, 95% CrI = $[-0.37, 0.18]$). In Experiment 3, larger groups reduced faster than smaller ones ($\beta = -1.22$, 95% CrI = $[-2.06, -0.29]$), but the faster reduction did not fully make up for the longer initial starting point, and was not robust to a sensitivity analysis (Figure A4F and Table A63).

While thin 6-person games showed a flatter reduction trajectory than thicker 6-person games in Experiment 2 (Tables A10-A12), there was no reliable effect of game thickness on reduction in Experiment 3 (Table A13).

The reduction patterns of description lengths is paralleled by how long matchers took to make selections; across conditions, matchers selected faster in later conditions (Figure A9), and the correlation between speed and description length was consistent across experiments (Figure A10).

Aggregating across experiments with a mega-analytic model, however, suggested that larger games were associated with steeper reduction ($\beta = -0.36$, 95% CrI = $[-0.51, -0.2]$) from a more verbose starting point ($\beta = 2.12$, 95% CrI = $[1.5, 2.75]$) than smaller games, and thin games had shallower reduction curves ($\beta = 0.79, 95\%\text{CrI} = [0.04, 1.52]$) than thicker games. Overall, then, smaller games used shorter descriptions than larger games across various time points in the experiment, and thinner games reduced less than thicker games.

Table 2.1: Examples from 6-player groups in Experiment 3 of successful descriptions for the same image across repetitions. Describers are indicated with an asterisk. More example descriptions are in Appendix Tables A1 and A2.

6-person thick game	
Rep 1: 4/5 correct	
A*	sitting down no legs showing
C	no arms?
B	Bunny ears?
A*	legs showing to one side no arms
C	kinda like kneeling?
A*	no feet
A*	yes kneeling
Rep 2: 5/5 correct	
A*	sitting down with no feet showing. legs to one side
A*	no arms
E	Swaddled up like a bb
C	cute bb
D	I think that's the one that looks like a ghost to me, like he's wearing a sheet. (?)
Rep 3: 5/5 correct	
A*	sitting down no arms showing legs to one side
E	the bb
B	Swadled baby?
F	Baby?
A*	the bb
Rep 4: 5/5 correct	
A*	sitting down with no arms legs to one side
Rep 5: 5/5 correct	
A*	sitting no arms and legs to one side. I will call them Kevin
D	<3 Kevin
C	bb kevin
Rep 6: 5/5 correct	
A*	kevin the baby
E	yess kevin

6-person thin game	
Rep 1: 5/5 correct	
L*	man with no arms or legs
O	🤔
P	🤔
L*	slight protuberance on the right
Rep 2: 5/5 correct	
N*	Can't see any arms. Imagine wrapped in a blanket completely.
N*	Armless and legless
N*	Burrito with a head
M	😂
O	😂
P	🤔
Rep 3: 5/5 correct	
O*	burrito
Rep 4: 5/5 correct	
Q*	burrito
Rep 5: 5/5 correct	
M*	burrito
Rep 6: 5/5 correct	
P*	And our last one! Anyone else hungry after this? I quite fancy a burrito!
L	😂
Q	✅
N	😂😂😂
Q	😂✅
M	✅
Q	😂
N	✅✅✅

2.2.4 Larger groups make greater use of matcher contributions

As a final measure of group performance, we examined the back-and-forth interactions between the describer and the group of matchers. Matchers use their chat contributions to actively provide feedback, ask questions, offer alternative descriptions, and seek clarification about the describer's referring expressions. Example transcripts from successful games, one in the 6-thick condition and one in the 6-thin condition, are shown in Table 2.1. Additional examples are in the Appendix Tables A1 and A2. Overall, we found that larger groups displayed a higher proportion of trials where at least one matcher produced utterances (Figure A6A, $\beta = 0.79, 95\%\text{CrI} = [0.58, 0.98]$), which declined across repetition blocks ($\beta = -0.8$, $95\%\,\text{CrI} = [-0.97, -0.62]$). On an individual level, a matcher in a larger group was more likely to make contributions than a matcher in a smaller group, although each contribution tended to be shorter (Figure A7, Tables A18, A20). The length of matcher interjections also decreased over time, especially for large groups (Figure A6D, $\beta = -0.41$, $95\%\,\text{CrI} = [-0.72, -0.11]$) consistent with the need for early matcher involvement in establishing referential conventions. Emoji use in Experiment 3 followed similar trends (Figure A8). Overall, describers in larger groups receive more total input from matchers, suggesting larger groups may require greater participation by matchers to reliably establish common ground.

2.2.5 Descriptions converge faster in groups with thicker channels

In the previous sections, we examined three metrics of communicative performance in groups of different sizes and interaction structures. We confirmed that groups in all conditions replicated the classic patterns of increasing accuracy and decreasing description length. We also found some initial evidence that larger groups may struggle to improve performance in the absence of thick communication channels. Here, we aim to better understand the mechanisms that allow describers to use shorter descriptions without sacrificing accuracy. In particular, we explore the hypothesis that

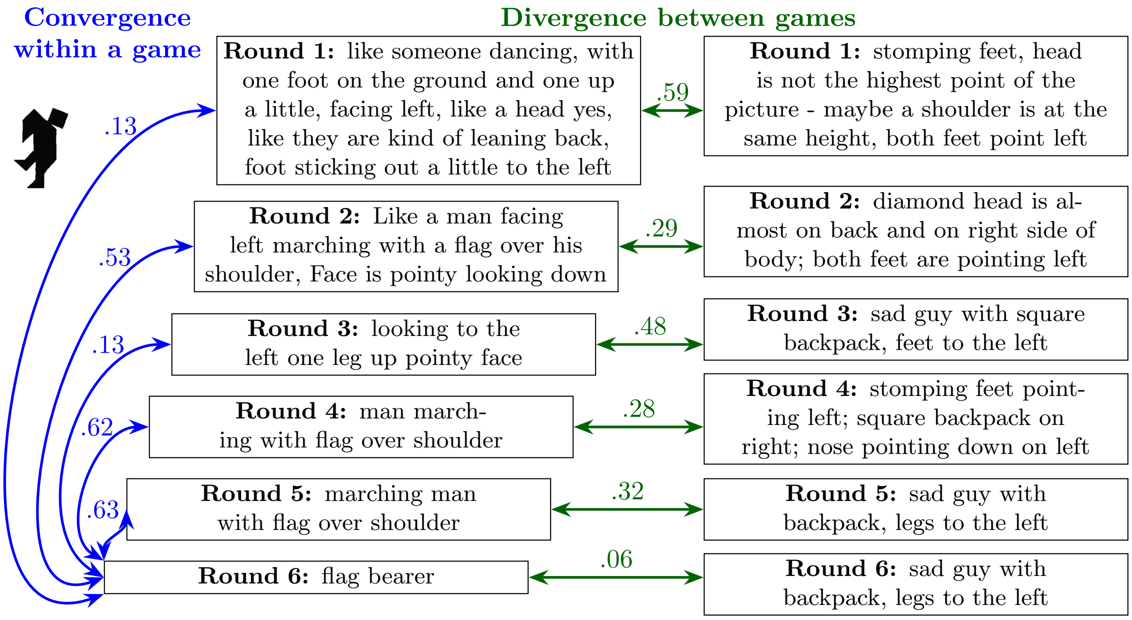


Figure 2.3: Example utterances describing the shown tangram figure produced by two 3-player games in Experiment 1. To measure convergence within a game (blue), we measured the cosine similarity between SBERT embeddings of descriptions and the embedding of the round 6 utterance (taken to be the convention). Higher cosine similarity indicates more similar meaning. To measure divergence between games (green), we measured the similarity between embeddings of utterances from the same round across games.

interaction structure and group size affect performance through a *convention formation* process (Clark & Wilkes-Gibbs, 1986). Under a recent model of convention formation (Hawkins, Franke, et al., 2023), groups are able to leverage their shared history to coordinate on stable expectations about how to refer to particular images. This model makes specific predictions about how interaction structure affects the ability to coordinate, in terms of the available feedback.

First, due to heterogeneity in the group – 6 individuals who may have diverging conceptualizations — a rational describer should provide a strictly more detailed initial description to hedge against multiple possible misunderstandings, as we previously observed. Second, all groups should display the characteristic dynamics of conventions: *stability*, or convergence within group, and *arbitrariness*, or divergence to multiple equilibria across groups. Third, convergence should be faster when a single individual is consistently in the describer role and when matchers are able to freely respond in natural language, as describers are able to aggregate feedback about the effectiveness of their own utterances from block to block and also immediately correct specific misunderstandings within a given trial.

To assess the dynamics of describer descriptions, we examine the *semantic similarity* of descriptions within and across games. We quantified description similarity by concatenating describer messages together within a trial and embedding this description into a high-dimensional vector space using SBERT. SBERT is a BERT-based sentence embedder designed to map semantically similar sentences to embeddings that are nearby in embedding space. Semantically meaningful comparisons between sentences are made by taking pairwise cosine similarities between the embeddings (Reimers & Gurevych, 2019).

To measure stability, or convergence within groups, we compared utterances from blocks one through five to the final (block six) description for the same image from the same game. To measure arbitrariness, or divergence across groups depending on group-specific history, we compared utterances produced by different describers for the same image in the corresponding blocks. Figure 2.3 illustrates these two measures with example utterances and their within-game and between-game cosine similarities.

We modeled semantic convergence with a mixed effects linear regression model predicting the similarity between a block 1-5 utterance and the corresponding block 6 utterance as a function of the earlier block number and condition (Figure 2.4A-C; Tables A25-A29). All conditions showed some convergence toward a conventional "shorthand" for the picture, but the speed of convergence was affected both by group size and channel width. First, we found that smaller groups reached stable descriptions faster than larger games. In Experiment 1, initial similarity was invariant across group size ($\beta = -0.008$, 95% CrI $= [-0.021, 0.005]$), but smaller groups converged faster (Figure 2.4A, $\beta = -0.008$, 95% CrI $= [-0.011, -0.005]$). In Experiment 3, 6-person thick games started off further from their eventual convention than 2-person thick games ($\beta = -0.069$, 95% CrI $= [-0.113, -0.025]$) but closed the gap over time (Figure 2.4C, $\beta = 0.009$, 95% CrI $= [0.001, 0.017]$, this effect was not robust to sensitivity analysis, Figure A5C and Table A68). Second, thicker games tended to converge

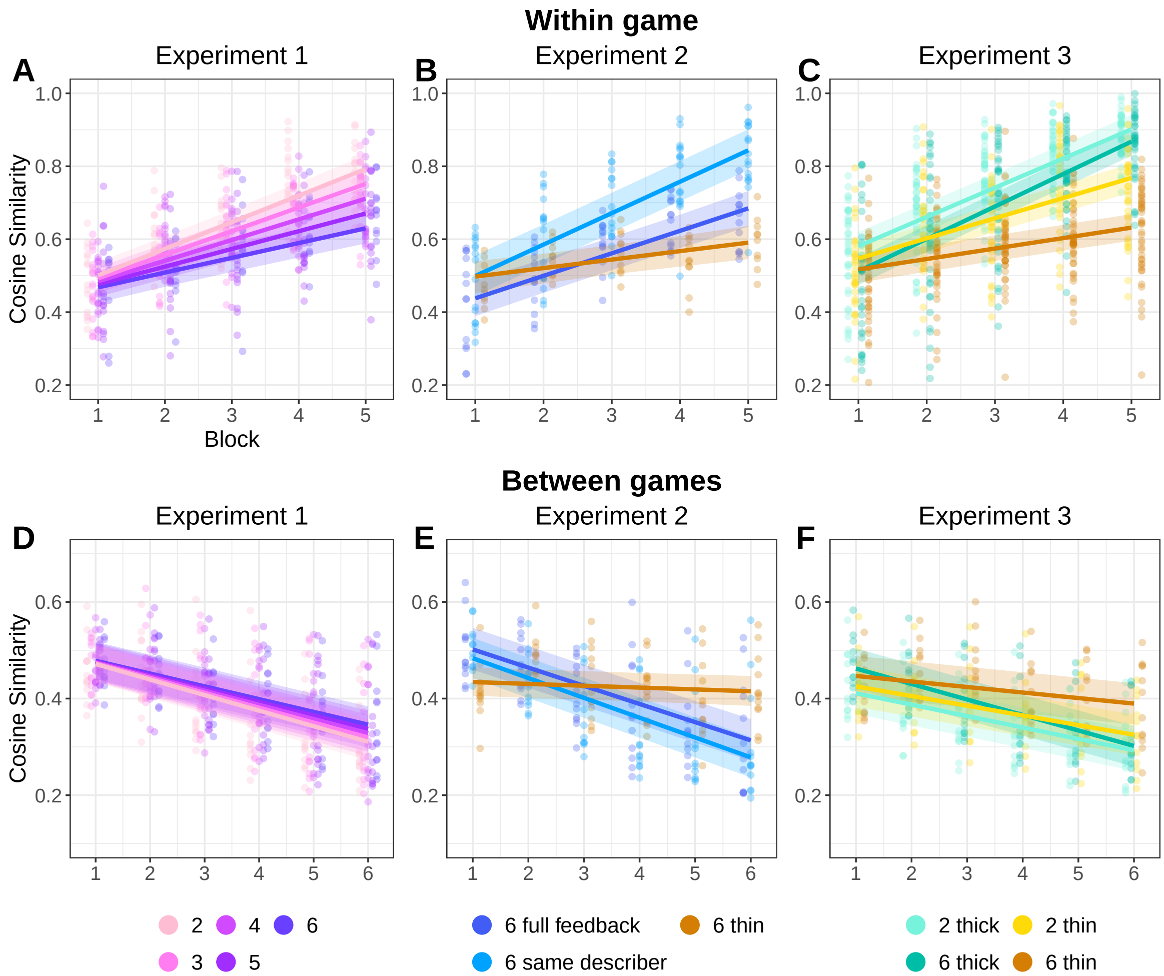


Figure 2.4: Language similarity results measured with pairwise cosine similarity between embeddings of two utterances. (A-C). Convergence of descriptions within games as measured by similarity between an utterance from block 1-5 to the block 6 utterance in the same game for the same image. (D-F). Divergence of descriptions across games as measured by the similarity between two utterances produced for the same image by different groups in the same block. For all, small dots are per game, per block means, and smooth lines are predictions from model fixed effects with 95% credible intervals. Y-axes are truncated, and a few outliers points are not visible.

faster than thin games (Figure 2.4B-C). In Experiment 3, small thin games started off slightly further from their convention than small thick games, and this gap widened over time ($\beta = -0.025$, 95% CrI = $[-0.033, -0.017]$). Finally, the combination of thin interaction structure and larger group hindered convergence more than either factor individually. Beyond the generally slower convergence in thin games, 6-person thin games showed substantially slower convergence even compared to 2-person thin games in Experiment 3 ($\beta = -0.035$, 95% CrI = $[-0.047, -0.025]$).

Pooling across experiments in a mega-analysis confirms this pattern. Thin games converge less than thick games overall ($\beta = -0.016$, 95% CrI = $[-0.025, -0.008]$), and *large* thin games are especially slow to converge ($\beta = -0.007$, 95% CrI = $[-0.01, -0.004]$). Across games, convergence towards the last utterance was driven by cumulative increasing similarity between pairs of utterances in adjacent blocks (Figure A12D-F, Tables A40-A44). In early rounds, descriptions could change substantially between rounds, but by later rounds, many descriptions had already reduced and solidified and varied little round to round. In summary, we found that stable descriptions emerged earlier if the group was smaller, or if the group had a thick interaction structure.

2.2.6 Games with thicker channels diverge from one another more quickly

While groups may initially overlap in their descriptions, including details of shapes or body parts, we predicted that their descriptions would become increasingly dissimilar as groups increasingly adapt to their own idiosyncratic shared history. To test this effect, we constructed a mixed-effects linear regression model predicting the cross-game similarity between a pair of utterances for the same image. A decrease in the similarity between different groups descriptions occurred in every condition, indicating increasing arbitrariness and group-specificity of descriptions (Figure 2.4D-F, Tables A30-A34). However, different game sizes and interaction structures revealed very different strengths of divergence.

First, smaller games used more group-specific language. In Experiment 1, smaller games diverged more quickly than larger games ($\beta = 0.001$, 95% CrI = $[0.001, 0.002]$). In Experiment 3, 2-person thick games started off more dissimiliar than 6-person thick games, although 6-person games diverged faster and eventually approached the dissimilarity levels of 2-person thick games (Table A34). Second, thicker interaction structure was associated with stronger group-specific divergence. In Experiment 3, 2-person thin games diverged more slowly than 2-person thick games ($\beta = 0.004$, 95% CrI = $[0.002, 0.005]$). As with the convergence patterns, large games with thin interaction structures had the flattest trajectories, as thinness and largeness compounded. In Experiment 3, 6-person thin games diverged even less than 2-player thin games (Figure 2.4F, $\beta = 0.017$, 95% CrI = $[0.015, 0.019]$), and in Experiment 2, 6-person thin games barely diverged at all (Figure 2.4E, $\beta = -0.004$, 95% CrI = $[-0.006, -0.001]$). A mega-analytic model confirms this pattern: thin games differentiate less between groups ($\beta = 0.005$, 95% CrI = $[0.004, 0.007]$) and large thin groups differentiate even less ($\beta = 0.004$, 95% CrI = $[0.004, 0.005]$).

As a complement to the embedding analysis, we also examined the frequency of a few classes of words in the descriptions. Literal geometric words (ex. square, triangle, etc) and words for body parts (leg, arm, etc) are common early in games, but decline over repetition in most conditions, to be replaced by more abstract descriptions that do not contain these classes of words (Figure A2). The 6-person thin condition, however, retains a higher level of literal geometric and body part words, along with high levels of positional words (above, left, below, etc) and posture words (kicking, standing, seated, etc), with a lower level of utterances that do not contain any of these classes of words.

2.3 Discussion

From classrooms to boardrooms, human communication often takes place in multi-party settings. However, experimental research rarely focuses on such settings, largely due to practical obstacles. In the current work, we asked how convention formation processes, typically studied in dyadic reference games, unfold in larger groups and under varying interaction structures. Across 3 online experiments and 11 experimental conditions, we varied multiple features of interaction structure including group size, modality of matcher contributions, and degree of group coherence. All conditions replicated classic dyadic phenomena: increasing accuracy and efficiency, semantic convergence within games, and differentiation of descriptions between groups. However, we also found that the interaction structure substantially affects how rapidly groups develop partner-specific conventions. Small groups may be able to successfully form conventions under limited feedback, but larger groups require thicker interaction structure. Multi-player groups may therefore reveal key factors which are masked in purely dyadic settings.

Increasing efficiency, for example, has often been taken as an index of group-specific convention formation (Brennan & Clark, 1996; Clark & Wilkes-Gibbs, 1986; S. O. Yoon & Brown-Schmidt, 2014, 2019a). In our work, however, we observe distinct patterns for measures of raw utterance length compared to the dynamics of semantic content. In Experiment 3, thin 6-person games showed much less group-specific divergence despite comparable accuracy and efficiency. This gap raises the possibility that it is possible to become more efficient and accurate without negotiating a unified group-specific label. Instead, they may be relying more strongly on the group's priors (Guilbeault et al., 2021). Thus, we encourage measures of semantic content (and not just performance) when evaluating convention formation. The transcripts for these games provide a rich dataset for exploring different ways language is used to form referential conventions.

The causal mechanisms driving group size effects remain unclear. There are many differences between a 2-person group and a 6-person group that could plausibly lead to different outcomes. For example, in a dyad, producers can tailor their utterances to the one matcher, but in large groups, producers must balance the competing needs of different comprehenders (Schober & Clark, 1989; Tolins & Fox Tree, 2015; S. O. Yoon & Brown-Schmidt, 2019a). These effects likely vary with the knowledge state and the communication channels available to comprehenders (Fox Tree & Clark, 2013; Horton & Gerrig, 2002a, 2005). Further work digging into the language used and the interactions between participants might unearth plausible mechanisms for how differences in group size and interaction structure influence outcomes, pointing towards future experimental conditions.

Even within the boundaries of the repeated reference game paradigm, there is a high-dimensional space of possible experiments. We sampled only a few points along a few salient dimensions. In our experiment 3, we grouped some factors together in order to run more games in each condition: a fully factorial design would have been too expensive to power adequately. We instantiated a "thin" channel by limiting matchers to 4 discrete utterances (emojis), but there are other possible restrictions that could be placed on the channel, such as rate-limited typing or explicit time pressure. Future work could explore other dimensions of the interaction structure, introducing pre-existing relationships or familiarity among group members, alternative incentives involving competition and power, or alternative referential targets involving more complex concepts.

A particularly important dimension shaping interaction is the modality of communication, including whether whether the participants use oral or written language, whether they are co-present in the same space, and whether they have visual access to each others' faces and gestures. Distinct modalities carry distinct affordances and norms. In this work, we relied on a text-based chat modality without allowing co-presence or visual access.

We suspect that the general pattern of effects we see, in terms of group size and coherence, are likely to extend to other modalities. However, different modalities may allow for different strategies that may be more or less sensitive to group size, describer rotation, or different levels of matcher contributions. For instance, in face-to-face oral settings, it may be easier for describers to continuously talk until interrupted, or to monitor the comprehension of individual group members from their facial expressions.

In conclusion, narrowly focusing on the settings that are easy to study in the lab – dyads with rich communication channels – can lead to theories that mispredict how interactions play out in multi-party groups. By studying common ground and coordination across a wider range of interaction structures, we can develop a more nuanced understanding of the obstacles that stand in the way of successful communication and how groups can overcome them. This understanding can inform the design of policies and collaborative platforms that promote effective communication in various contexts, from small-scale conversations to large-scale civic discourse. As remote work and online communication become increasingly prevalent, it is increasingly crucial to understand how the structure of group communication environments shapes the effectiveness of human communication.

2.4 Materials and Methods

Our iterated reference task was implemented with Empirica (Almaatouq et al., 2021), a React-based web development framework for real-time multi-player tasks. Our experiments were designed sequentially and pre-registered individually.[1] We followed the pre-registered analysis plan for each experiment, although accuracy models were not explicitly specified until Experiment 3, and linguistic analyses were only verbally described starting with Experiment 2b. Results from some pre-registered models are omitted from the main text for brevity but are shown in the Appendix. Exploratory mega-analytic models pooling across the three experiments were not pre-registered.

All materials, data, and analysis code is available at `https://github.com/vboyce/multiparty-tangrams`.

2.4.1 Participants

This research was covered by the Stanford IRB under protocol 20009 "Online investigations of language learning". Participants were recruited using the Prolific platform. All participants self-reported as fluent native English speakers on Prolific's demographic prescreen. Experiment 1 took place between May and July 2021, Experiment 2 between March and August 2022, and Experiment 3 in October 2022. Each participant took part in only one experiment and was blocked from participating in subsequent experiments. As games with more participants tended to run longer, we paid participants different rates based on group size, with the goal of a consistent $10 hourly rate. Participants were paid $7 for 2-player games, $8.50 for 3-player games, $10 for 4-player games, and $11 for 5- and 6-player games. When one player occupied the describer role for the entirety of a 6-player game, they were rewarded an additional $2 bonus. Across all games, participants could earn up to $2.88 in performance bonuses.

A total of 1319 people participated across the 3 experiments. We recruited enough participants for 20 games in each condition in experiments 1 and 2 and 40 games per

[1] Experiment 1: `https://osf.io/cn9f4` for the 2-4 player groups, and `https://osf.io/rpz67` for the 5-6 player data run later. Experiment 2: same describer at `https://osf.io/f9xyd`, full feedback at `https://osf.io/j5zbm`, and thin at `https://osf.io/k5f4t`. Experiment 3: `https://osf.io/untzy`

condition in experiment 3. However, due to attrition in filling the games initially and due to participants dropping out of the games, we ended up with fewer games in some conditions. For logistical reasons of matching participants into real-time games, we had to recruit participants in fairly large batches, and so did not have precise control to add new games to replace games that did not fill or had participants drop out early. A breakdown of number of games and participants in each condition is shown in Table A3 along with further discussion of recruitment logistics.

2.4.2 Materials

The same 12 tangram images, drawn from Hawkins, Frank, et al. (2020) and Clark & Wilkes-Gibbs (1986), were used every block. These images were displayed in a 4×3 grid with the order randomized across participants to disincentivize spatial descriptions such as “top left,” as the image might be in a different place on the describer’s and matchers’ screens. To reduce cognitive load from visual search, the locations were fixed for each participant across trials.

2.4.3 Procedure

The experimental procedure was very similar across the three experiments. We first describe the procedure used in Experiment 1 and then describe the differences in later experiments.

Experiment 1 Participants were directed from Prolific to our custom web application, where they were presented with a consent form and a series of instruction pages explaining the protocol. After finishing the instructions, they needed to pass a quiz to proceed. They were then directed to a “waiting room” lobby. Once the lobby filled to the required number of players, the game began. One lobby was filled before another was started; if a participant was waiting for 5 minutes, that lobby timed out, and the participant was paid without completing the experiment. Due to technical constraints with assigning participants to lobbies and games, only games of a single experimental condition could be active at a time. Thus, different conditions were run on different days or times of day.

One of the participants was randomly selected to begin in the role of describer, and the other participants were assigned to the role of matchers. On each trial, the describer saw a fixed array of tangrams with one tangram (privately) highlighted as the *target*. They were given a chat interface to communicate the target to the matchers, who were asked to determine which of the 12 images was the referential target. All participants were free to use the chat box to communicate at any time, but matchers could only make a selection after the describer had sent a message. Once a matcher clicked, they could not change their selection. There was no signal to the describer or other matchers about who had already made a selection. We recorded what all participants said in the chat, as well as who selected which image and how long they took to make their selections.

Once all matchers had made a selection (or a 3-minute timer ran out), participants were given feedback and proceeded to the next trial. Matchers only received *binary* feedback about whether they had chosen correctly or not; that is, matchers who made an incorrect choice were not shown the correct answer (see Figure A1 for example feedback). The describer saw which tangram each matcher selected, but matchers did not see one another's selections. Matchers got 4 points for each correct answer; the describer got points equal to the average of the matchers' points. These points were translated into performance bonuses at the end of the experiment (1 point = 1 cent bonus). After the describer had described each of the 12 images as targets, in a randomized sequence, the process repeated with the same set of targets, for a total of 6 such repetition blocks (72 trials).

The same person was the describer for an entire block, but participants rotated roles between blocks. Thus, over the course of the 6 blocks, participants were describers 3 times in 2-player games, twice in 3-player games, once or twice in 4 and 5-player games, and once in 6-player games. Rotating the describer was chosen in this first experiment to keep participants more equally engaged (the describer role is more work), and to provide a more robust test of our hypotheses regarding efficiency and convention formation. After the game finished, participants were given a survey asking for optional demographic information and feedback on their experience with the game.

If a participant disconnected from the experiment, the game would stop.

Experiment 2 Experiment 2 consisted of three different variations on Experiment 1, all conducted in 6-player games. Each of these conditions differed from the Experiment 1 baseline in exactly one way. In the *same describer* condition, one person was designated the describer for the entire game, rather than having the describer role rotate. In the *full feedback* condition, all participants were shown what all others had selected as well as the identity of the correct target. This condition was similar to previous dyadic work, such as Hawkins, Frank, et al. (2020), where the correct answer was indicated during feedback. In the *thin* condition, we altered the chatbox interface for matchers. Instead of a textbox, matchers had 4 buttons, each of which sent a different emoji to the chat. Matchers were given suggested meanings for the 4 emojis during the instruction phase. They could send as many emojis as desired; for instance, they might initially indicate confusion, and later indicate understanding. In addition, for the thin condition, we added notifications that appeared in the chat box marking the time when each player had made a selection.

Experiment 3 The thin channel condition in Experiment 3 was the same as the thin condition in Experiment 2. The thick condition combined the two coherency-enhancing variations from Experiment 2: the same participant remained in the describer role throughout, and full feedback was given about the correct answer and what all other players had selected. Across both conditions in Experiment 3, notifications were sent to the chat to indicate when a participant had made a selection. For experiment 3, game lobbies worked slightly differently, and 5 minutes after the first participant had joined the lobby, the game started if there were at least two participants. Correspondingly, in experiment 3, games did not stop if a player disconnected, instead if there were at least two players still active, the game continued, swapping a player into the role of describer if necessary to continue the game.

2.4.4 Data pre-processing and exclusions

Participants could use the chat box freely, which meant that the chat transcript contained some non-referential language. The first author skimmed the chat transcripts, tagging utterances that did not refer to the current tangram. These were primarily pleasantries (“Hello”), meta-commentary about how well the task was going, and bare confirmations or denials ("ok", "got it", "yes", "no"). We excluded these utterances from our analyses. Note that chat lines sometimes included non-referential words in addition to words referring to the tangrams ("ok, so it looks like a zombie", "yes, the one with legs"); these lines were retained intact.

In Experiments 1 and 2, games did not start if there were not enough participants and ended if any participant disconnected. In Experiment 3, games started after a waiting period even if they were not entirely full and continued even in the event that a participant disconnected (with describer role reassigned if necessary), unless the game dropped below 2 players. The distribution of player counts in games that were initially recruited to be 6 player games is shown in Figure A3. The realities of online recruitment and disconnection meant that the number of games varied between conditions. We excluded incomplete blocks from analyses, but included complete blocks from partial games (See Table A3). When skimming transcripts to tag non-referential utterances, we noticed that one game in the 6-player thick condition had a describer who did not give any sort of coherent descriptions, even with substantial matcher prompting. We excluded this game from analyses.

2.4.5 Modelling strategy

We fit all regression models in brms (Bürkner, 2018) with weakly regularizing priors. We were unable to fit the full pre-registered mixed effects structure in a reasonable amount of time for some models, so we included the maximal hierarchical effects that were tractable. All model results and priors and formulae are reported in the Appendix. Models of accuracy used by-group random intercepts only, models of word count used full mixed effect structure, and models of S-BERT similarities used by-group and by-target random intercepts as applicable (see Figure A11). Models of matcher accuracy were logistic models with normal(0,1) priors for betas and sd. Models of describer efficiency were run as linear models with an intercept prior of normal(12,20), a beta prior of normal(0,10), an sd prior of normal(0,5) and a random-effect correlation prior of lkj(1). For all of the models of SBERT similarity, we used linear models with the priors normal(.5,.2) for the intercept, normal(0,.1) for betas, and normal(0,.05) for sd. As an additional post-hoc analysis, we ran mega-analytic models combining data across all experiments. For these models, we grouped the 3 thin-ish conditions (2c, and the two thin conditions of experiment 3) as one level, and coded the rest of the conditions as thick-ish. Game size was coded as a continuous measure (2 through 6). The priors for the mega-analytic models were the same as for the per-experiment models described above.

As a sensitivity analysis, we re-ran the primary models on the subset of the data from games that a) completed all 72 trials and b) had the full complement of players the entire time (relevant to 6-player experiment 3 games where games could start or continue with fewer players). Discrepancies are mentioned in the results, and these analyses are depicted in Figures A4 and A5 and Tables A54-A73. We also needed to decide how to handle dropout in Experiment 3, as some of the 6-player games did not retain all 6 players for the entire game. Our decision was to follow an intent-to-treat analysis and treat data as missing completely at random. Note that this choice underestimates differences between 2-player and (genuine) 6-player games by labeling some smaller groups as 6-player groups. We do not know exactly what leads some participants to drop out, but it is possible that some factors may be random (ex. connection issues) and others may be correlated with performance (ex. frustration because group is struggling).

We do not know whether groups that start and continue at the full size differ from games where some participants drop out. This is potentially an issue across all experiments; in experiments 1 and 2, groups stopped playing if anyone dropped out, and in experiment 3 they kept playing as a smaller group. The number of games in each condition and rates of dropoff are shown in Table A3 and Figure A3.

Chapter 3

Preschoolers can form conventional pacts with each other to communicate about novel referents

3.1 Introduction

Learning a language requires learning not only the content of that language, but also how to use the language to communicate. One case study for language use is referential communication, the ability to describe a target so an interlocutor can pick it out from a set of possibilities. Adults show sensitivity to both the visual context and their audience during referential communication, calibrating the description they provide to their beliefs about the interlocutor's knowledge state.

Iterated reference games provide an important paradigm for studying referential communication. In these games, one player repeatedly describes a set of abstract shapes to a partner so they can identify the target images (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986; Hawkins, Frank, et al., 2020; Krauss & Weinheimer, 1964). Over repetition, features of the initial descriptions are conventionalized as each pair comes to agree on a shared understanding of how to label each image. Success at this task requires mastery of a number of linguistic and communicative skills, including producing adequate initial descriptions, monitoring for comprehension, asking for clarification, and appropriately using the shared conversation history to inform later referring expressions. Studying how children play iterated reference games can provide insight into the developmental trajectory of the ability to produce referential expressions in order to achieve joint understanding.

One influential early study suggested that 4-5-year-old preschoolers struggle with child-child referential communication (Glucksberg et al., 1966). In their paradigm, one child was given a set of 6 blocks in a specific order. Their task was to describe the image on each block so their partner could pick out their corresponding block. As children described and selected blocks, they stacked them on pegs. While 4-5-year-old children succeeded on practice trials with familiar shapes and visual access to each other's blocks, children failed on critical trials where the blocks had abstract drawings and there was no visual access. Even after multiple rounds with the same images, children were not able to correctly order the blocks. Glucksberg et al. (1966) attributed children's communicative failures to their production of ego-centric descriptions that did not account for the other child's perspective.

Similar experiments with older children indicated a gradual improvement through adolescence both for initial accuracy and for the increase in accuracy across repetitions. Still, even the 9th grade sample was noticeably worse than the adult college student sample (Glucksberg & Krauss, 1967; Krauss & Glucksberg, 1969). Given that even teenagers had difficulties with the task, the complex stacking paradigm and the large number of potential targets may have posed task demands that prevented accurate measurement of children's abilities.

More recently, though no evidence has directly contradicted Glucksberg et al. (1966), a number of other studies have revealed early emerging skills in the preschool years that support the use of referential communication. Preschool-aged children are faster to select a target when it is referred to in a consistent way (Graham et al., 2013; Matthews et al., 2010) and 6-year-olds are more likely to use consistent referential expressions when their partner is consistent (Köymen et al., 2014), suggesting that young children are sensitive to the norm of consistent descriptions in cooperative communication. Further, preschool-aged children adapt the informativeness of their referential expressions based on the visual content available to their interlocutor (MATTHEWS et al., 2006; Nadig & Sedivy, 2002; Nilsen & Graham, 2009). By age 5, children integrate information about other's perspectives into their comprehension

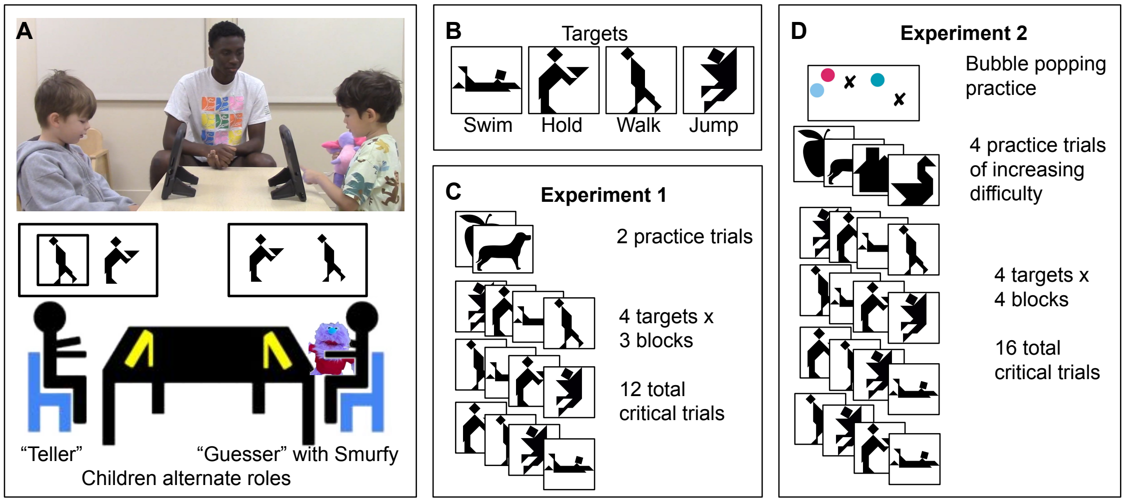


Figure 3.1: Experimental setup and procedure. Panel A shows the experimental setup. Panel B shows the 4 possible targets; names for targets are for cross-reference with later figures only. Panel C shows the procedure for Experiment 1; within critical blocks targets were ordered randomly. Panel D shows the procedure for Experiment 2.

of utterances (San Juan et al., 2015). Overall, by late preschool, children can can reason about others' perspectives to communicate more effectively.

While children are sensitive to others' perspectives, they still struggle to appropriately tailor the specificity of their utterances to the visual context, often resulting in under-informative utterances (Leung et al., 2024; Matthews et al., 2012). When children must describe one of two very similar images, 4 and 5-year-olds sometimes neglect to mention the relevant features, although they do better when playing in an interactive game than when not (Grigoroglou & Papafragou, 2019). By 5 years old, children are sensitive to over- and under-informative utterances, asking for clarification on some under-informative utterances and taking longer to make selections on over-informative utterances (Morisseau et al., 2013).

A common thread among many developmental studies is that children perform better when the cognitive load of the task is reduced, i.e. when there are fewer possible referents (Abbot-Smith et al., 2015; San Juan et al., 2015). These findings paired with evidence of emerging communicative skills in young children suggest that tailored referential expression production is possible in children, but it is likely to be masked when the task demands are too high.

Since Glucksberg et al. (1966), few studies have revisited the question of whether children can successfully communicate in an iterated reference game. In one study, 8-10-year-olds exhibited adult-like patterns of increasing accuracy, increasing speed, and shorter descriptions across repetitions, but children's accuracy was still far below adult performance and highly variable between dyads (H. P. Branigan et al., 2016). 4-6-year-olds successfully used conventionalized gestures with their partners in an iterated reference game where children could only use gestures to communicate (Bohn & Frank, 2019). 4-8-year-old children succeeded at playing an iterated reference game with a parent using a simple, child-friendly tablet-based task (Leung et al., 2024). In this task, even 4-year-olds had an initial accuracy above 80%, which rose to above 90% in later repetitions (Leung et al., 2024). Together, these findings provide evidence of young children's ability to communicate about novel referents under the right conditions.

Given the task demands in Glucksberg et al. (1966) and work showing that children can succeed in a less demanding paradigm, here we revisit the question of child-child referential communication. In the present study, we re-examine young children's ability to establish effective referring expressions with each other in an iterated reference game using a simplified tablet-based paradigm. Across an initial study and a replication including a total of 51 pairs of 4-5-year-old children, we found that preschool-aged children were successful in an iterated reference game, suggesting that children's capacity to construct effective referring expressions in novel contexts emerges earlier than once claimed.

3.2 Experiment 1

3.2.1 Methods

Our goal for Experiment 1 was to test young children's ability to coordinate to produce descriptions of abstract shapes that their partner could understand. Young children can be very sensitive to task demands and cognitive load (Carruthers, 2013; Keen, 2003; Turan-Küçük & Kibbe, 2024), so we adapted the experimental framework from Leung et al. (2024), and further simplified it by reducing the total pool of targets and the number of trials. This experiment was pre-registered at anonymized link.

Participants

4 and 5-year-old children were recruited from a university preschool during the school day. Children played with another child from the same class. Experiment 1 was conducted between June and August 2023. Pairs of children were included in analyses if they completed at least 8 of the 12 critical trials. We had 19 games that completed 12 critical trials, and 1 game that completed 11 critical trials. Of the 40 children, 21 were girls, and the median age was 57 months, with a range of 48-70 months.

Materials

For the target stimuli, we used four of the ten tangram images from Leung et al. (2024), chosen based on visual dissimilarity (Figure 3.1B). We coded the matching game using Empirica (Almaatouq et al., 2021), hosted it on a server, and then accessed the game on tablets that were locked in a kiosk mode so children could not navigate away from the game.

Procedure

Once a pair of children agreed to play the game, a research assistant took them to a quiet testing room. Children were introduced to a stuffed animal "Smurfy" who wanted to play a matching game. Children sat across a table from each other, each with a tablet in front of them (Figure 3.1A). On each trial, one child was the "teller" and saw a black box around one of two images on their screen and was asked to "tell Smurfy what they see" in the black box. The "guesser" saw the same two images in a randomized order and tried to select the described image to help Smurfy make a match. When the guesser selected an image, both children received feedback in form of a smiley or frowny face and an excited or disappointed sound. After each trial, children switched roles. Children passed Smurfy back and forth to keep track of whether they were the "guesser" or "teller" on a given trial.

Children completed two warm-up trials with black and white images of familiar shapes, followed by 3 blocks of the 4 target images (Figure 3.1C). Targets were randomly paired with another of the critical images as the foil.

The experimenters running the game did not volunteer descriptions, but they did scaffold the interaction, prompting children to describe the images, and sometimes repeating children's statements (especially when utterances were inaudible or the child did not respond immediately; this aspect of the procedure was modified in Experiment 2). The entire interaction was video-recorded.

Data processing

Children's selections and the time to selection were recorded from the experiment software. Children's descriptions were automatically transcribed from the video using Whisper (Radford et al., 2022) for the first pass and then hand-corrected by experimenters. Transcripts were hand-annotated for when each trial started, who said each line, and what referential descriptions were used. We excluded trials where the "teller" did not produce a description, or where all description was unintelligible and impossible to transcribe. After exclusions, we had 231 trials remaining.

Statistical analyses were run in brms (Bürkner, 2018) with weakly informative priors. We report estimates and 95% credible intervals. The experimental set-up, analysis code, de-identified transcripts, and performance data for both experiments is available at anonymized repo.

3.2.2 Results

Accuracy and speed

Our primary measure of interest was whether children could accurately communicate the intended target. To test for changes in accuracy over time, we fit a Bayesian mixed effects logistic regression predicting accuracy.[1] Children's accuracy was above chance (Odds Ratio: 3, ($95\%\,\text{CrI} = [1.14, 8.09]$)), and their accuracy slightly increased over the game (OR of one trial later: 1.17, ($95\%\,\text{CrI} = [1.03, 1.39]$), Figure 3.2). This level of accuracy is generally in-line with accuracies from 4-year-olds playing with their parents in Leung et al. (2024) and indicates that children can understand and succeed at the task.

[1] correct.num∼ trial.num + (trial.num|game) + (1|target)

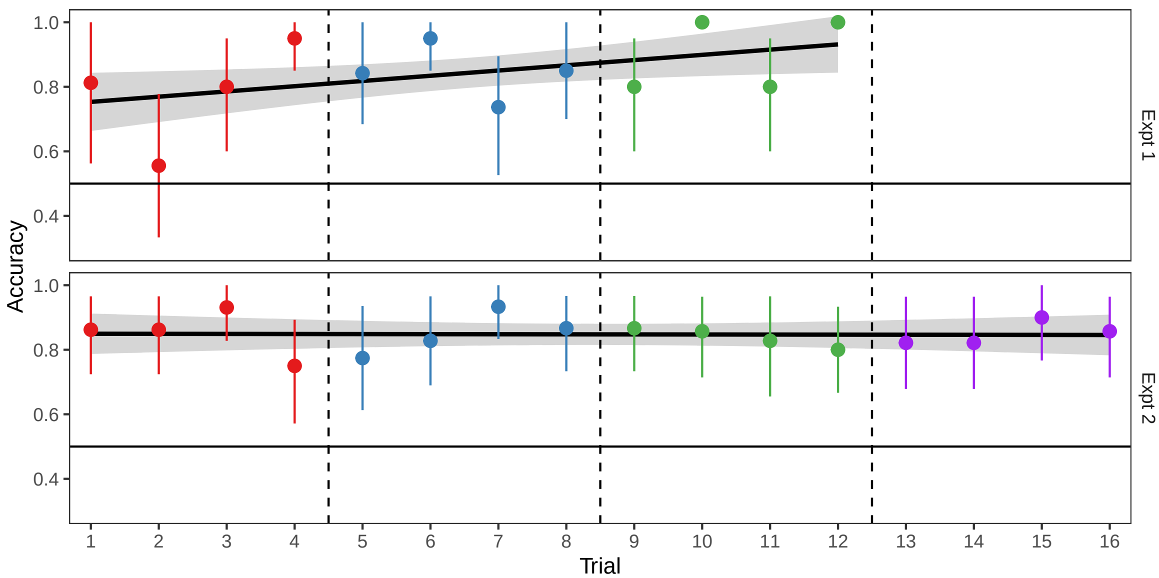


Figure 3.2: Children's accuracy at selecting the correct target over time. Error bars are bootstrapped 95% CIs with a linear trend line overlaid.

As another measure of children's performance, we looked at how long children spent on each trial. We ran a Bayesian mixed effects linear regression predicting the time to selection in seconds.[2] The first critical trial averaged 27.48, (95% CrI = $[20.48, 34.53]$) seconds, and children got faster over time (-1.22, (95%CrI = $[-1.9, -0.53]$) seconds / trial). Children were able to achieve the same accuracy in less time, suggesting that they were becoming more efficient at completing the task.

Description length

In iterated reference games with adults, description lengths usually shorten over repeated references (Boyce et al., 2024; Clark & Wilkes-Gibbs, 1986; Hawkins, Frank, et al., 2020). We were curious if children's descriptions would display the same trend, so we ran a Bayesian mixed effects linear regression predicting the number of words in the description the "teller" produced.[3] On the first critical trial, descriptions averaged 3.66, (95% CrI = $[2.56, 4.74]$) words, and description length was relatively stable over

[2]time.sec∼ trial.num + (trial.num|game) + (1|target)
[3]words∼ trial.num + (trial.num|game) + (1|target)

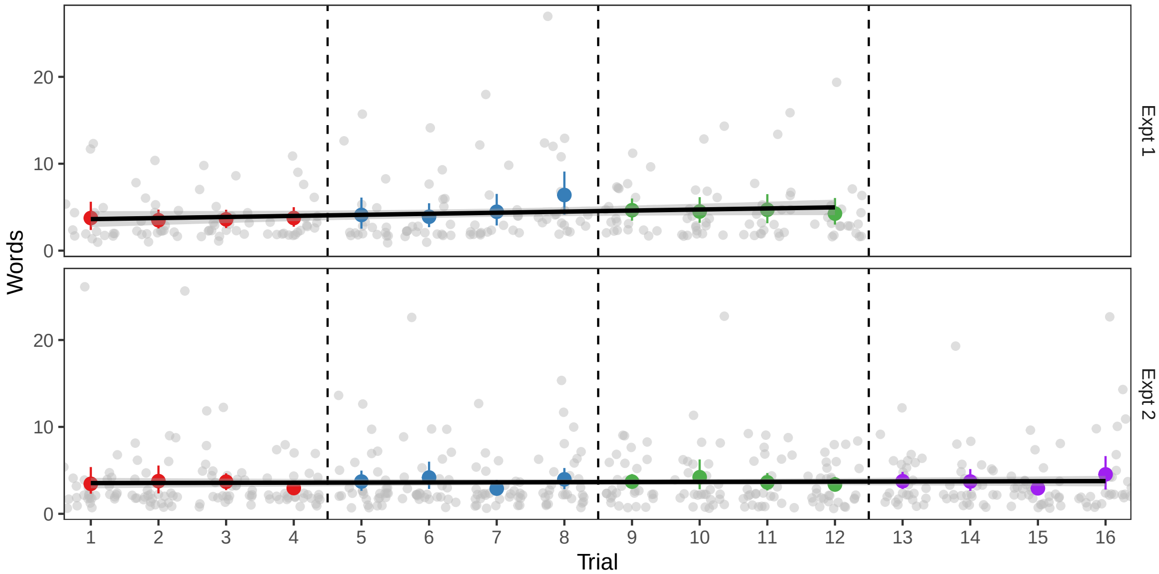


Figure 3.3: Length of description produced by the teller each trial. Grey dots are individual data points, colored dots are per trial means with bootstrapped 95% CIs.

time (change of 0.12, (95% CrI = $[-0.04, 0.28]$) words per trial, Figure 3.3). Thus, children's increasing speed was not from shorter utterances, but instead from some combination of improved task understanding, faster utterance planning, and faster decisions of what to select.

Some examples to illustrate the variety of effective descriptions children employed are shown in Table 3.1 (these specific examples are from Experiment 2, but both experiments had similar distributions of descriptions).

Convergence

While description length is often used as a proxy for measuring convention formation, it does not capture semantic overlap between utterances. Boyce et al. (2024) introduced a more sensitive measure of semantic convergence that compares the content of utterances using word embeddings to trace how similarities within and across games change over time. With only 3 blocks of descriptions, we do not expect semantic similarity for descriptions of a given target to show any meaningful change over time. However, we can test for a more coarse measure of sensitivity to partner: whether

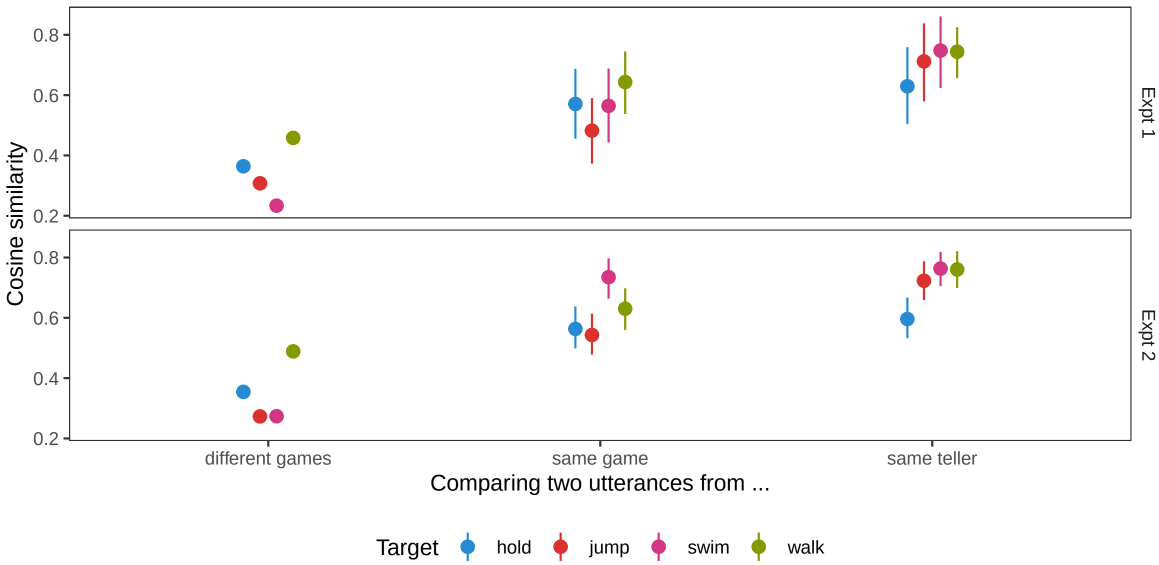


Figure 3.4: Semantic similarity between pairs of descriptions from different sources. Dots are means and lines are bootstrapped 95% CIs.

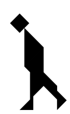
children's utterances are more like their partner's than children in other games'. If children are fully ego-centric (as suggested by Glucksberg et al., 1966), their choices of descriptions would be independent from their partners.

Following the methods of Boyce et al. (2024), we embedded each description in a semantic vector space using S-BERT (Reimers & Gurevych, 2019), and then used the cosine between embeddings as a measure of semantic similarity.

We compared the semantic similarities between descriptions of the same target based on who produced the description (Figure 3.4). We used a Bayesian mixed effects linear regression to predict similarity.[4] Utterances were more similar if they came from the same partnership (0.222, (95% CrI = $[0.184, 0.26]$)). Utterances were slightly more similar still if they came from the same person (0.135, (95% CrI = $[0.077, 0.193]$)), which is expected since children are likely to be fairly consistent with themselves. However, children used descriptions that were much more similar to their partner's than to other children's (Figure 3.4), indicating sensitivity to their partner's expressions.

[4] sim∼ same_game + same_speaker + (1|target)

Table 3.1: Example descriptions children successfully used to identify different target images in Experiment 2.

• person • a person holding a sandwich • a people carrying a box of dirt • a monster • someone holding a plate and giving it to a restaurant and has watermelon	
• vampire • hopping • a person flying • a person • a kite • a triangle with a head on it with feet • somebody skydiving, not in the airplane	
• racecar • airplane • alligator • a person fell down • a boat • a person that's in a race car that has one triangle and two triangles	
• person • a person walking • a person looking down • a people, but it doesn't have any arms	

3.2.3 Discussion

In Experiment 1, we adapted the paradigm of Leung et al. (2024) for pairs of children, taking an already simple set-up and making it shorter. Our goal was to see if young children were at all able to provide adequate descriptions, so children received a lot of scaffolding around the experimental interaction. Sometimes, this scaffolding included experimenters echoing children's descriptions, which could potentially influence children's responses. In Experiment 2, we repeated the same paradigm, with a tighter experimental script and a larger sample size.

3.3 Experiment 2

3.3.1 Methods

As Experiment 2 was very similar to Experiment 1, we focus on the changes made compared to Experiment 1. Experiment 2 was pre-registered at anonymized link.

The biggest change between the experiments was increasing the number of repetitions of target stimuli from 3 to 4 (from 12 to 16 trials). The greater number of trials in Experiment 2 made it possible to look for changes over time that could be indicative of convergence to shared descriptions within a game and divergence between games.

Participants

Experiment 2 was run between March and August of 2024, at the same preschool as Experiment 1. No children participated in both experiments. 30 pairs of children completed all 16 critical trials, and 1 pair of children completed 10 critical trials. Our target age range was 4 and 5-year-olds, but one older 3-year-old was unintentionally included. Of the 62 children, 30 were girls, and the children had a median age of 56 months, and a range of 45-69 months.

Materials

The same 4 critical images were used as in Experiment 1. In response to some children struggling with the abrupt switch from familiar to non-nameable shapes, we introduced more practice trials for Experiment 2. We used a total of 4 practice trials to provide a gradient from familiar shapes to less recognizable, blockier shapes (Figure 3.1D).

Procedure

The procedure was much the same as Experiment 1(Figure 3.1D). We added an initial “bubble popping” exercise to give children practice tapping the tablet appropriately (this was an issue for some children in Experiment 1). The experimental script was fully written out and memorized by experimenters so children all received the same instructions. We wrote up contingency statements that the experimenter could use to prompt children who were not giving descriptions or making selections. Experimenters helped with game mechanics such as whose turn it was to tell and who should press the screen, but avoided contributing or repeating any content about the images or the descriptions.

Data processing

Data were processed in the same way as Experiment 1. After excluding trials where children did not give a description or where the experimenter echoed a child's description, we had 466 trials total.

3.3.2 Results

Accuracy and speed

In Experiment 2, children's accuracy was above chance (Odds Ratio: 5.95, (95%CrI = $[3.07, 11.89]$)) and relatively stable over time (OR of one trial later: 1.01, (95% CrI = $[0.94, 1.09]$), Figure 3.2). The first critical trial averaged 22.06, (95%CrI = $[15.86, 28.58]$) seconds, and children got faster over time (-0.7, (95% CrI = $[-0.99, -0.41]$) seconds / trial). Children were initially faster in Experiment 2 than Experiment 1, possibly due to the increased number of practice trials and pre-training on how to press the screens. Taken together, we find more evidence that children can successfully communicate with each other about these abstract shapes, and do so with increasing efficiency.

Description length

The average length of descriptions on the first trial was 3.44, (95% CrI = $[2.23, 4.75]$) words and description length was relatively stable over time (change of 0.02, (95%CrI = $[-0.05, 0.09]$) words / trial, Figure 3.3). This finding is comparable to Experiment 1, again finding that children produce short utterances without much change in length over time.

Convergence

As a coarse measure of partner-sensitivity, we repeated the semantic analysis from Experiment 1. Utterances were more similar if they came from the same partnership (0.27, (95% CrI = $[0.243, 0.297]$)) and were slightly more similar if they came from the same person (0.097, (95% CrI = $[0.059, 0.132]$)).

As Experiment 2 had 4 blocks, we examined whether descriptions were converging semantically toward the final description. We compared the utterances from the first three blocks to the descriptions in the last block using a Bayesian mixed effects linear regression predicting similarity.[5] Over the first three blocks, descriptions became increasingly similar to the last block description (0.042, (95% CrI = $[0.007, 0.078]$)). Descriptions were more similar if they came from the same child, which is expected as a sign of internal consistency (0.067, (95% CrI = $[0.006, 0.127]$)). Although over time descriptions did get more similar to the last block utterance, the semantic distance between adjacent block utterances was relatively constant (0.009, (95% CrI = $[-0.026, 0.044]$)).

Partnerships often diverged from one another as groups focused on distinct aspects of the image. We tested whether descriptions in different games diverged over time using a Bayesian mixed effects linear regression.[6] As the games progressed, descriptions to the same target from different games became slightly less similar (-0.013, (95% CrI = $[-0.018, -0.008]$)), which indicates that games are converging to different conventions. These patterns of increasing similarity within games and increasing divergence between games qualitatively match the patterns found for adults (Boyce et al., 2024).

3.4 Joint analysis

As the two experiments were similar to one another, we re-ran models pooling the data across the two experiments, using experiment number as a random effect. Pooling the two experiments, children's accuracy was above chance (OR: 4.64, (95% CrI =

[5] sim∼ earlier_block.num + same_speaker + (1|game1) + (1|target)
[6] sim∼ block.num + (1|target)

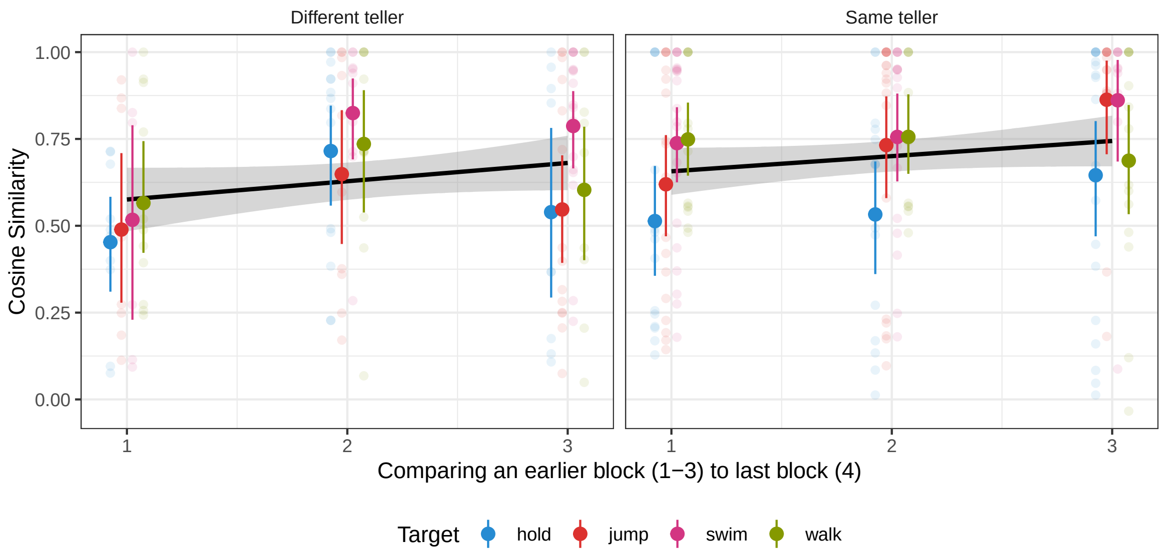


Figure 3.5: Semantic similarity between descriptions from earlier blocks (1-3) and the last block in Experiment 2. Heavy dots are means with bootstrapped 95% CIs; light dots are individual values.

$[1.42, 12.16]$)) and accuracy numerically increased over the course of the game, although the credible interval included 0 (OR of one trial later: 1.04, (95% CrI $= [0.98, 1.12]$)). Descriptions produced by tellers averaged 3.62, (95% CrI $= [0.09, 7.38]$) words on the first trial, and description length was relatively stable over time (change of 0.04, (95% CrI $= [-0.02, 0.11]$) words per trial). Utterances were more similar if they came from the same partnership (increase in cosine similarity: 0.257, (95% CrI $= [0.234, 0.279]$)) and were slightly more similar still if they came from the same person within the partnership (0.109, (95% CrI $= [0.078, 0.14]$)).

We might expect that whether a description was successful influences whether the same description, or a variant of it, is employed in future rounds. Intuitively, successful descriptions can be copied and built upon, while unsuccessful descriptions should be replaced by a fresh attempt. To test whether accuracy is predictive of similarity to future descriptions, we ran a post-hoc Bayesian linear model predicting similarity to the next block description in terms of accuracy.[7] Descriptions that elicited a correct response were more similar to the next block description (0.145,

[7] sim∼ earlier_block.num × correct + same_speaker × correct + (1|game1) + (1|target) + (1|expt)

(95% CrI = $[0.055, 0.236]$)) with no substantial interaction with block number or whether both descriptions came from the same teller. This pattern of results is consistent with the expectation that children are more likely to stick to their own description or repeat the other child's description if it was previously successful.

3.5 General discussion

Prominent early studies claimed that young children cannot overcome their egocentrism to coordinate with each other in reference games (Glucksberg et al., 1966). However, more recent developmental work has found that young children show emerging communicative and pragmatic sensitivity, especially when cognitive demands are low. Here, we revisited the question of preschoolers' performance in child-child reference games, using a scaffolded paradigm to reduce extraneous task demands.

Across 2 experiments and 51 pairs of 4 and 5-year-old children, we tested how well children could produce referential expressions that allowed their partner to find a matching abstract shape. Children varied substantially in what sorts of descriptions they produced, but overall accuracy was high (85%), indicating that children were generally able to produce adequate descriptions. Additionally, children's utterances showed signs of converging toward conceptual pacts. While this task is substantially scaled down relative to measures used for adult competence, it does suggest that the relevant communication skills are present at least in rudimentary form by the end of the preschool years.

Unlike adults, children did not display an increase in accuracy or a shortening of referential expressions over the course of the game. Still, our findings show that descriptions became increasingly similar to descriptions in the last block and that successful utterances were more similar to future utterances, suggesting that children are adapting their descriptions as the game unfolds. These null findings are likely a result of initially high accuracy in the first block and initially short utterances that leave little room for reduction.

It is unclear to what extent the uniformly short descriptions we observed are a product of the simplified task or children's behavioral differences from adults. In this case, the low number of options and relatively easy-to-describe shapes may have obviated the need for long initial descriptions. Indeed, adult controls in Leung et al. (2024) used shorter initial descriptions than adults in studies with larger arrays of harder to distinguish images (Boyce et al., 2024; Hawkins, Frank, et al., 2020). However, young children may also struggle to produce longer descriptions, and young children may be more willing to take guesses when adults would seek additional clarification. Especially in light of other work suggesting that conceptual pact formation and reduction in utterance length sometimes decouple in adults (Boyce et al., 2024), further empirical work on the factors driving verbosity in reference games is warranted.

The generalizability of our results is limited by the target population, the target images, and the task structure. We sampled a convenience population of children at a university nursery school. The set of tangram images may be easier to distinguish and have higher codability than other target images used in adult reference games. We specifically targeted children's abilities to construct referring expressions that can be jointly understood, so children were provided scaffolding around taking turns and talking to their partner. Thus, children's performance should be taken as a proof-of-concept about ability, rather than a claim about how generally children spontaneously demonstrate these abilities.

In the broader picture of language acquisition, there is debate over the timing of the emergence of communicative and pragmatic abilities relative to the acquisition of grammar and meaning. On one side, children seem to learn literal semantics far before they display an understanding of some pragmatic implicatures (Huang & Snedeker, 2009; Noveck, 2001); on the other, sensitivity to communicative intent is an early emerging skill that develops in parallel with linguistic knowledge and may bootstrap language learning (Bates, 1974; Bohn et al., 2019; Tomasello, 2008). Our current findings are most consistent with a gradual development of children's communicative and linguistic skills, where the skills emerge early and then are refined over time, as children's cognitive capacities increase. At 4-5 years old, children are already able to establish novel referential conventions with one another as part of their broader ability to communicate and coordinate.

Chapter 4

Idiosyncratic but not opaque: Linguistic conventions formed in reference games are interpretable by naïve humans and vision–language models

4.1 Introduction

When a teen says about someone that "he's got rizz," what does this mean? The idea that teen slang is arbitrary and opaque to outsiders (i.e., older generations) is enough of a cultural touchstone that late night comedy shows have segments about it. Teens are far from the only ones to have in-group naming conventions; many communities form stable linguistic conventions including professional jargon, regionalisms, and of course slang.

Iterated reference games are commonly used as a model system for studying the formation of linguistic conventions. In these games, a describer tells their partner how to sort or match a series of abstract images (e.g., Clark & Wilkes-Gibbs, 1986; Hawkins, Frank, et al., 2020). Over repeated rounds of referring to the same targets, pairs develop conventionalized nicknames for the target images. These nicknames are often partner-specific, in that different pairs develop different nicknames for the same targets. When describing the targets to a new person, describers return to more elaborated descriptions, indicating an expectation that prior conventions are not appropriate descriptions to use for new matchers (Hawkins et al., 2021; Wilkes-Gibbs & Clark, 1992; S. O. Yoon & Brown-Schmidt, 2019a). This partner specificity is generally interpreted to be efficient, implying that conventions are not understandable to those outside of the convention formation process.

For one-shot reference games, the choice of referring expression can be described as a process of recursive inference, in which speakers reason about how listeners will interpret their utterance and vice versa. This process is described in Bayesian pragmatics models such as the Rational Speech Acts model (RSA) (Frank & Goodman, 2012; Goodman & Frank, 2016). The Continual Hierarchical Adaptation through Inference model (CHAI) builds on RSA by adding rules for how agents update their belief distributions after each interaction, to account for the dynamics of repeated interaction in iterated reference games (Hawkins, Franke, et al., 2023). In models like CHAI, both speaker and listener agents are cooperatively reasoning about each other, but related models can be used when the interaction is one-sided, such as when a naïve listener is overhearing a conversation. A key factor permitting variation in referring expressions is lexical uncertainty; RSA models incorporating lexical uncertainty (Bergen et al., 2016; Potts et al., 2015) treat listeners as Bayesian agents who jointly infer the meaning of a speaker's utterance and their model of the speaker's lexicon. Models incorporating lexical uncertainty have been used to model children's word learning (Bohn et al., 2022) as well as person- and group-level differences (Hawkins, Goodman, et al., 2020; Schuster & Degen, 2020). Usually models assume that the person-to-person variation in lexica is small, so the agent is learning a general meaning and a hierarchical set of tweaks to account for slightly different word extensions based on speaker and circumstance, rather than learning arbitrary menaings for each person (Hawkins, Franke, et al., 2023; Schuster & Degen, 2020; but see Misyak et al., 2016).

Conventions in iterated reference games are formed between people without a salient shared group identity prior to the interaction, but people do not use the conventions with new partners. How opaque are the temporary linguistic conventions created in reference games: are they opaque like “rizz” or interpretable like “roundabout”? From a theoretical perspective, measuring this opacity can help inform the development of models like CHAI that aim to account for lexical change in conversation. We attempt to answer this question here.

We consider opacity of references to be a measure of the semantic distance between the referring expression and the target. Transparent expressions (those with low opacity) have signifiers and referents that are semantically close, such that anyone with the same general lexicon can identify the appropriate referent given the signifier. In contrast, expressions that are opaque have signifiers and referents that are semantically distant in the lexicon, such that the relations are arbitrary and inaccessible without additional clues such as the partner-specific conversation history. We note that properties of utterances such as groundedness or compositionality and properities of the target such as iconicity and naemability may load on opacity but are not the same as opacity.

One option for measuring the transparency of referring expressions is to use vision–language models to operationalize a shared semantic space for both language and images. Computational methods have enabled the embedding of various stimuli (including images and text) into high-dimensional feature spaces; these embeddings have properties which suggest that they are reasonable approximations of humans’ semantic spaces, including similarity in representational geometries (e.g., Grand et al., 2022; Muttenthaler & Hebart, 2021). Indeed, embeddings from neural network models have been used as a form of semantics in a range of reference game scenarios (e.g., Gul & Artzi, 2024; Ji et al., 2022; Kang et al., 2020; Le et al., 2022; Ohmer et al., 2022). In particular, such embeddings can be treated as the default context- and speaker-independent lexicon, since they are not updated to account for convention formation within an iterated reference game.

A second option for measuring the transparency of referring expressions is to measure how often naïve humans, who were not part of the group who formed the convention, can correctly associate the target referent with the referring expression. Prior work with naïve matchers has been limited and has focused on the role of conversational history. In this work, naïve matchers tend to do better the more their observation history resembles that of the original conversation—when hearing descriptions in order instead of in reverse order (Murfitt & McAllister, 2001), when listening to the entire game instead of starting in the third round (Schober & Clark, 1989), and when seeing yoked trials from a single game rather than trials sampled across 10 games (Hawkins, Sano, et al., 2023). In these studies, naïve matchers had worse accuracy than in-game matchers, but their performance was still far above chance, suggesting that the convention–target relationship is not purely arbitrary. In fact, even when pairs of participants try to obfuscate their meaning, overhearers can still do quite well at identifying the target referents (Clark & Schaefer, 1987). Nonetheless, receiving more context from an interaction—and in particular having that context be in order—is beneficial to matchers. Except for the shuffled condition of Hawkins, Sano, et al. (2023), these studies do not address how opaque descriptions are when they are presented without any prior context. Such an approach would be important to have truly naïve matchers that lack even the context of trial order, providing a clearer understanding of when these expressions are opaque and when they are merely idiosyncratic.

In the current work, we measure the opacity of the referring expressions created in iterated reference games. We use conversations from Boyce et al. (2024), who created a large corpus of iterated reference games that were played online using a chatbox for communication. This corpus is made up of 6-round iterated reference games using the same 12 target images. Games varied in how large the describer–matcher groups were (2–6 participants) and how "thick" the communication channels were (for example, if matchers could send back messages or just emoji). The varied conditions within a consistent framework allowed us to test how the opacity of referring expressions varies depending on the conditions the referring expressions came from.

We use both human experiments and models to assess when and why expressions are opaque or understandable to outside observers. We first present a computational approach using a vision–language model to measure the semantic similarities between referring expressions and their targets, and we validate our model against naïve human matchers (Experiment 1). We then use both naïve human matchers and the model to compare the opacity of referring expressions across different game conditions and time points (Experiment 2). Finally, we address the role of conversation history by comparing naïve matcher performance on game transcripts presented in order versus

Figure 4.1: Experimental setup. Naïve matchers read transcripts from trials in reference games from Boyce et al. (2024) and selected which image they thought was being described. Matchers recieved bonus payments for correct selections.

out of order (Experiment 3).

4.2 Task setup

4.2.1 Materials

We drew our referring expressions from Boyce et al. (2024), excluding utterances that were marked as not containing referential content. For our naïve matcher experiments, we sampled different subsets of this corpus. Within the subsets, we excluded transcripts that contained swear words or crude or sexual language. For the computational model, we used the entire corpus, and pre-processed the text by concatenating all the referential messages sent by the describer for a given trial.

4.2.2 Experimental procedure

We recruited English-speaking participants from Prolific. On each trial, participants saw the full transcript from that trial, containing all the chat messages marked by whether they were from the speaker or a listener. Participants selected the image they thought was the target from the tableau of 12 (Figure 4.1). Participants received feedback on whether they were right or wrong on each trial. Except when the specific viewing order was part of the experimental manipulation, we randomized the order of trials, subject to the constraint that the same target could not repeat on adjacent trials. The task was implemented in jsPsych (de Leeuw et al., 2023). We paid participants $10 an hour plus a bonus of 5 cents per correct response. All our experimental code is at this anonymized repo.

4.2.3 Computational models

We used the Contrastive Language-Image Pretraining model (CLIP; `clip-vit-large-patch14`) as a comprehender model for our domain (Radford et al., 2021). CLIP is a vision-language model that uses a text transformer and a vision transformer to embed text and images into the same space, trained to maximize the similarity between representations of images and their English captions. It is a natural choice for reference games, as the model is trained to estimate the correspondence between images and phrases in natural language. We ran CLIP for the concatenated describer utterances and all 12 tangram shapes. For each utterance, we computed probabilities for each tangram shape using logit scores from CLIP. The simplest way to do this is simply taking the softmax of the logits. However, tangram shapes are outside of the training distribution for the model, perhaps explaining why it favored some images over others regardless of the content of the text.

To improve the performance of base CLIP, we trained a set of readout models to assign probabilities to images using CLIP's logits as features. Models were trained to maximize task performance (i.e., to assign high probability to the target tangram given the concatenated describer utterance). We compared four types of models: random forest, logistic regression, multi-layer perceptron (MLP), and gradient-boosted tree. Classifiers were implemented in the `scikit-learn` and `XGBoost` libraries (Chen & Guestrin, 2016; Pedregosa et al., 2011). Each readout model was evaluated using 10-fold cross-validation, where the model was trained on 90% of the data and evaluated on the remaining 10%. Table 4.1 shows the cross-validated accuracy of different readout models, as well as the performance of CLIP with no readout. The MLP with two hidden layers of size 100 performed the best on held-out data; in subsequent analyses, we use the MLP trained on all the data.

Table 4.1: Cross-validated accuracies for classifiers. Standard deviations in accuracy across the 10 folds are shown in parentheses. Best performance within each model class is underlined, and best overall performance is bolded.

Classifier	Accuracy
Random baseline	0.08
CLIP without readout	0.31
Logistic regression	
No penalty	<u>0.50 (0.01)</u>
L2 penalty	0.50 (0.01)
Random forest	
10 estimators	0.46 (0.02)
50 estimators	0.51 (0.02)
100 estimators	0.52 (0.02)
500 estimators	<u>0.52 (0.02)</u>
Gradient-boosted tree	
10 estimators	0.48 (0.02)
100 estimators	<u>0.51 (0.02)</u>
Multi-layer perceptron	
1 × 32-dim hidden layer	0.50 (0.01)
1 × 100-dim hidden layer	0.52 (0.01)
1 × 512-dim hidden layer	0.53 (0.02)
1 × 1028-dim hidden layer	0.53 (0.02)
2 × 32-dim hidden layers	0.51 (0.02)
2 × 100-dim hidden layers	**<u>0.55 (0.02)</u>**

4.3 Experiment 1

Our CLIP-MLP computational model was optimized for task accuracy. To validate whether this objective also results in human-like response patterns, we conducted a calibration experiment to determine if model-assigned target probabilities were aligned with the probabilities that naïve human matchers would choose particular target images across a range of utterance-target pairs.

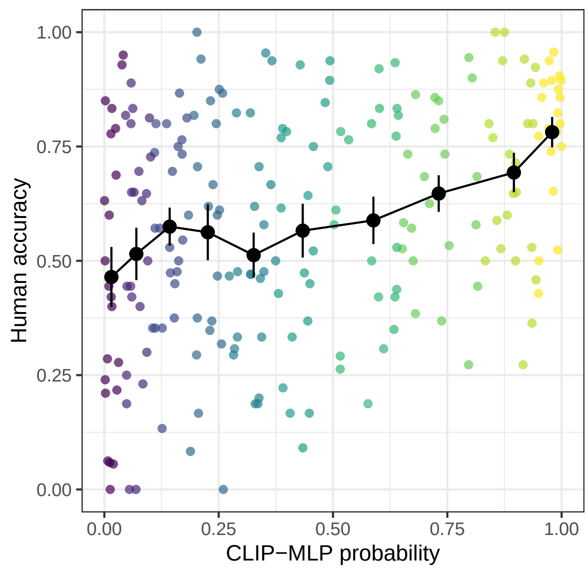


Figure 4.2: Correlation between human accuracy and CLIP-MLP probability of target in Experiment 1. Small points are individual descriptions, colored by decile of CLIP-MLP probability, large points and error bars are the bootstrapped mean and 95% CI across descriptions for each decile.

4.3.1 Methods

We first obtained target probabilities from our CLIP-MLP model for all utterances from Boyce et al. (2024). We then used stratified sampling to select 217 trials by dividing model-predicted probabilities into deciles and choosing approximately 22 utterances per decile, spanning the 12 different possible target images. We recruited 61 participants who each saw 64 trials randomly sampled from the 217 tested trials. On average, each trial was seen by 18 participants. This experiment was pre-registered at this anonymized link.

4.3.2 Results and discussion

We obtained human accuracies on each trial by dividing the number of participants who selected the target by the total number of participants who saw the trial (Figure 4.2). There was a modest but significant positive correlation between model-predicted probabilities and human accuracies ($r = 0.33$ [0.21, 0.45]). This result suggests that

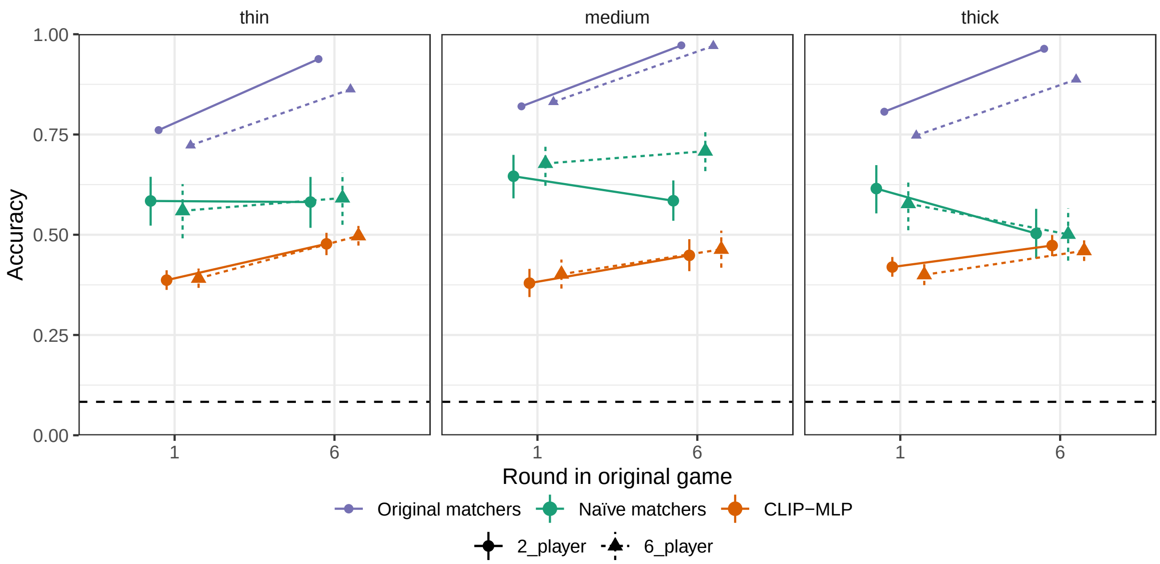


Figure 4.3: Accuracies for naïve human matchers and the CLIP-MLP model for Experiments 2a and 2b, grouped by the source of the referential description. Facets are the communication thickness of the original game and x-axis is when in the game the transcript came from. Point estimates and 95% CrI are predictions from the fixed effects of logistic and beta regressions. Bootstrapped mean accuracy from the original matchers is included as a ceiling, and random chance as a baseline.

model predictions were calibrated to human response patterns, albeit not perfectly. It is possible to use these calibration results to tune model predictions to better approximate human responses; we leave this approach for future work. Nonetheless, the observed positive correlation suggests that our computational model carries some signal about human accuracies, validating its use in subsequent experiments as a computational comparison.

4.4 Experiment 2

As a starting point for examining the opacity of referring expressions, we focused on referring expressions from the first and last rounds of reference games. Based on the idea that conventions form across repeated communication, later-round utterances should be more opaque. To test this hypothesis, we ran a recognition experiment

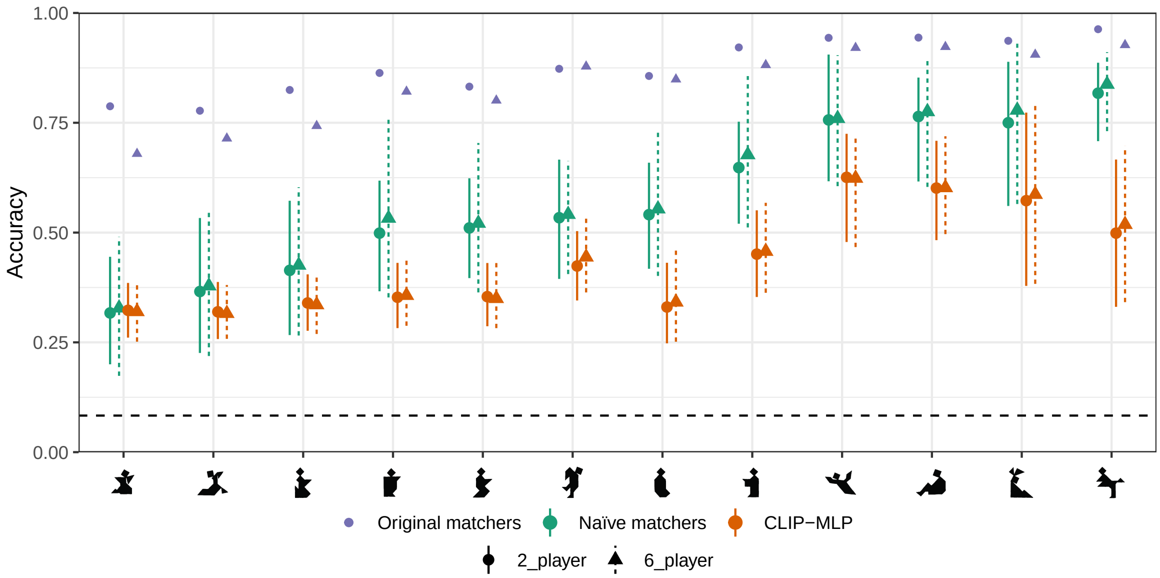


Figure 4.4: Accuracies for naïve human matchers and the CLIP-MLP model for Experiments 2a and 2b, split out by target image. Point estimates and 95% CI are predictions from the fixed effects and by-tangram random effects of logistic and beta regressions, bootstrapped across conditions. Bootstrapped mean accuracy from the original matchers is included as a ceiling, and random chance as a baseline.

including descriptions from games of different sizes and communication thicknesses. Based on the patterns of cross-game similarity in Boyce et al. (2024), we expected that smaller and thicker games, whose descriptions diverged fastest, would have more idiosyncratic and opaque conventions than larger groups with thinner communication channels.

4.4.1 Methods

Experiment 2a

To establish a baseline of how well naïve matchers could understand descriptions without context, we ran a 2 × 2 within-subjects experiment, drawing target transcripts from 2- and 6-player games from Experiment 1 of Boyce et al. (2024) and from the first and last blocks of these games. These games had medium-thick communication channels, where matchers could send text messages to the chat, the describer role rotated each round, and matchers received limited feedback. We recruited 60 participants who each saw 60 trials (15 in each of the 4 conditions). Overall, participants saw 774 transcripts from 40 games. This experiment was pre-registered at this anonymized link.

Experiment 2b

After observing limited condition differences in Experiment 2a, we ran a follow-up experiment on descriptions from Experiment 3 of Boyce et al. (2024), where the communication channel thicknesses were more extreme. Here, we used a 2 × 2 × 2 within-subjects design, drawing our transcripts from the first and last rounds of thick and thin, 2- and 6- person games. In the "thick" condition, matchers could send text messages to the chat, one person was the describer for the whole game, and matchers received feedback on everyone's selections. In contrast, in the "thin" condition, matchers could only communicate by sending 4 emoji, the describer role rotated, and matchers recieved limited feedback. As the emoji did not have referential content, we did not include them in the transcripts shown to naïve matchers. For experiment 2b, we recruited 60 participants who each saw 64 trials (8 in each of the 8 conditions). Overall, participants saw 2392 transcripts from 163 games. This experiment was pre-registered at this anonymized link.

4.4.2 Results

Experiment 2a

For Experiment 2a, we ran a Bayesian mixed-effects logistic model of naïve matcher accuracy in `brms` (Bürkner, 2018).[1] For this and other models, we report the model estimate and 95% credible interval. Overall, naïve matchers were right 62% of the time, far above the $1/12 = 8.3\%$ expected by random chance (OR = 1.93, (95% CrI = $[1.05, 3.62]$)). There were no large effects of condition (Figure 4.3 middle panel). Participants tended to be less accurate at descriptions from the last round (OR of last round = 0.77, (95% CrI = $[0.53, 1.1]$)). There was no clear effect of original

[1] correct∼ group_size × round + trial_order + (group_size × round|correct_tangram) + (group_size × round + trial_order|workerid)

group size (OR of 6-player game = 1.15, (95% CrI = $[0.89, 1.47]$)), but there was an interaction between round and group size (OR = 1.49, (95% CrI = $[1.06, 2.1]$)). Later transcripts from larger games were easier to understand, but earlier transcripts from smaller games were easier to understand. Much of the variation in accuracy was driven by the target image, which accounted for more variation than participant differences (standard deviation of image distribution = 0.98, (95% CrI = $[0.63, 1.51]$); SD of participant distribution = 0.64, (95% CrI = $[0.42, 0.88]$)). Some images were much easier to identify as the target than others (Figure 4.4).

Experiment 2b

For Experiment 2b, we ran a similar Bayesian mixed-effects logistic model.[2] Naïve matchers were above chance (OR = 1.81, (95% CrI = $[1.06, 3.08]$), Figure 4.3). As in Experiment 2a, there were not substantial effects of condition. Last-round descriptions had slightly lower accuracy (OR of last round = 0.64, (95% CrI = $[0.47, 0.85]$)), but there was an interaction with thickness, where for thin games, last round descriptions were less opaque (OR = 1.55, (95% CrI = $[1.02, 2.33]$)). Again there was strong variation based on target image (0.81, (95% CrI = $[0.51, 1.28]$)), which exceeded by-participant variation (0.62, (95% CrI = $[0.43, 0.83]$)).

Additional predictors

We considered the accuracy of the in-game matchers from Boyce et al. (2024) and the length of the description as post-hoc predictors. In both experiments, in-game accuracy was predictive of naïve matcher accuracy (Expt 2a OR = 3.33, (95% CrI = $[2.45, 4.53]$), Expt 2b OR = 2.39, (95% CrI = $[1.88, 3.03]$)). The log number of words in the description was not predictive in Experiment 2a (OR = 1.05, (95% CrI = $[0.94, 1.17]$)), but longer descriptions were slightly beneficial in Experiment 2b (OR = 1.1, (95% CrI = $[1.01, 1.2]$)).

The pattern of which conditions became more opaque in later rounds resembled the pattern of which conditions produced descriptions that diverged the most

[2] correct∼ group_size × thickness × round + trial_order + (group_size × thickness × round|correct_tangram) + (group_size × thickness × round + trial_order|workerid)

in semantic space in Boyce et al. (2024). As a post-hoc test of whether opacity might be related to semantic divergence from other descriptions, we used the mean semantic similarity between an utterance and other utterances in the same condition as an additional predictor of accuracy.[3] Similarity to other utterances was strongly predictive of increased accuracy in both experiments (Expt 2a: OR = 12.49, (95% CrI = $[4.7, 33.64]$), Expt 2b: OR = 14.75, (95% CrI = $[5.93, 36.87]$)) and was more predictive for the last round descriptions (Expt 2a: OR = 3.49, (95% CrI = $[1.09, 10.92]$), Expt 2b: OR = 4.78, (95% CrI = $[1.61, 14.25]$)). While exploratory, this analysis suggests that referring expressions that are further from shared semantic priors (i.e., more idiosyncratic) are harder for naïve matchers to understand.

4.4.3 Model results

As a computational comparison, we used the probability the CLIP-MLP model assigned to the correct target as our dependent measure and fit a Bayesian mixed-effects beta regression on the descriptions from Experiment 2.[4] The CLIP-MLP model was far above chance, but had lower accuracy than the human participants (OR = 0.6, (95% CrI = $[0.45, 0.82]$)). The strongest predictor of accuracy was later round (OR = 1.32, (95% CrI = $[0.94, 1.83]$)), but even this was uncertain. There was substantial by-target image variation (SD = 0.46, (95% CrI = $[0.27, 0.76]$)).

In additional models, we checked the effect of in-game matcher accuracy, length of the description, and semantic divergence. CLIP-MLP had higher accuracy when in-game matcher accuracy was higher (OR = 1.52, (95% CrI = $[1.35, 1.71]$)), and when descriptions were shorter (OR for log words = 0.85, (95% CrI = $[0.82, 0.9]$)). The model may perform poorly on long descriptions because they are further from the model's training distribution of image captions. A description's semantic similarity to other descriptions was predictive of higher accuracy (OR = 11.85, (95%CrI = $[6.51, 21.06]$)), especially for last round utterances (OR = 3.46, (95%CrI = $[1.65, 7.36]$)), in line with the human results.

[3]Semantic similarity was operationalized as cosine similarity between S-BERT embeddings (Reimers & Gurevych, 2019), the measure of semantic distance used in Boyce et al. (2024).

[4]correct$\sim$ group_size $\times$ thickness $\times$ round + (group_size $\times$ thickness $\times$ round|correct_tangram)

4.4.4 Discussion

Overall, naïve human matchers were fairly accurate overall, but less accurate than matchers in the original game, consistent with prior work. The computational model was less accurate, but still far above chance. The largest source of variability in accuracy was from target images, and whether earlier or later utterances were more opaque varied by game condition. The level of semantic divergence from other expressions was strongly predictive of the opacity of the expression.While this analysis was not prespecified, it still provides some suggestion that descriptions that were closer to shared semantic priors were also more interpretable.

4.5 Experiment 3

The experience of naïve matchers in Experiment 2 differed from in-game matchers in several ways; any of these could explain differences in accuracy. In-game matchers received descriptions from a consistent group, in the order they were created, and were the intended audience of the descriptions. In Experiment 3, we focused on the role of context and group-specific interaction history to tease apart some of these differences. One question of interest was how much seeing the entire the conversation history in order would increase the interpretability of later round descriptions.

Additionally, we were interested in how quickly naive matchers process the linguistic descriptions and how this might relate to the oredictability and informativity of the descriptions. Word predictability (often measured as surprisal) is a predictor of language processing times. The iterated reference game transcripts present an interesting test case because expressions are likely to be globall uncommon (and thus unpredictable), but may be licenced and unsurprising given the immediate visual context and conversation history. Thus, another question of interest is how rapidly comprehenders are able to adapt to the changing context.

4.5.1 Methods

We compared naïve matchers in yoked and shuffled conditions. In the yoked condition, naïve matchers saw all the descriptions from a single game in the order they originally

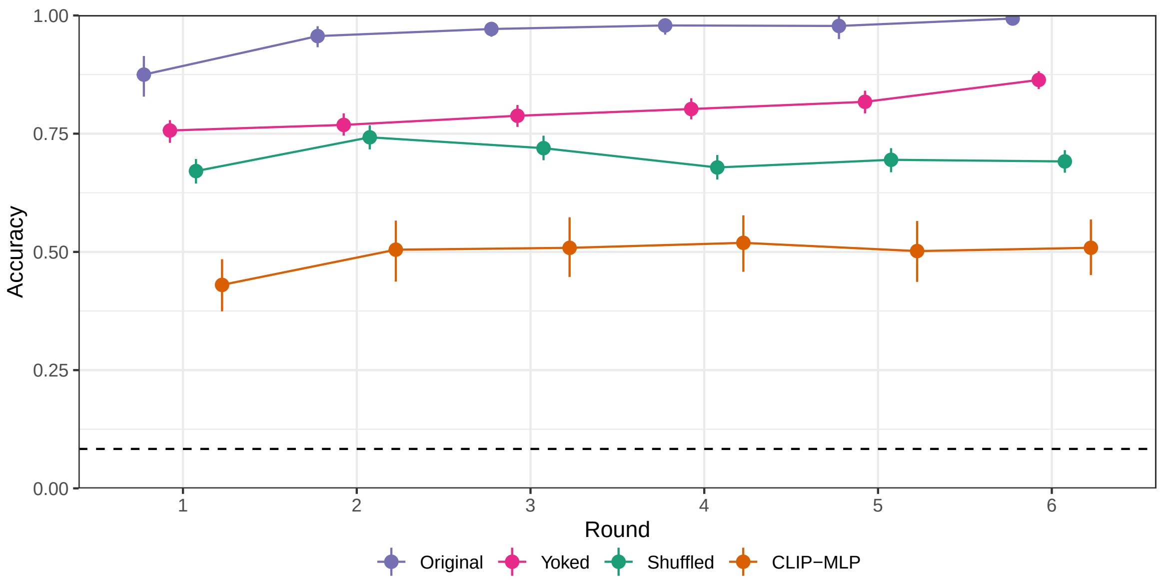


Figure 4.5: Accuracies for Experiment 3. Error bars are bootstrapped 95% CIs.

occurred. In the shuffled condition, naïve matchers saw all the descriptions from a single game in a randomized order.[5]

Because some descriptions are already fairly comprehensible in isolation, we focused on games that showed strong group-specificity. We hand-picked 10 games from Boyce et al. (2024) on the basis of high in-game matcher accuracy, strong patterns of descriptions shortening over repetition, and the use of idiosyncratic or non-modal referring expressions. Thus, these games showed the hallmarks of strong conventionalization to terms that were more likely to be opaque to outsiders.

We recruited 196 participants (99 in the yoked condition and 97 in shuffled) who each saw all 72 trials of one of the 10 games. This experiment was pre-registered at this anonymized link. Participants read the transcripts in a modified self-paced reading procedure where they uncovered the text word-by-word (revealed words stayed visible); only after uncovering the entire transcript could participants select an image.

[5]For clarity, we note this is the same yoking but a different shuffling than that used in Hawkins, Sano, et al. (2023).

4.5.2 Results and discussion

Accuracy

One primary question of interest was how much seeing the conversation history unfold in order would help participants interpret descriptions, especially those from later rounds.

We compared accuracy across the yoked and shuffled conditions with a Bayesian mixed-effects logistic regression.[6]. The descriptions were more transparent when they were presented in a yoked order (OR = 2.2, (95% CrI = $[1.63, 3]$), Figure 4.5). In the shuffled condition, there was no main effect of round number (OR for one round later = 0.99, (95% CrI = $[0.95, 1.02]$)), but there was a marginal interaction where the benefit of the yoked condition decreased for later rounds (OR for one round later = 0.94, (95% CrI = $[0.89, 1]$)). This was offset by matchers in both conditions improving at the task over time (OR for one trial later in matcher viewing order = 1.02, (95% CrI = $[1.02, 1.02]$)). In the yoked condition round and trial number were aligned, so an improvement over time could be either from matcher practice or from descriptions being easier to understand. In the shuffled condition, matcher practice effects did not correlate with position in the original game.

Comparing to the performance of in-game matchers, we separated out the benefits of seeing the descriptions in order versus being a participant in the group.[7] There was a benefit to seeing the items in order (OR = 2.24, (95% CrI = $[1.63, 3.04]$)) and a larger benefit to being a participant during the game (OR = 4.35, (95% CrI = $[2.77, 6.89]$)). The benefit of seeing the items in order waned in later blocks (OR = 0.94, (95% CrI = $[0.89, 1]$)), but the benefit of being in the game did not (OR = 1.06, (95% CrI = $[0.95, 1.18]$)). In all cases, there was a baseline improvement over trials (OR = 1.02, (95% CrI = $[1.02, 1.02]$)). As a caveat, we note that in-game matchers and naïve matchers may have varied from each other in terms of effort and time spent on the task, and thus the comparison should be interpreted cautiously.

The accuracy of the CLIP-MLP model was worse than the shuffled human results,

[6] correct$\sim$ orig_repNum $\times$ condition + matcher_trialNum + (1|gameId) + (1|correct_tangram) + (1|workerid)

[7] correct$\sim$ orig_repNum $\times$ order + orig_repNum $\times$ setting + matcher_trialNum + (1|gameId) + (1|correct_tangram) + (1|workerid)

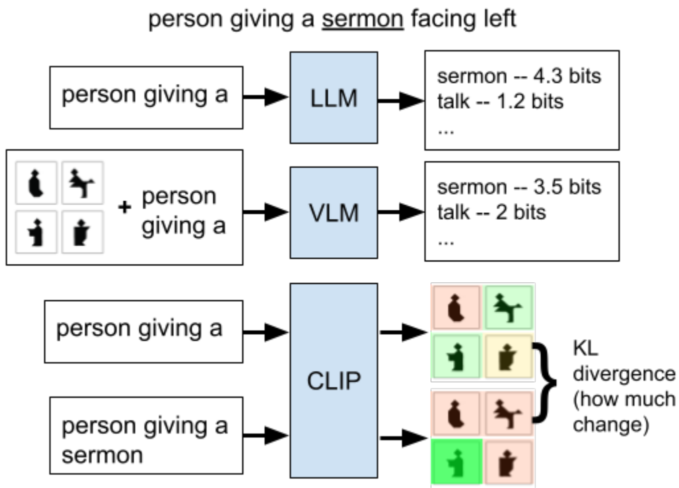


Figure 4.6: Schematic of informativity predictors used in the incremental reading time analysis.

and did not change across rounds (OR for one round later = 1.02, (95% CrI = $[0.97, 1.07]$)). The larger difference between naïve human and CLIP-MLP accuracies in Experiment 3 than Experiment 2 suggests that the shuffled ordering still provides useful context that helps matchers understand the conventions. This history was not available to the CLIP-MLP model which saw every description as a one-shot task.

Incremental reading times

Our second question was whether comprehenders would rapidly adapt to the task, which we addressed by considering different potential predictors of word-by-word reading times that varied on whether the took in local context. As a baseline, we included word length and log frequency. We also considered three measures of a word's informativity: **1)** surprisal from a base large language model (LLM, LLama 3.1 8B) (Grattafiori et al., 2024), **2)** surprisal from a vision language model (VLM, Llama 3.2 11B)(Grattafiori et al., 2024), conditioned on grid of 12 possible targets, and **3)** the word-wise change (operationalized as KL divergence) in predicted target distribution from the CLIP-MLP model discussed above (Figure 3). This last measure represents the task-relevant information content of a word based on how much it shifts the model's distribution over possible targets. We residualized the measures of

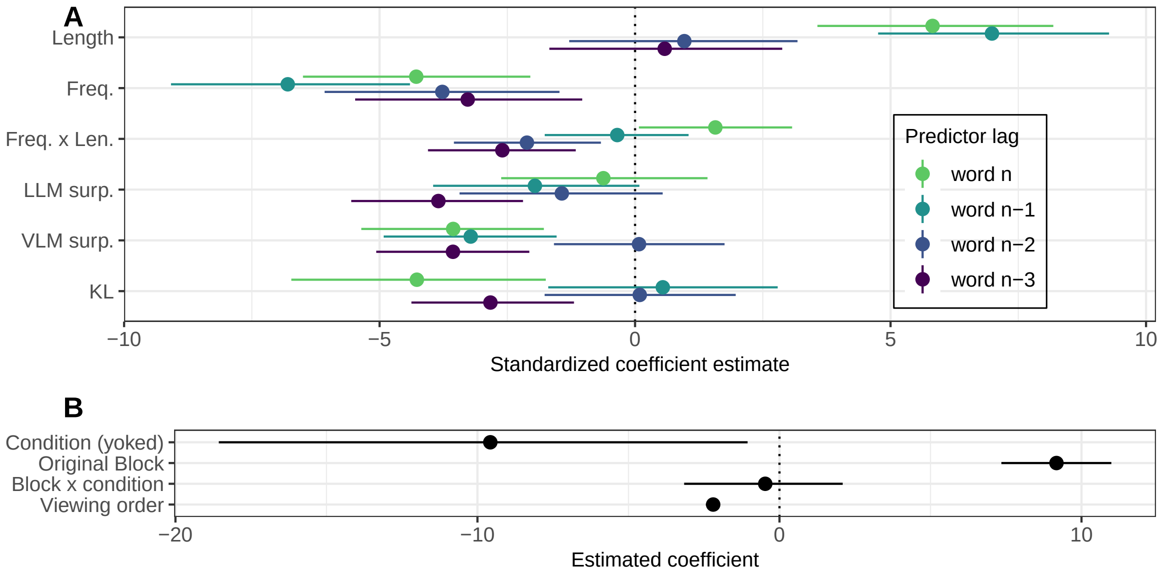


Figure 4.7: Estimates and 95% credible intervals for coefficients in the incremental processing model. Word-level predictors are shown in panel A, colored by spillover position. Condition effects are shown in panel B.

informativity and normalized all word-level predictors, and then modeled RTs as a function of these predictors for the current and 3 prior words, in addition to condition-level predictors (Figure 4.6). [8]

Model results are shown in Figure 4.7. As expected, shorter and more frequent words were faster to read. Unexpectedly, current word LLM surprisal was not predictive of reading time, and both VLM surprisal and KL divergence were predictive of shorter reading times, in contrast to the typical positive relationship between surprisal and RTs. LLM surprisal, VLM surprisal, and KL all measure different sources of information with increasing task-specificity, and may have different impulse response functions, leading to different patterns of lag.

Overall, participants tended to read faster in the yoked condition than the shuffled condition, indicative of context having a facilitative role on reading time. Participants also tended to get faster over the course of the task, which could be either adaptation

[8]The model formula was *RT* $\sim$ *[word-length* $\times$ *frequency + LLM + VLM + KL] + original-block* $\times$ *condition + order-viewed + (1 | game) + (1 | target) + (1 | participant)* Bracketed predictors were included for current word and 3 prior words to account for spillover.

to the task, or an increasing comprehension of the descriptions (which would be supported by their increasing accuracy over time). Participants tended to read more slowly per word for descriptions coming from later original blocks; however; this measure is confounded both by viewing order being correlated with block for the yoked condition, and by reduced representation of short descriptions in the reading time data, due to the need for spillover predictors.

The reading time paradigm we used is noisy and prone to spillover effects. For user interface reasons, we did not require a button press after the last word to enable target selection, so we do not have RTs for the final words. Additionally, our analytic approach meant that we only considered words where we had predictors for the three previous words (to account for spillover); however, this does mean that we do not have usable data for utterances that were 4 words or fewer. We do think we were getting some signal as the RTs were correlated with word length and word frequency, but unfortunately, we are unable to interpret the measures of information because of the noisy paradigm.

4.6 General Discussion

Real-world conventions vary in whether they are opaque to outsiders ("rizz") or interpretable even to those who don't produce them ("roundabout"). Convention formation in the real world is difficult to study, so iterated reference games are a method for operationalizing convention formation for experimental study. In reference games, conventions are partner-specific: different groups' evolving conventions follow different paths through semantic space. Despite the number of studies on iterated reference games, few studies have examined whether the descriptions are interpretable by outsiders.

Across multiple experiments, we found that naïve human matchers were far above chance at identifying the targets, and our computational model was also above chance. For both humans and models, more variation was explained by the target image than the round or game condition the descriptions came from, suggesting that conventionalization was not the primary driver of how difficult an expression was to interpret.

Even for games selected for strong conventionalization, naïve matchers had high accuracy overall, although this accuracy was further increased if they saw the conversation history in order. Exploratory analyses also suggested that more idiosyncratic descriptions were more opaque. Our findings are consistent with a lexical uncertainty approach, where expressions that are closer to overall priors are easier to understand, and groups that have fewer people and thicker channels are more able to break away from these priors and have conventions that drift farther apart in semantic space.

This work suggests that conventions formed within a small group may still be fairly comprehensible to those outside the group, as the semantics of these conventionalized expressions may still be relatively close to the semantics of the image. Hence, such conventions may not be completely arbitrary, and are instead strongly related to the image and the shared general lexicon, although it is possible that greater arbitrariness and opacity may be obtained with more extended interaction.

While this first attempt to combine incremental sentence processing methods with descriptions from iterated reference games was not very informative, it points out that this genre of methods could be used to address questions of how both visual and conversational context influence influence word predictability and in turn the speed at which words are integrated in online sentence processing. Further work could refine the measures using more sensitive techniques such as eye-tracking, analyses that better account for time course, and measures of information that take prior informativity into account. Beyond reading time measures, one could also consider indexing informativity more directly with a gated paradigm that allows for periodic guesses of the target to provide human measures of incremental task-informativity.

Limiting the generality of our findings, our experimental and computational results were only on a specific set of iterated reference game transcripts and images. Some images may lend themselves to more transparent descriptions because they are more iconic, with a narrower prior over different ways they could be conceptualized, or they may be further from competitors within this pool of images. Future work sampling across larger sets of images (such as Ji et al., 2022) could probe image-level factors.

Nonetheless, this work has demonstrated the utility of adopting a broader perspective on convention comprehension. In particular, the use of computational modelling allowed for a means to estimate semantics under lexical uncertainty without requiring symbolic semantic representations. Future work could capitalize on this approach to better understand semantic dynamics underlying convention formation, as well as provide further quantitative investigations of pragmatics models like RSA and CHAI. These directions will help us to better understand the nature of reference and conventions, and how humans navigate the complex and ever-evolving landscape of communication.

Conclusion

This dissertation started with the questions of robustness and generalizability of the phenomena associated with convention formation in iterated reference games, with a goal of understanding whether these phenomena were "special", necessitating their own theoretical explanations, or whether they could be accounted for as non-central examples within broader-coverage theories. Our approach was to run iterated reference game experiments over a wider set of parameters, to probe how well empirical evidence lined up either with the special verbal theories or a wider terrain of smoother variability.

Synthesizing across experiments, one observation is that the central findings of the iterated reference game literature are quite robust. People very frequently and naturally create shorthand names for targets as a way to effectively communicate about them, and which nicknames are used vary widely, with some popular concepts shared across many groups, but also a long tail of idiosyncratic, yet effective terms. Initial descriptions are usually long, evolving over time into shorter expressions. With "thick" communication channels, similar to those used in much of the literature, but mediated by text instead of speech, the trends are very consistent with prior findings from the in-person dyadic literature. People are just good at communicating, and good at coordinating with each other.

The interesting part is that some of the classic patterns occur even under more limiting circumstances, even beyond what seems to be predicted from the informal theories. For instance, at least for pairs, people do just fine without much in the way of backchannel. Even for larger groups of 6 people without backchannel and with limited knowledge of performance, accuracy increases and utterances shorten. The descriptions produced under these circumstances are so effective that even naïve matchers get them right more than 50% of the time on 12-way choice. In one view, this is robust evidence for our ability to produce and interpret language effectively. In another, it means that maybe common ground and audience design aren't so important if people can succeed without very much of it?

Similarly, we find robust evidence of the path-dependency of what referring expressions are used. Many different descriptions can be effective, and dyads find many of them. Children display similar patterns in miniature – pairs can use many different conceptualizations effectively. However, most of them can be understood reasonably well by others, and many of them are reasonably transparent to models of language statistics. Thus, while the formation of referring expressions may be ideosyncratic and depend on the partner, the interpretation of the expression is less dependent. It is as if people start out pluripotent in the set of associations they will accept and the referring expression process mostly is of picking one association, with rarer instances of creating new associations beyond the shared pool.

Finally, a wider range of circumstances complicates the use of reduction as a proxy for partner-specific convention formation. In larger groups with constrained communication channels, the degree of group-specificity of referring expression seems to uncouple from the length of the utterances. In a different uncoupling, children do show some partner-specificity but do not show a decrease in expression length.

Overall, we do find quantitative patterns where more communication and more context generally leads to better performance and faster conventionalization. It's easy to come up with a theory that predicts that directionally. The hard part is determining *how much* more supportive some circumstances should be than others.

So, what should we think about human communicative capacities? Some of the ideas that people are really attuned to conversation with a partner and say just the right things for their partner to understand probably aren't true, at least in the strong form. Which is not at all to say that people aren't great at communicating – we find instead that the communication abilities are wide and flexible. This work opens up new questions: how do we explain the path-dependence of people's utterances, if it isn't driven by necessity and efficiency? How much of the path-dependency and referring expression length are driven by production constraints? How do we model the generation process that occasionally leads to very idiosyncratic descriptions and more often results in less idiosyncratic descriptions? How do these patterns relate to iconicity and group dynamics?

While the work presented in this dissertation pushes forward our empirical understanding of adaptive language use in iterated reference games, there are a set of limitations that should be noted. Some of these limitations apply more generally to the literature as a whole, and thus point to opportunities for future advancement.

First, it must be noted that the generalizability of the work is limited by the generalizability of the stimuli, the participant populations, and the paradigm. We used a limited set of 12 vaguely humanoid tangrams in Chapters 2 and 4 and a different set of 4 vaguely humanoid tangrams in Chapter 3. The literature as whole has more variability, so we have no reason to believe that our results are contingent on the stimuli, but features of stimuli are very likely to influence the results via their levels of iconcity, affordances for describing strategies (pose, geometry, body parts), and levels of codeability and discriminability. We are limited in our ability to assess variation due to stimuli because of the small number of stimuli that we sampled.

Our studies were all conducted in English using convenience populations of adults recruited for Prolific and children recruited from a university preschool. From the literature we have some evidence that the similar patterns of key convention phenomena do occur in French and Dutch as well. To the extent that the paradigm is accessing some general communicative and linguistic abilities, we should expect the general patterns to generalize across demographic factors, languages, and cultures. However, different languages may have different affordances for how to identify unknown referents and and how to shorten and develop nicknames. Typological variation in language might lead different parts of speech to be more common in reduced utterances, for instance.

As we transition from documenting that humans can use language to communicate under these circumstances and become increasingly consise to trying to model how humans do this, features of the language itself may become more relevant. One blind spot in the literature is with signed languages, which have some different affordances related to iconicity in that modified meanings can be expressed either with additional signs or with modifications to the head signs (Ferrara et al., 2025).

From a social perspective, we might suspect that social distances, social hierarchies, and social norms could all constrain the interactions and patterns of language use in reference games. We put people in an artificial game environment with specific communication goals, but participants are interacting with people and bring their social knowledge with them. Thus, there are unmeasureable expectations about how conversation works and unmeasured goals for social connection that could be influencing all of our measures. One could imagine that people with closer connections or different social mores might show some different patterns and quantitative variation. We might be somewhat reassured by general robustness of the underlying patterns to modalities with different social norms (in-person, by phone, and by web-chat) and the fact that some studies don't find differences between friends and strangers. However, while suggestive of some generalizability, we have not tested how social factors are influencing game play. Shared society and culture might be particularly relevant to our naïve matcher results since guessing the target may depend on shared conceptual interpretations and priors over what the images could represent.

Another concern with this work is that we do not know the causal mechanisms behind the dependent variables that are measured. Thus, we also do not know what moderators are relevant and what dimensions we expect results to generalize over. Current statistical models used to assess reduction (or disfluency rate, or speed, etc) have the issue that they are fit to a specific dataset and can't be compared across datasets and situations, so it's difficult to assess their quantitative generality (Yarkoni & Westfall, 2017). Developing formal theory, even at a computational level, is difficult as many dynamics and equations could suggest the same directional outcome (van Dongen et al., 2025). It is particularly complicated here where the measurable outcomes are interrelated and are not the actual target of interest. To think about a generating model, we would need to determine what latent constructs are core invariants and which variables are affected by implementational details.

A final consideration is how ecologically valid the experimental paradigm is. To what extent are iterated reference games a useful model of natural human communication that captures something relevant about convention formation? There are various disanalogies between normal conversation and reference games. In real life, many interactions are more than just about reference and have a more open set of possible referents. This may be an acceptable simplification, and in other work, we do find suggestions that referential reduction and alignment can occur in coordination games (Boyce & Frank, 2023; Mankewitz et al., 2021). In real life, many of our interactions are with friends or other people who we interact with regularly, unlike in reference games where participants are generally strangers. Thus, the amount of shared history and the timeline of accumulating shared history is quite dissimilar between friendships and reference games. It is possible that the differences in type and distribution of shared context could lead to different types of convention formation. I do not think iterated reference games are a complete model of all linguistic convention formation, but I do think it is a sufficiently good analogy, especially for more temporary conventions between people who do not have years-long history together. Iterated reference games certainly can tell us something about how people use language to do a task that has an atypical distribution of communicative needs.

In ongoing work, I am using a couple of new approaches to partially address some of the limitations discussed here. The first method is to aggregate and harmonize data from multiple existing datasets to get some generalizability and answer questions that cannot be answered in a satisfactory way with only one dataset. The other is to use tools from natural language processing to explain semantic and lexical changes in referring expressions over the process of conventionalization.

One question that currently cannot be adequately answered by the data presented here is what is the functional form predicting the number of words used as a function of repetition. We could try to fit that model, but we would not be learning anything generalizable since we only have 6 time points and the entire dataset shares the same set of targets and some other paradigm properties. It's not even clear that repetition is the right predictor versus something more like how much to reduce based on what informational profile the last utterance had. However, by using a larger set of data with more variability in experiment length, number of options, paradigm details, and stimuli, we might be able to more meaningfully test a set of different functional form relations and have some confidence that we were not just overfitting to irrelevant design choices.

Other questions best addressed across multiple datasets include the robustness and generalizability of the key phenomena. Different papers report somewhat differing sets of outcomes and operationalize outcomes differently even when the same construct is of interest. Thus, we might know the directionality of effects is consistent across set-ups, but we don't know how much variability there is, either based on moderators, or just in general.

One approach to pooling effects and studying moderators is meta-analysis, which combines like effect size measures across different exact experimental implementations. However, given the different measures of interest and different ways of measuring change over time, for iterated reference games, the numerical results are not easy to compare. Instead, we are working on harmonizing raw data across different datasets, in order to do mega-analytic statistical measures where the individual data points all go into the same regressions, with data source as hierarchical random effects.

In addition to studying potential moderators of interest, pooling data across datasets also allows us to check for robustness to factors that we have posited as unlikely to be crucial. For instance, while the conditions reported in Chapter 2 all shared a text-based chat interface and the same set of target stimuli, other experiments from other researchers have used face to face oral paradigms and line drawing stimuli. Thus, applying a mega-analytic approach across multiple datasets allows for estimating the effect from these moderators. In addition to getting robust estimates and measures of generalizability, this could also potentially inform future experimental design, both by points out gaps in the experimental space that has been explored, but also identifying what dimensions of experimental variation seem to matter most meaningfully. For instance, one could imagine that type of stimuli has more interacting effects on number of words whereas written versus spoken might just cause an overall decrease without any interactions. One might then decide to one the more convenient modality but focus on varying experimental stimuli more systematically.

The other future direction we are pursuing is applying NLP tools to try to get a better grasp on the linguistic patterns in iterated reference games. We believe that the changes observed are probably operating more on semantics and information of the words than on metrics like length. While we don't have the models we would need for a generative model of iterated reference game utterances, we do have the tools to more closely look at semantic changes using embeddings. For instance, we might be able to fine tune a model on utterance – selection pairs and use it as a proxy for matchers to then do attribution of information to words within the utterance. Determining what the information profile of utterances is and whether informativity is a strong predictor of what does or doesn't get reduced is a way we can use the tools now available to answer scientific questions about the process of reduction. Since this process involves model training, we need as much language from reference games as we can get. The more data and data diversity we have, the better and more robust a model we can tune.

There are many possible approaches to understanding linguistic communication. In the dissertation work, we have built on many previous studies of language from a variety of sub-disciplines. We've drawn from the study of conversation, from sentence processing, and from computational linguistics. Beyond language research, we've also drawn on cognitive science more broadly and developmental psychology. Advances in computational methods and experimental methods have enabled this work, and further theoretical and technical connections will lead to further insights. Understanding language deeply requires explaining patterns of language on different time scales and levels of abstraction. I look forward to continuing to explore language use in its duality as a communicative modality and a mental process.

Appendix A

Feedback

Figure A1 shows the type of feedback shown to participants in different conditions.

Semantic content of descriptions

As an exploratory, qualitative analysis of the describer's language, we performed dictionary searches to classify whether the describers utterances on each trial contained the following types of words.

Geometric/literal words: squar*, triangle, triangular, diamond, shape, trapez*, angle, degree, parallel*, rhomb*, box, cube, line, white, black

Words for body parts: face, head, heads, back, shoulder, shoulders, arm, arms, leg, legs, foot, feet, body, knee, knees, toe, toes, hand, hands, body, butt, heel, heels, ear, ears, nose, neck, chest, hair

Positional words: right, left, above, below, under, over, top, bottom, behind, side, beneath

Posture words: kick*, crouch*, squat*, kneel*, knelt, stood, stand*, sit*, sat, lying, walk*, facing, fall*, looking, lean*, seat*, laying

Qualitatively, descriptions that contain none of these words are usually shorter, more abstract, more conventionalized descriptions. Figure A2 shows the fraction of trials where the speaker used these words. Generally, the more concrete word categories become less frequent over time, although this effect seems weaker in larger, thinner games.

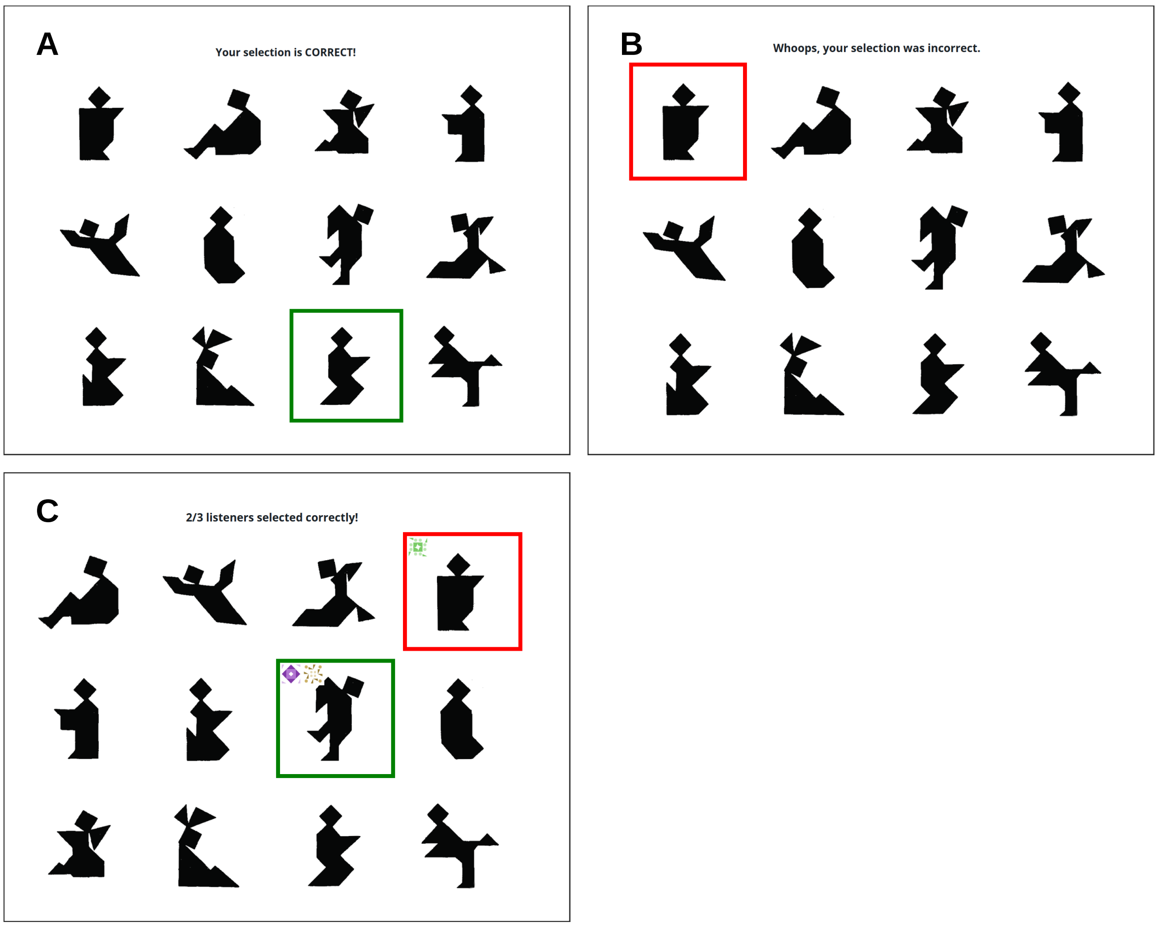


Figure A1: Feedback shown to participants. In low-coherence games, matchers saw feedback as shown in A and B. In all games, describers saw feedback like in C; icons were associated with players and indicated who selected which. In high feedback games, matchers saw grids like C, but with text as in A and B.

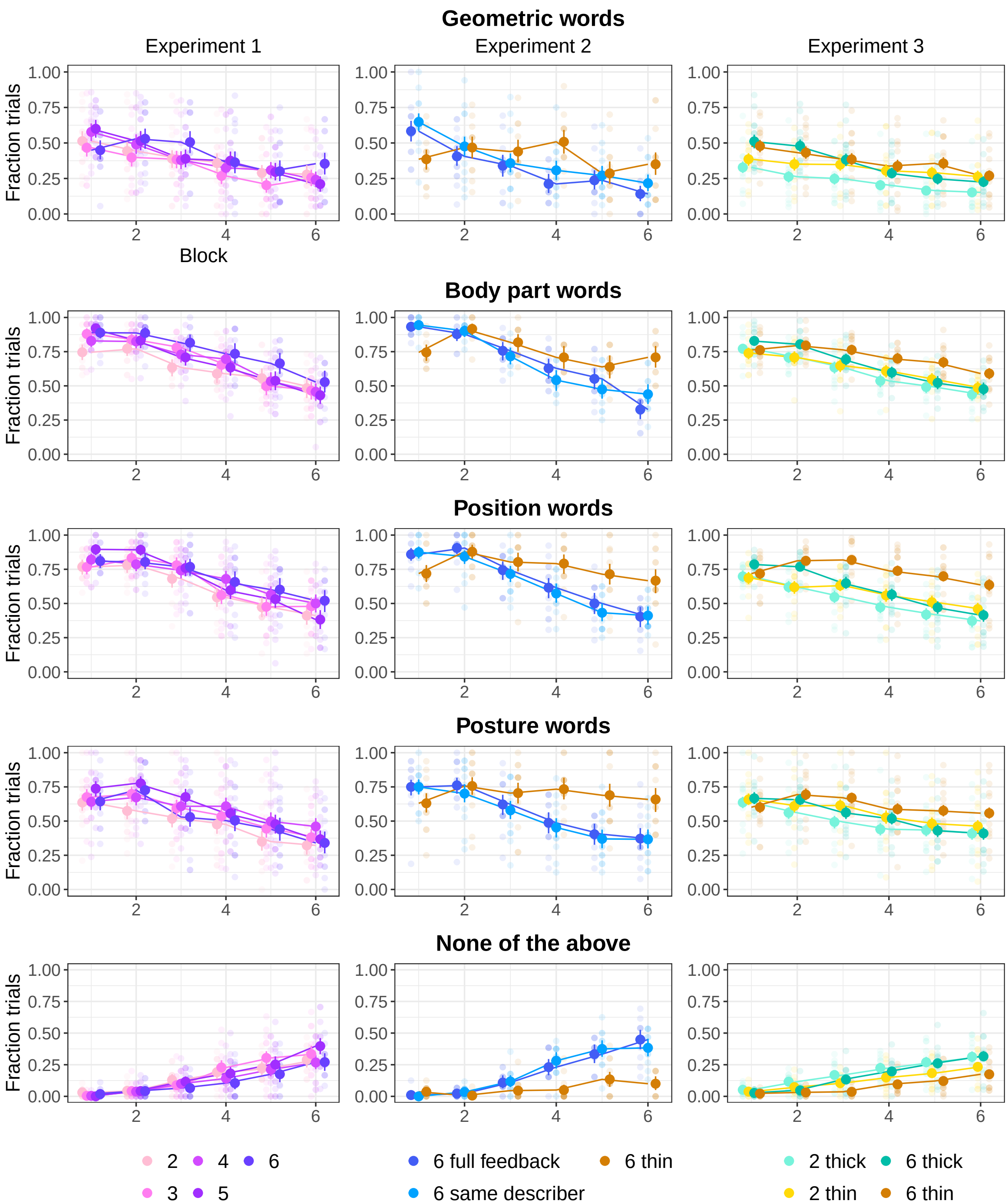


Figure A2: Fraction of trials where the describer used certain types of words. See text for word lists. Faint dots are per tangram, larger dots and error bars are means and boostrapped 95% CIs.

Additional game transcripts

Table A1: Additional examples from 6-player thick games for the same image across repetitions. Describers are indicated with an asterisk.

6-person thick game	
Rep 1: 5/5 correct	
Q*	oh no. okay. this one is similar to the one i called minnesota, so like pretty simple, with just a distinct “head” diamond and then a body beneath
Q*	the body has sort of sloping shoulders
P	BEAN SHAPED BODY?
O	with a small triangle on the bottom right?
Q*	yeah i think so! bean is a good description lol
Q*	bean body
Rep 2: 5/5 correct	
Q*	bean body!!
Rep 3: 5/5 correct	
Q*	bean body!
Rep 4: 5/5 correct	
Q*	bean body!
Rep 5: 5/5 correct	
Q*	you guys are doin’ great. :) bean body for this one
Rep 6: 5/5 correct	
Q*	bean body for this one

6-person thick game	
Rep 1: 2/5 correct	
V*	This shape hasn’t got much sticking out. Small triangle pointing right, tilted square on top
Y	Is the triangle on the right at the bottom?
V*	Long tall shape, not far off symmetrical.
V*	Yes triangle bottom right
W	Are their two triangles on it?
X	does it look like a rabbit?
V*	Not the rabbit
X	does it look like someone on their back kicking a square soccer ball
V*	It’s like one block like a gemstone
V*	but square on top
V*	Not on back
V*	It’s the one with the least sharp corners I think
V*	One long gem shape with small triangle bottom right, tilted square on top
Rep 2: 4/5 correct	
V*	Long tall shape. Little triangle sticking out bottom right. Diamond on top
Rep 3: 3/5 correct	
V*	Long tall hexagon, triangle bottom right, diamond on top
Rep 4: 4/5 correct	
V*	Long hexagon
V*	little triangle bottom right
V*	Diamond top
Rep 5: 4/5 correct	
V*	Long hexagon.
V*	Diamond on top
V*	little triangle bottom right
Rep 6: 4/5 correct	
V*	Long hexagon
V*	Triangle bottom right

Table A2: Additional examples from 6-player thin games for the same image across repetitions. Target image is same as Table A2. Describers are indicated with an asterisk.

6-person thin game	
Rep 1: 2/5 correct	
J*	Looks like a skinnier candle but with a shadow at the bottom
K	🤔
H	✅
J*	The top square looks like the flame
J*	It has a triangle leading to that flame
I	❌
Rep 2: 3/5 correct	
K	🤔
I*	the simplest image in the whole, no much issues
J	🤔
L	🤔
I*	1 square as the head
L	🤔
J	🤔
H	🤔
K	🤔
I*	candle with more missing chunks
Rep 3: 1/5 correct	
L*	looks like a candle with more chunks missing
J	🤔
I	🤔
L*	The bottom part faces the right and forms a triangle
L*	The top is a square
I	✅
J	❌
Rep 4: 3/5 correct	
H*	Candle one with shadow
H*	The straight one not the one with bits missing
L	🤔
I	😂
L	🤔
H*	Lower case j but the bottom swings to the right
I	✅ ✅ ✅ ✅
Rep 5: 0/5 correct	
K*	Candle with chunks missing on the right
I	✅ ✅ ✅
Rep 6: 4/5 correct	
G*	candle shape with a chunk of the bottom left missing and added to the bottom right instead
K	🤔 🤔
H	✅
I	✅

6-person thin game	
Rep 1: 2/5 correct	
A*	Square on top
A*	At the bottom of the base part of it sticks out to the right
A*	The base has 8 faces (8 sides of the shape)
C	✅
A*	The square balances on a triangular point of the base
A*	It is a thick base shape, the bottom curves right
D	✅
Rep 2: 5/5 correct	
B*	tombstone and its shadow with a rotated square on top
D	✅
F	✅
C	🤔
Rep 3: 5/5 correct	
D*	tombstone casting a shadow (sorry i've pinched that from before)
C	🤔
B	😂
E	🤔
C	🤔
E	🤔 🤔
D*	square on top sitting on top of a triangle which is on top of a square, with 5 side shape at the bottom
Rep 4: 4/5 correct	
C*	looks mummified
C*	person with no limbs
D	✅
B	🤔
C*	square atop, chunky with triangle bottom right
Rep 5: 5/5 correct	
F*	Gravestone casting shadow. Square on top
D	✅
Rep 6: 5/5 correct	
E*	tombstone
D	✅

Recruitment and number of games

The number of games in each condition varied due to the realities of online recruitment for real-time games. We had control over how many participants we recruited on

Prolific. Because of the real-time nature of the games, we relied on the fact that most Prolific workers who accepted the task would rapidly open it and go to our experiment site, where they would read the instructions and be assigned into games. Due to technical details about how games filled, we only ran one condition of games at any one time.

In experiments 1 and 2, games would only start if the game was full. Thus, if for example, we recruited 28 people, and 27 people went to the experiment when we were running a 4-person game, three people would be unable to make a complete game, and we also wouldn't have groupmates available for a 28th person who opened the experiment late.

We ran games in large batches to try to minimize the fraction of people who wouldn't be matched into a game. If we got many fewer (complete) games than anticipated, we could recruit another batch of participants, taking the expected attrition rate into account. However, the logistics of needing participants to all open the study at roughly the same time to match into games meant that it was infeasible to recruit fewer than 20 or so participants at a time, so we could not precisely achieve the targeted number of games. Thus, the number of games varied somewhat across conditions.

In addition to not having control over exactly how many games started, some games also ended prematurely when participants disconnected. These partial games were more common in larger groups. We speculate it could be due either to just having more people who might have connection issues or need to leave the game, or more frustration causing people to quit the game (sometimes very early in the game).

Both of these factors led to a low yield rate in terms of completed games per participants recruited, especially in experiment 2 where all the games were 6-player. To try to maximize the data collected per money spent on recruitment, we made adjustments for experiment 3. In experiment 3, we allowed games to them to start and continue with fewer than 6 people. Thus, while many of the "6-player" games did not have 6 players the entire game, we at least have data from the entire course of the game.

In experiment 3, the 6* player games did not all have 6 players, both because games continued as participants dropped out and because if there weren't enough

Table A3: The number of games in each experiment and condition. Complete games finished all 6 blocks; partial games ended early due to disconnections, but contributed at least one complete block of data. 6* indicates that some games started with fewer than 6 players or continued with fewer than 6 players after participants disconnected.

Experiment	Players	Complete	Partial	Total Participants
1: baseline	2	15	4	38
1: baseline	3	18	2	60
1: baseline	4	19	2	84
1: baseline	5	17	3	100
1: baseline	6	12	6	108
2: same describer	6	15	3	108
2: full feedback	6	13	4	102
2: thin	6	10	6	96
3: thin	2	35	3	76
3: thin	6*	44	0	235
3: thick	2	39	3	84
3: thick	6*	38	2	222

players after 5 minutes of waiting, the game would start with whoever was there. All analyses use "intent to treat" and call these 6 player games.

In Figure A3, the number of games goes up in some cases because only complete blocks (where the describer said something every trial) are analysed. If there was initial confusion and a describer missed a trial, that block was excluded.

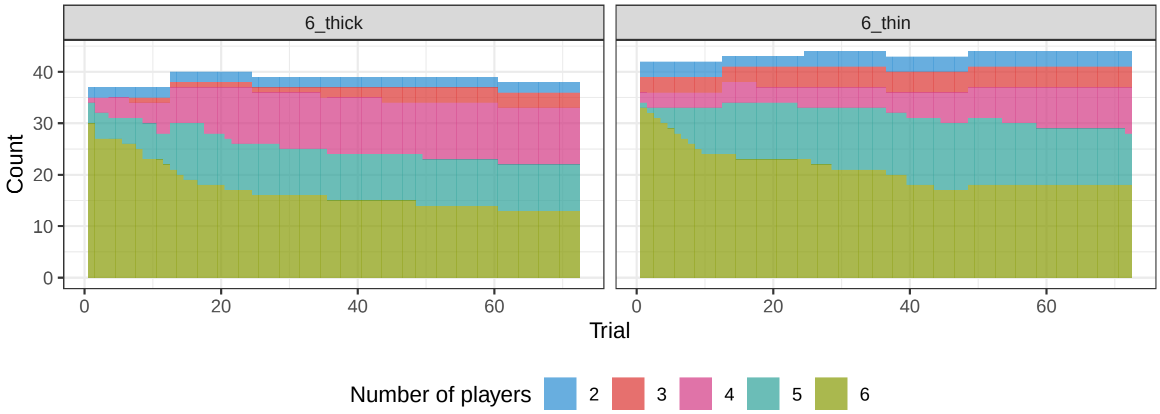


Figure A3: Number of players during 6 thin and 6 thick games in experiment 3. Blocks that were incomplete were excluded, so if a describer said nothing during a trial, that block was excluded.

Sensitivity analyses

As (post-hoc) sensitivity analyses, we re-ran our main analyses on the subset of games that completed all 6 blocks with a full set of players. This is a check on whether any of the results are driven by a) partial games that were included in early blocks but not later blocks or b) experiment 3 games that started without a full set of players or had players drop out.

Versions of main paper Figures 2 and 4 with just these complete games are shown here in Figures A4 and A5. The full model output for the sensitivity analyses is in Tables A54-A73.

The only qualitative differences in the results between these models and those on the full dataset are in experiment 3. Note that large experiment 3 games were the ones most likely to be excluded, both for incompleteness and for not having the full 6 players throughout.

In the accuracy model, the block:gameSize interaction is not robust, and there is a marginal effect of block:gameSize:thin instead. The main effect of gameSize is also weaker. In the reduction model, the block:gameSize interaction is not robust. In the convergence model, the earlier:gameSize interaction is not robust.

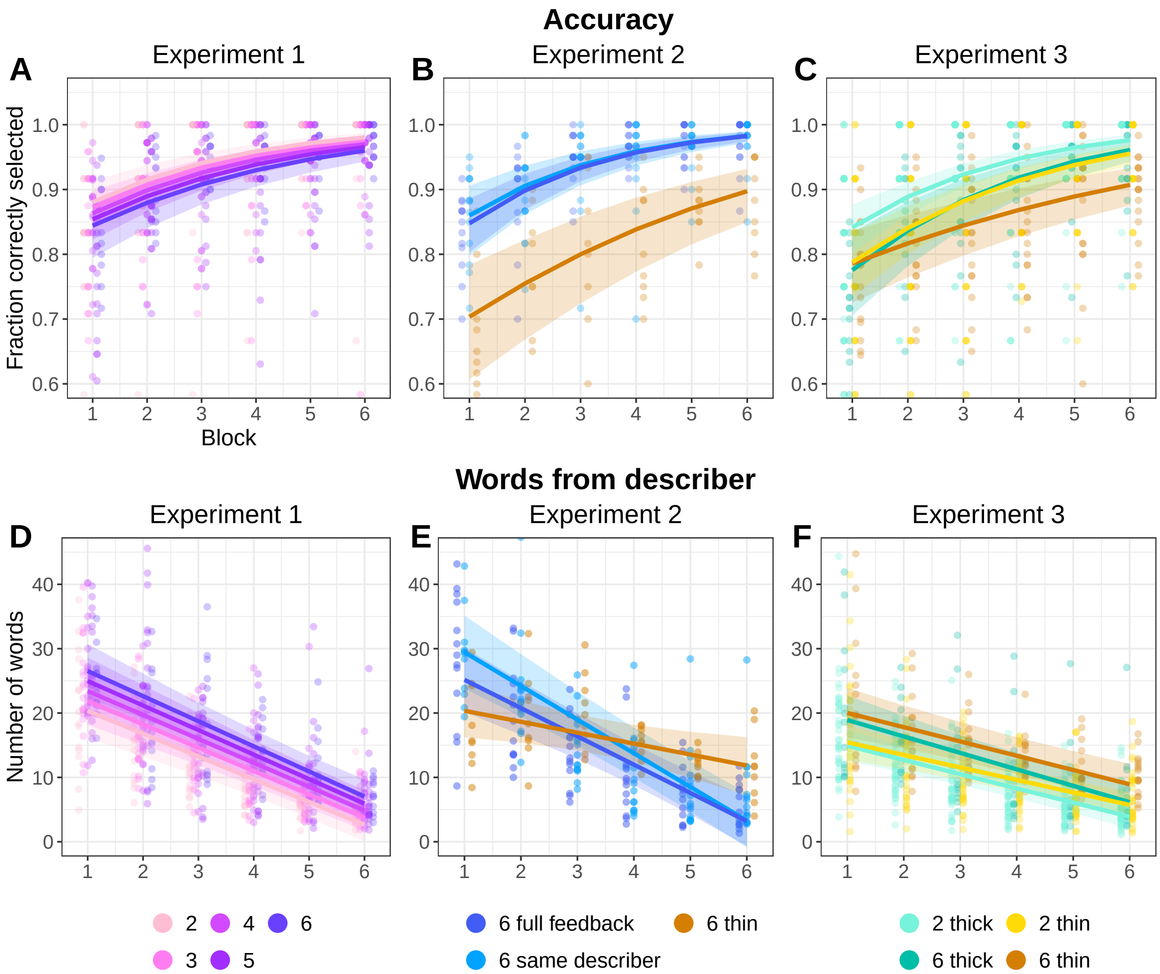


Figure A4: Equivalent of Main text Figure 2, with only full complete games. Behavioral results across all three experiments. (A-C). Matcher accuracy at selecting the target image. (D-F). Number of words produced by the describer each trial. For all, small dots are per game, per block means, and smooth lines are predictions from model fixed effects with 95% credible intervals. Y-axes are truncated, and a few outliers points are not visible.

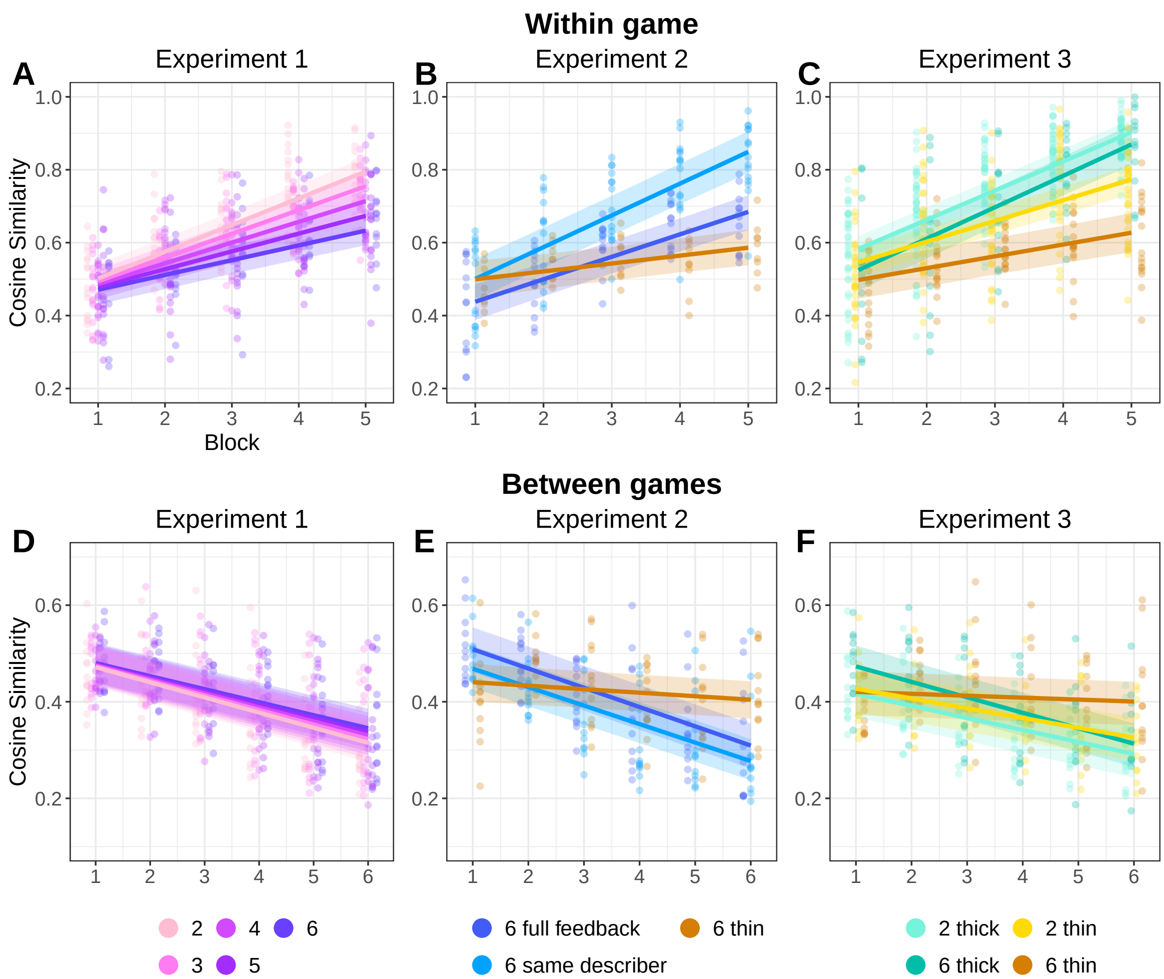


Figure A5: Equivalent of Main text Figure 4 with only full complete games. Language similarity results measured with pairwise cosine similarity between embeddings of two utterances. (A-C). Convergence of descriptions within games as measured by similarity between an utterance from block 1-5 to the block 6 utterance in the same game for the same image. (D-F). Divergence of descriptions across games as measured by the similarity between two utterances produced for the same image by different groups in the same block. For all, small dots are per game, per block means, and smooth lines are predictions from model fixed effects with 95% credible intervals. Y-axes are truncated, and a few outliers points are not visible.

Matcher contributions

We can think about matcher utterances either at the level of one matcher or at the level of a group. In larger groups, the total number of matchers who could contribute is higher, so matcher utterances might be more common. On a per-matcher basis, having more group-mates could potentially increase the amount a given matcher contributes (more people to potentially have lateral conversations with) or it could decrease it (others are already asking the questions you were going to).

We analyzed matcher utterances both ways – on a game level and on an individual level. We broke it down into two steps: 1) are there matcher contributions on a given trial and 2), for trials with matcher contributions, how long are the contributions.

Matchers' contributions declined over the course of the game. The use of emoji in the thin games is not directly comparable to matcher language use in thick games, since some emoji usage (such as the green checkmark) is most likely equivalent to non-referential matcher language ("got it" etc.) that was excluded. The higher rate of emoji use versus referential language thus could be due to its non-equivalence, a lower level of accuracy in thin games, or matchers having a lower threshold for sending emojis compared to writing out clarifications.

We note a deviation from the pre-registration here in the analysis of the emojis. In the pre-registration we said we would "analyse the distribution of emoji's produced as a function of block and its relation to accuracy and speaker utterance length." We did not do this beyond the visualization shown here.

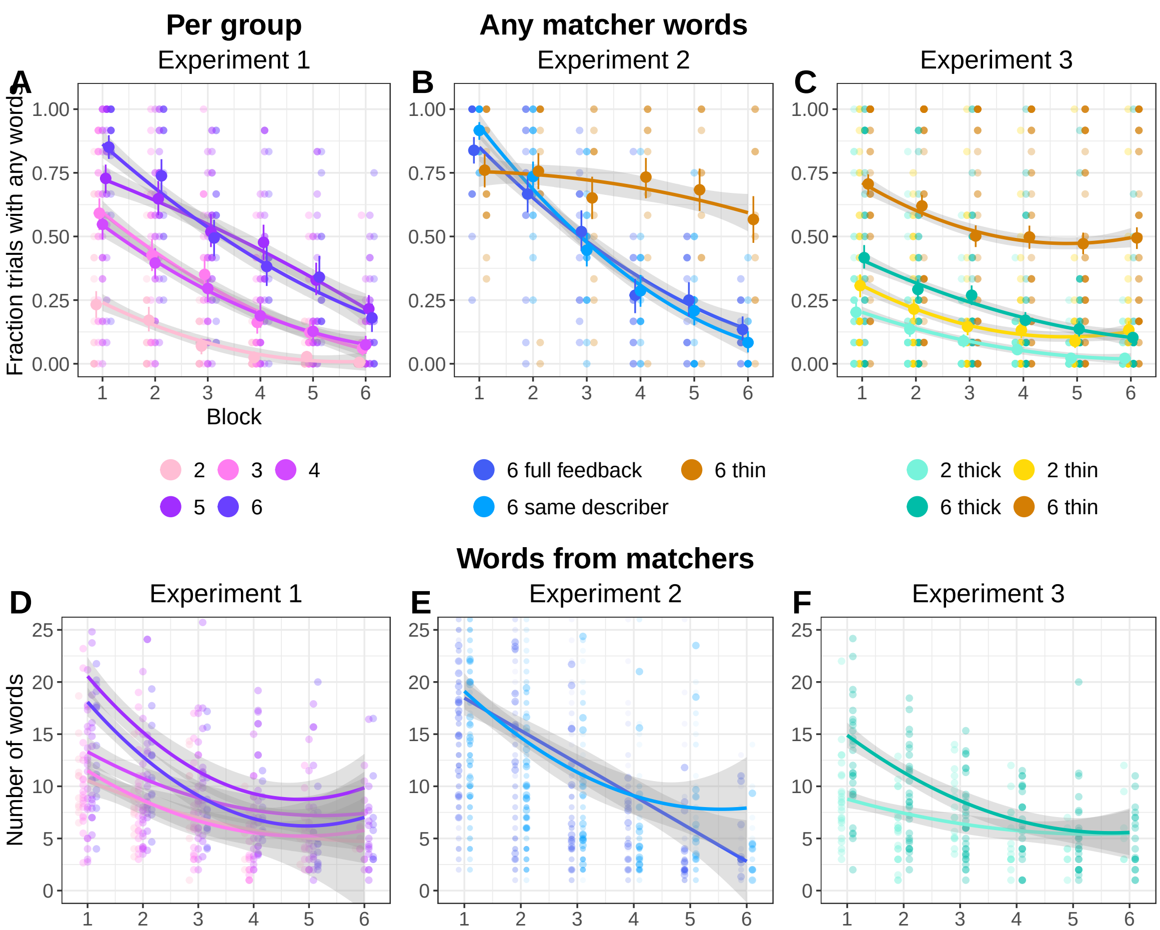


Figure A6: Matcher contributions at the group level. A-C: Fraction of trials where any matcher said anything that was referential. Dots are per game averages. Smooths are binomial fit lines. D-F: On trials where at least one matcher contributed, the number of words of referential language produced by matchers. Dots are per game averages. Smooths are quadratic fit lines. Y-axis is truncated, and a few outliers points are not visible.

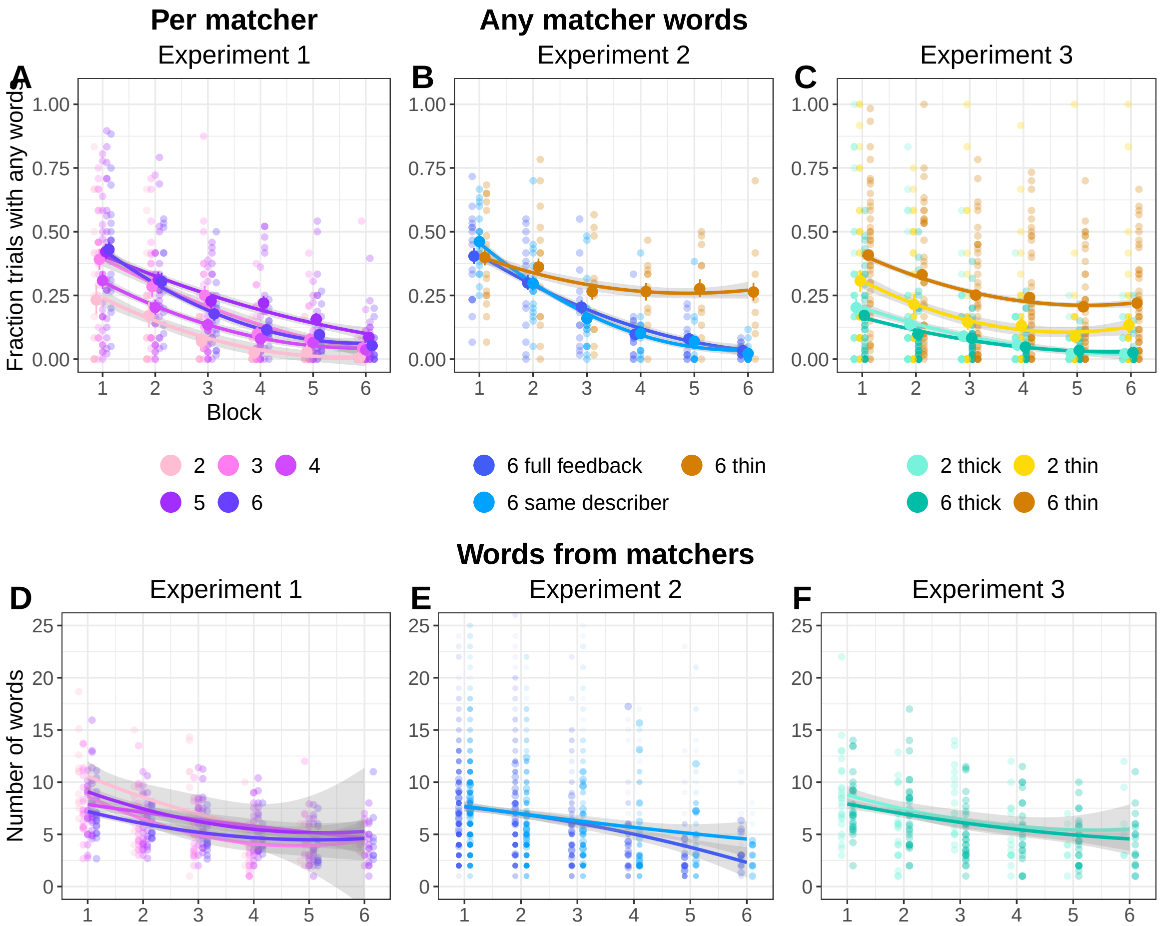


Figure A7: Matcher contributions at the individual level. A-C: Fraction of trials where an individual matcher said anything that was referential. Dots are per game averages. Smooths are binomial fit lines. D-F: On trials where a matcher contributed, the number of words of referential language that matcher produced. Dots are per game averages. Smooths are quadratic fit lines. Y-axis is truncated, and a few outliers points are not visible.

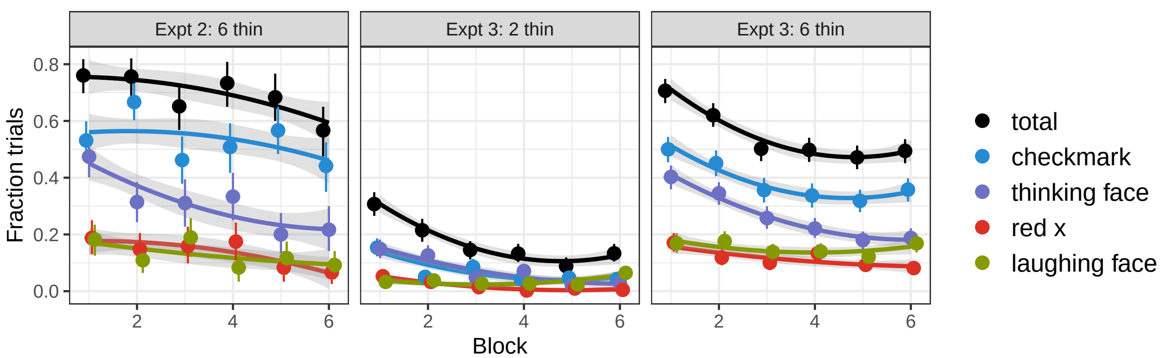


Figure A8: Fraction of trials on which at least one matcher produced the labelled emoji. Fraction of trials when any emoji was produced are shown in black. Dots are per condition, per block estimates with 95% bootstrapped CIs. Smooths are binomial fit lines.

Reaction times

As the number of words produced by the speaker each trial decreases over repetition, so does the time each trial takes. In Figure A9, the time to selection (per-matcher) is shown in panels A-C, and the time per trial (equivalent to how long the slowest matcher took) is shown in panels B-D.

Matchers sped up over time across conditions, although thin large groups may not have speed up as much. Larger groups appear to be a bit slower at the level of individual matchers, which compounds to overall slower games as there are more matchers to wait for.

We also confirm in Figure A10 that there is a strong relationship between the number of words from the describer and the length of a trial.

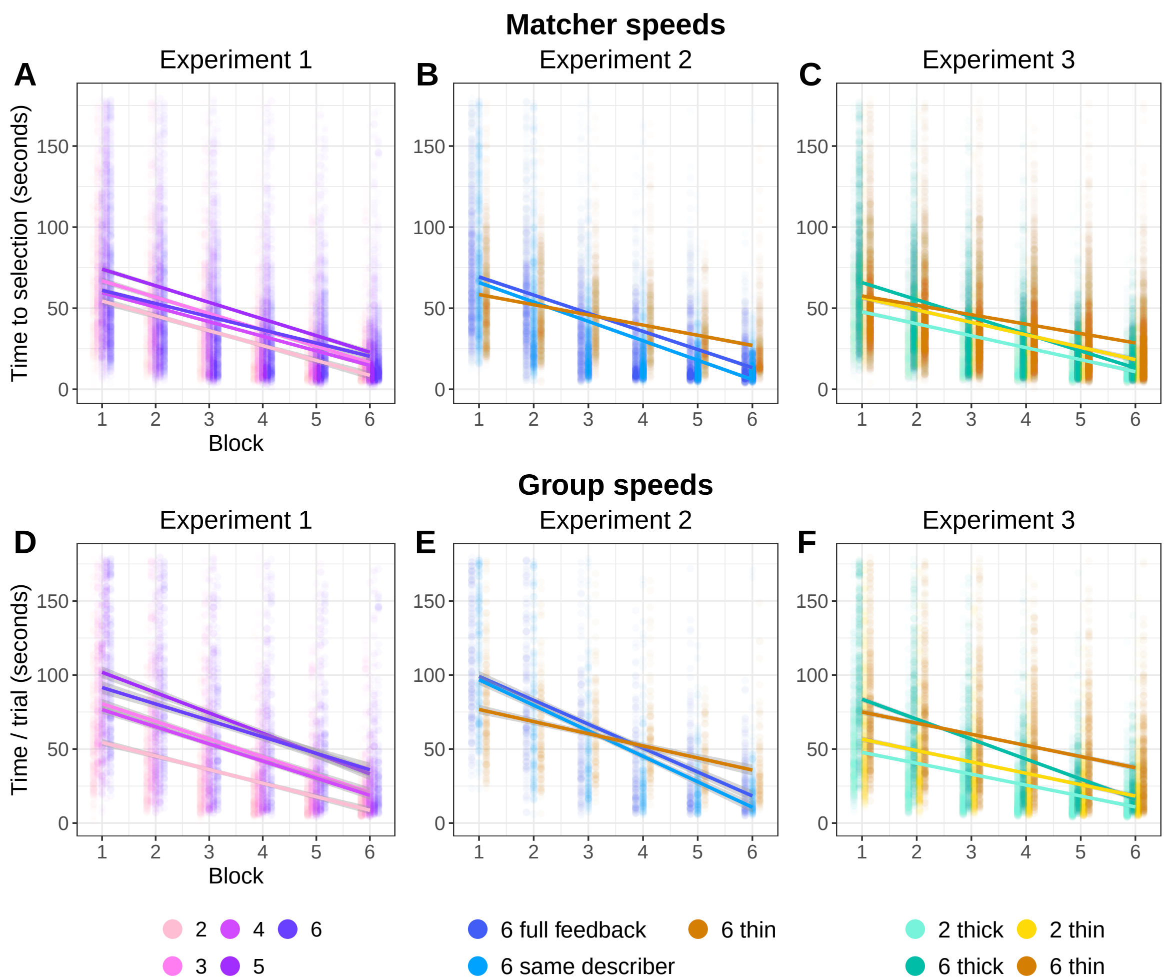


Figure A9: Behavioral results across all three experiments. (A-C). Individual matcher time to selection. (D-F). Time to end of trial for each group. For all, small dots are per game, per block means, and smooth lines are linear fits. Y-axes are truncated, and a few outliers points are not visible.

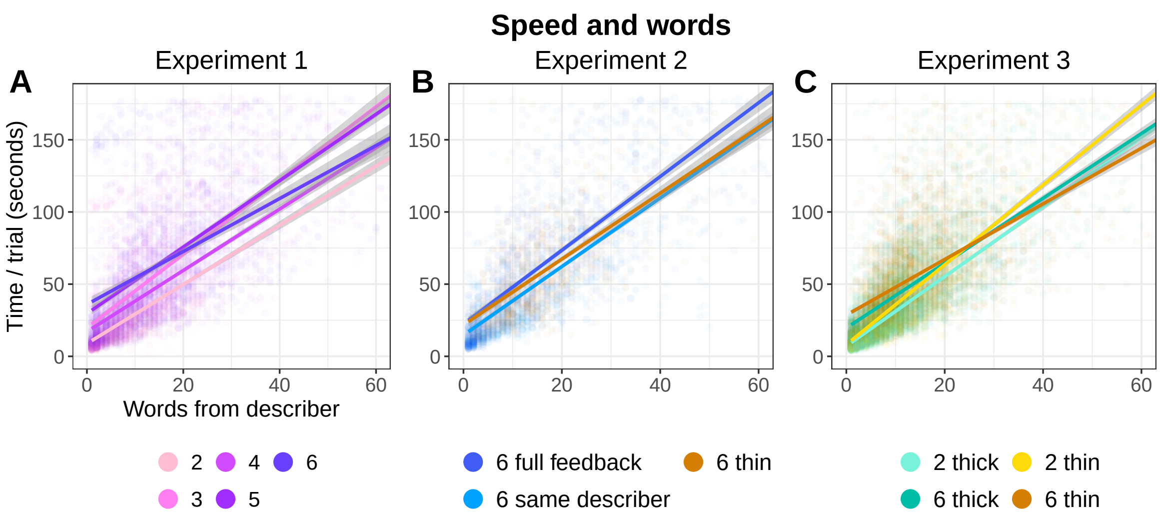


Figure A10: Relationship between number of words produced by describer and trial length. Across conditions, a similar pattern holds. X-axes are truncated, and a few outliers points are not visible.

Group level variation in models

One problem with precisely estimating the fixed effects of interest is that there is substantial variation at the game and tangram levels leading to wide confidence intervals on some of the fixed effects.

Models were fit with the maximum mixed effect structure for which the model would run in a reasonable amount of time.

As shown in Figure A11, the standard deviations for the grouping levels are substantial compared to the non-Intercept coefficient estimates. This means that the between-game or between-tangram variation is large compared to the effects of interest. This variation causes wide estimates as there is model uncertainty in how to apportion observed effects between group-level idiosyncrasies and population-level fixed effects.

This variability is diminished in the mega-analytic models which were fit with less extensive mixed effects (to aid in model convergence) and with more data.

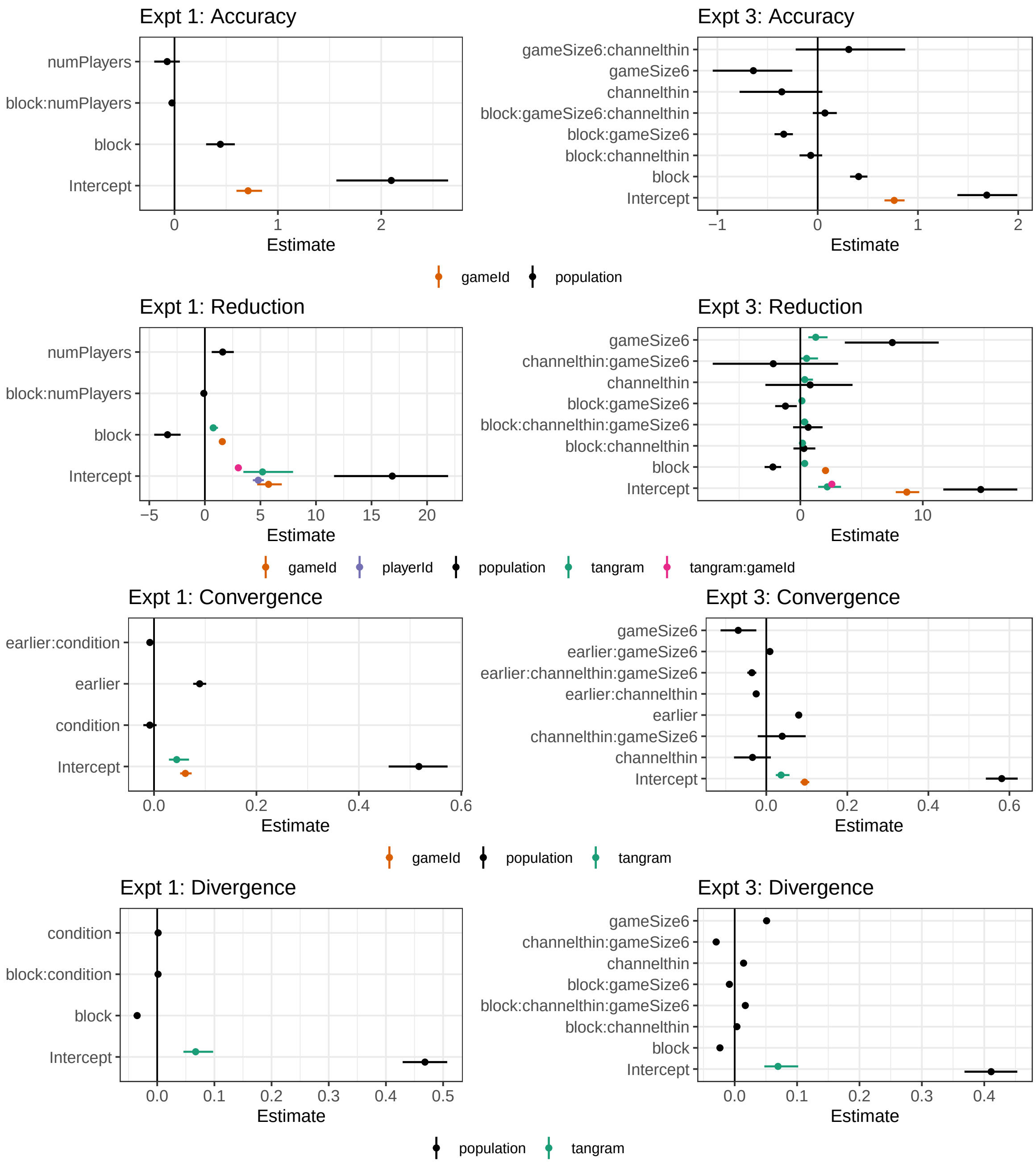


Figure A11: Comparison between the population-level coefficients (point estimate and 95% CI) and the group-level standard deviations (point estimates and 95% CIs) for the heirarchical effects.

Additional measures of convergence

The main text included the graph for convergence comparing utterances from blocks 1-5 to the utterance from block 6. Here we show two other measures of semantic shifts for descriptions for the same tangram in the same game: similarity to the first utterance and similarity to the next utterance.

Similarity to the first utterance is not very informative (but we pre-registered it). Similarity to the next utterance is what actually drives the convergence phenomena: pairs of utterances from adjacent blocks become closer together over time.

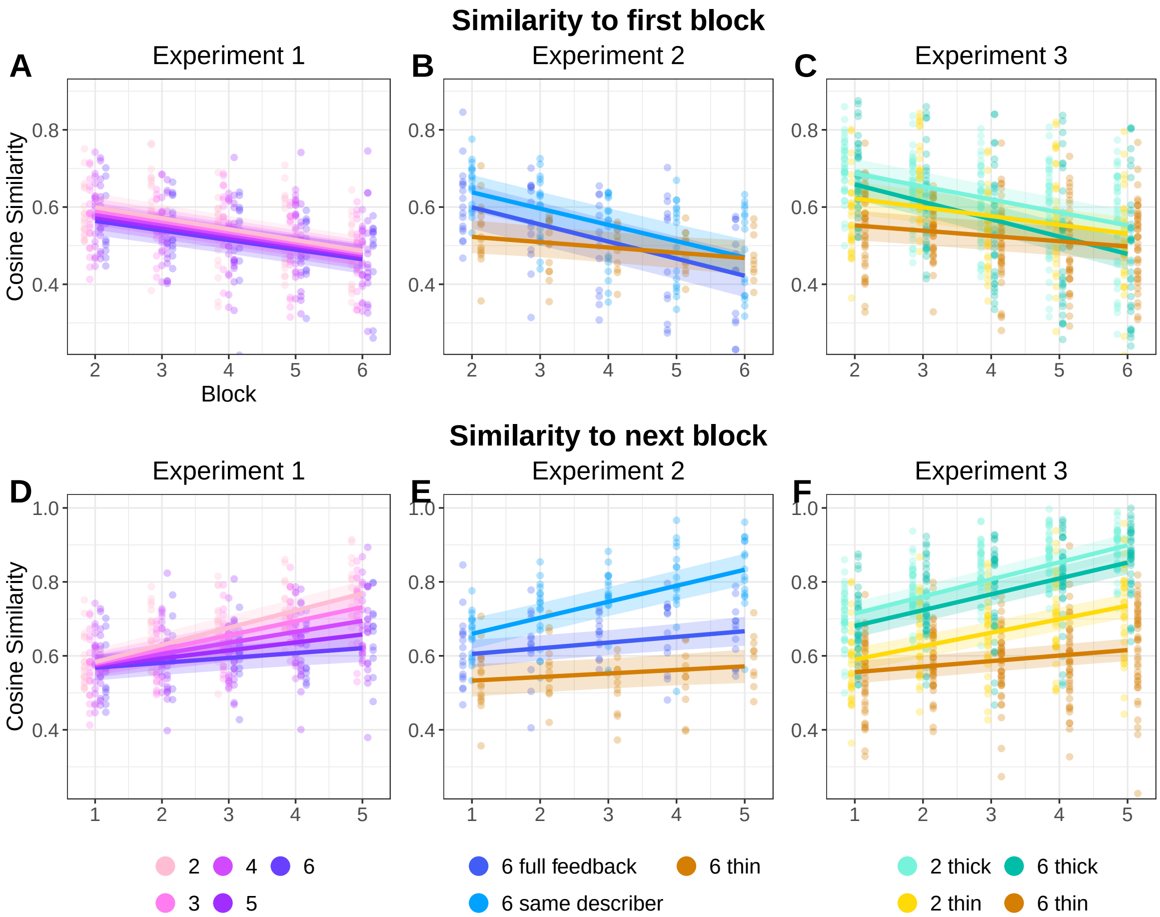


Figure A12: Additional measures of convergence and divergence. A-C is the similarity between utterances on a given block to the first block utterance for the same image, in the same game. Dots are per-game averages, smooths are quadratic. D-F is the similarity between utterances on a given block to the corresponding utterances in the next block. Dots are per-game averages, smooths are quadratic.

Distinctiveness of tangrams

An additional measure of convergence/divergence patterns is how different tangrams get described in the same game – as nicknames evolve, different tangrams get more different descriptions.

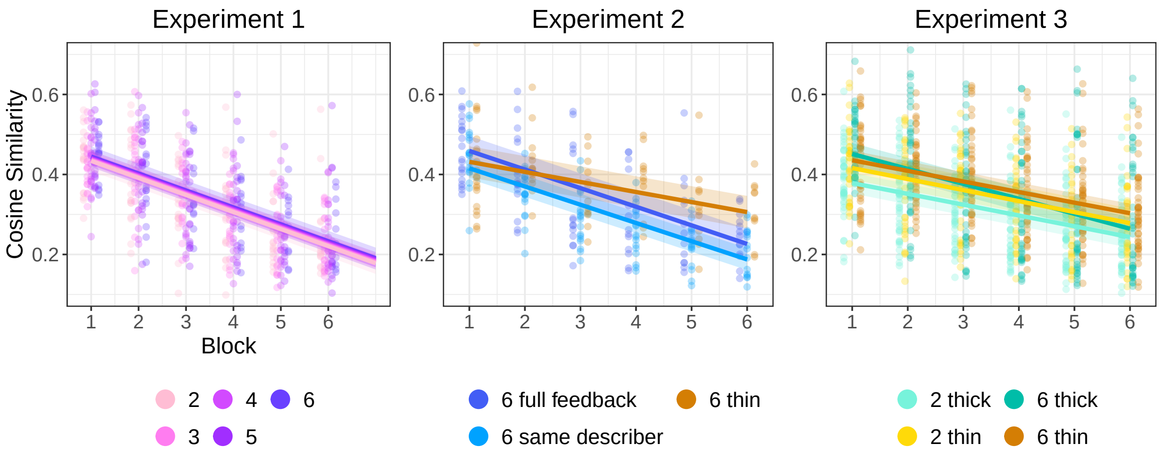


Figure A13: Divergence in descriptions of different tangrams. Cosine similarity between the descriptions of two different tangrams in the same block and group are shown. Dots are per-game averages, smooths are quadratic.

Summaries of model outputs

The following sections contain model outputs. All models were run using BRMS. We report the priors and pre-registration status for each group of models. Tables provide the individual model formulae and the point estimates and 95% credible intervals for the fixed effects.

Note that for all models, block was 0 indexed, so intercepts are what happened during the first block.

Accuracy models

Accuracy models were all run as logistic models with normal(0,1) priors for both betas and sd. This model was not explicitly included in the experiment 1 and 2 pre-registrations; it was included with more ambitious mixed effects (which did not run in a timely manner) in the experiment 3 pre-registration.

Table A4: Experiment 1 logistic model of matcher accuracy: correct.num ~ block × numPlayers + (1|gameId)

Term	Est.	95% CrI
Intercept	2.10	[1.57, 2.65]
block	0.44	[0.31, 0.58]
block:numPlayers	-0.02	[-0.05, 0.01]
numPlayers	-0.07	[-0.2, 0.05]

Table A5: Experiment 2: 6 same describer logistic model of matcher accuracy: correct.num ~ block + (1|gameId)

Term	Est.	95% CrI
Intercept	1.78	[1.4, 2.19]
block	0.45	[0.39, 0.52]

Table A6: Experiment 2: 6 full feedback logistic model of matcher accuracy: correct.num ~ block + (1|gameId)

Term	Est.	95% CrI
Intercept	1.35	[0.59, 2.06]
block	0.47	[0.39, 0.54]

Table A7: Experiment 2: 6 thin logistic model of matcher accuracy: correct.num ~ block + (1|gameId)

Term	Est.	95% CrI
Intercept	0.88	[0.64, 1.12]
block	0.23	[0.19, 0.28]

Table A8: Experiment 3 logistic model of matcher accuracy: correct.num ∼ block × gameSize × channel + (1|gameId)

Term	Est.	95% CrI
Intercept	1.69	[1.39, 1.99]
block	0.41	[0.32, 0.5]
block:channelthin	-0.07	[-0.18, 0.04]
block:gameSize6	-0.34	[-0.43, -0.25]
block:gameSize6:channelthin	0.07	[-0.05, 0.19]
channelthin	-0.36	[-0.78, 0.05]
gameSize6	-0.64	[-1.05, -0.25]
gameSize6:channelthin	0.31	[-0.22, 0.87]

Reduction models

Primary reduction model

Reduction models were run as linear models with an intercept prior of normal(12,20), a beta prior of normal(0,10), an sd prior of normal(0,5) and a correlation prior of lkj(1). This model was pre-registered for each experiment and run with the mixed effects structure as pre-specified.

Table A9: Experiment 1: words ∼ block × numPlayers + (block|tangram) + (1|playerId) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	16.87	[11.63, 21.89]
block	-3.36	[-4.56, -2.18]
block:numPlayers	-0.09	[-0.37, 0.18]
numPlayers	1.60	[0.62, 2.6]

Table A10: Experiment 2: 6 same describer: words $\sim$ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	29.65	[24.82, 34.49]
block	-5.31	[-6.35, -4.3]

Table A11: Experiment 2: 6 full feedback: words $\sim$ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	25.79	[20.97, 30.29]
block	-4.64	[-5.81, -3.53]

Table A12: Experiment 2: 6 thin: words $\sim$ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	20.3	[17.37, 23.53]
block	-2.1	[-3.37, -1.12]

Table A13: Experiment 3: words ∼ block × channel × gameSize + (block × channel × gameSize|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	14.74	[11.68, 17.72]
block	-2.24	[-2.92, -1.57]
block:channelthin	0.29	[-0.56, 1.23]
block:channelthin:gameSize6	0.64	[-0.59, 1.81]
block:gameSize6	-1.22	[-2.06, -0.29]
channelthin	0.80	[-2.85, 4.26]
channelthin:gameSize6	-2.21	[-7.16, 3.08]
gameSize6	7.51	[3.63, 11.3]

Extra reduction model

For experiment 1, we also pre-specified a model about whether the describer's correctness on the prior block (when they were a matcher) had an effect on how many words of description they produced. Priors were the same as for primary reduction model.

Table A14: Experiment 1: words ∼ block × numPlayers + block × wasINcorrect + (block|tangram) + (1|playerId) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	12.16	[6.48, 18.07]
block	-2.17	[-3.39, -1]
block:numPlayers	-0.22	[-0.5, 0.06]
block:wasINcorrect	0.24	[-0.24, 0.72]
numPlayers	2.09	[0.88, 3.3]
wasINcorrect	3.07	[1.67, 4.45]

Log reduction model

As an exploratory check, we reran some of the primary reduction models using the log number of words as the DV; otherwise model specifications and priors were the same.

Table A15: Experiment 1 log reduction: logwords $\sim$ block $\times$ numPlayers + (block|tangram) + (1|playerId) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	2.75	[2.48, 3.01]
block	-0.38	[-0.47, -0.29]
block:numPlayers	0.01	[-0.01, 0.03]
numPlayers	0.07	[0.02, 0.12]

Table A16: Experiment 3 log reduction: logwords $\sim$ block $\times$ channel $\times$ gameSize + (block $\times$ channel $\times$ gameSize|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	2.50	[2.32, 2.69]
block	-0.27	[-0.32, -0.22]
block:channelthin	0.06	[0, 0.14]
block:channelthin:gameSize6	0.03	[-0.06, 0.13]
block:gameSize6	-0.02	[-0.09, 0.04]
channelthin	0.06	[-0.17, 0.29]
channelthin:gameSize6	-0.13	[-0.44, 0.18]
gameSize6	0.45	[0.23, 0.67]

Matcher reduction models

These models were not pre-registered.

For the model of how often any matchers made contributions, the priors were normal(0,1) for both beta and sd.

For the model of how much was said on trials when matchers talked, the priors were the same as for the primary (describer) reduction model.

We ran these models both at the group level (did any matcher make a contribution, how many words of matcher contributions were there) and at the individual level (for a given matcher, did they make a contribution, and if so, how many words did they contribute).

Table A17: Experiment 1: group-level: is.words $\sim$ block $\times$ numPlayers + (1|gameId)

Term	Est.	95% CrI
Intercept	-2.67	[-3.54, -1.79]
block	-0.80	[-0.97, -0.62]
block:numPlayers	0.03	[-0.01, 0.07]
numPlayers	0.79	[0.58, 0.98]

Table A18: Experiment 1: individual-level: is.words $\sim$ block $\times$ numPlayers + (1|playerId)

Term	Est.	95% CrI
Intercept	-1.65	[-2.24, -1.05]
block	-0.82	[-0.96, -0.68]
block:numPlayers	0.05	[0.02, 0.08]
numPlayers	0.20	[0.07, 0.33]

Table A19: Experiment 1: group-level: words $\sim$ block $\times$ numPlayers + (block|gameId)

Term	Est.	95% CrI
Intercept	4.72	[0.09, 9.44]
block	-0.17	[-1.53, 1.3]
block:numPlayers	-0.41	[-0.72, -0.11]
numPlayers	2.07	[1, 3.12]

Table A20: Experiment 1: individual-level: words $\sim$ block $\times$ numPlayers + (block|playerId)

Term	Est.	95% CrI
Intercept	9.85	[8.3, 11.4]
block	-1.45	[-2.05, -0.86]
block:numPlayers	0.14	[0.02, 0.27]
numPlayers	-0.52	[-0.84, -0.21]

Initial utterance reduction model

These models were not pre-registered. They looked at describer reduction only on words that were produced prior to the first matcher message each trial. These models were only run on experimental conditions where matchers could contribute textual responses.

Reduction models were run as linear models with the same priors as the primary reduction model.

Table A21: Experiment 1: words ∼ block × numPlayers + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	18.66	[14.58, 22.71]
block	-3.56	[-4.54, -2.55]
block:numPlayers	0.27	[0.03, 0.5]
numPlayers	-0.33	[-1.14, 0.53]

Table A22: Experiment 2: 6 same describer: words ∼ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	18.06	[14.76, 21.44]
block	-2.49	[-3.19, -1.79]

Table A23: Experiment 2: 6 full feedback: words ∼ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	16.69	[13.41, 20.02]
block	-2.49	[-3.34, -1.62]

Table A24: Experiment 3: words ∼ block × gameSize + (block × gameSize|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	13.88	[11.62, 16.2]
block	-2.08	[-2.66, -1.47]
block:gameSize6	-0.65	[-1.43, 0.14]
gameSize6	5.13	[2, 7.95]

Linguistic content models

We ran a number of models predicting the cosine similarity between pairs of S-BERT embeddings of utterances. For all of these models, we used linear models with the priors normal(.5,.2) for intercept, normal(0,.1) for beta, and normal(0,.05) for sd.

These models were verbally described (but not formally specified) in the preregistrations for experiment 2 in the full feedback and thin conditions and for experiment 3, for looking at divergence between games, convergence within games (compared to first block, next block, and last block utterances), and divergence between tangrams within games.

Convergence within games: comparison to last round

This is the primary convergence metric presented in the main paper.

Table A25: Experiment 1: sim $\sim$ earlier $\times$ condition + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.517	[0.458, 0.573]
condition	-0.008	[-0.021, 0.005]
earlier	0.089	[0.076, 0.102]
earlier:condition	-0.008	[-0.011, -0.005]

Table A26: Experiment 2: 6 same describer: sim $\sim$ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.499	[0.444, 0.556]
earlier	0.086	[0.078, 0.094]

Table A27: Experiment 2: 6 full feedback: sim ∼ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.438	[0.389, 0.487]
earlier	0.062	[0.051, 0.072]

Table A28: Experiment 2: 6 thin: sim ∼ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.498	[0.453, 0.54]
earlier	0.023	[0.013, 0.033]

Table A29: Experiment 3: sim ∼ earlier × channel × gameSize + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.581	[0.542, 0.62]
channelthin	-0.034	[-0.08, 0.011]
channelthin:gameSize6	0.039	[-0.021, 0.097]
earlier	0.080	[0.074, 0.086]
earlier:channelthin	-0.025	[-0.033, -0.017]
earlier:channelthin:gameSize6	-0.035	[-0.047, -0.025]
earlier:gameSize6	0.009	[0.001, 0.017]
gameSize6	-0.069	[-0.113, -0.025]

Divergence across games

This is the divergence metric presented in the paper.

Table A30: Experiment 1: sim $\sim$ block $\times$ condition + (1|tangram)

Term	Est.	95% CrI
Intercept	0.468	[0.429, 0.507]
block	-0.035	[-0.038, -0.032]
block:condition	0.001	[0.001, 0.002]
condition	0.002	[0, 0.004]

Table A31: Experiment 2: 6 same describer: sim $\sim$ block + (1|tangram)

Term	Est.	95% CrI
Intercept	0.484	[0.442, 0.526]
block	-0.041	[-0.043, -0.039]

Table A32: Experiment 2: 6 full feedback: sim $\sim$ block + (1|tangram)

Term	Est.	95% CrI
Intercept	0.502	[0.46, 0.546]
block	-0.038	[-0.04, -0.035]

Table A33: Experiment 2: 6 thin: sim $\sim$ block + (1|tangram)

Term	Est.	95% CrI
Intercept	0.434	[0.406, 0.465]
block	-0.004	[-0.006, -0.001]

Table A34: Experiment 3: sim $\sim$ block $\times$ channel $\times$ gameSize + (1|tangram)

Term	Est.	95% CrI
Intercept	0.411	[0.368, 0.453]
block	-0.024	[-0.025, -0.023]
block:channelthin	0.004	[0.002, 0.005]
block:channelthin:gameSize6	0.017	[0.015, 0.019]
block:gameSize6	-0.008	[-0.01, -0.007]
channelthin	0.014	[0.01, 0.018]
channelthin:gameSize6	-0.030	[-0.035, -0.024]
gameSize6	0.051	[0.047, 0.055]

Divergence across tangrams

This is an additional metric comparing the similiarities between descriptions for different tangrams within a game. It measures how distinct the descriptions for different tangram images are.

Table A35: Experiment 1: sim $\sim$ block $\times$ condition + (1|gameId)

Term	Est.	95% CrI
Intercept	0.429	[0.382, 0.473]
block	-0.043	[-0.046, -0.039]
block:condition	0.000	[-0.001, 0.001]
condition	0.003	[-0.008, 0.014]

Table A36: Experiment 2: 6 same describer: sim $\sim$ block + (1|gameId)

Term	Est.	95% CrI
Intercept	0.416	[0.389, 0.443]
block	-0.046	[-0.048, -0.044]

Table A37: Experiment 2: 6 full feedback: sim ~ block + (1|gameId)

Term	Est.	95% CrI
Intercept	0.459	[0.422, 0.496]
block	-0.047	[-0.049, -0.044]

Table A38: Experiment 2: 6 thin: sim ~ block + (1|gameId)

Term	Est.	95% CrI
Intercept	0.432	[0.393, 0.471]
block	-0.025	[-0.028, -0.022]

Table A39: Experiment 3: sim ~ block × channel × gameSize + (1|gameId)

Term	Est.	95% CrI
Intercept	0.378	[0.352, 0.404]
block	-0.027	[-0.029, -0.025]
block:channelthin	-0.001	[-0.003, 0.002]
block:channelthin:gameSize6	0.011	[0.008, 0.015]
block:gameSize6	-0.010	[-0.013, -0.008]
channelthin	0.038	[-0.001, 0.082]
channelthin:gameSize6	-0.053	[-0.115, 0]
gameSize6	0.073	[0.035, 0.113]

Convergence to next

We also looked at how similar an utterance was to the next block utterance for the same image in the same group: this can be thought of as the derivative of the to-last comparison. (Although cosine similarities are not actually additive in the same way integrals are).

Table A40: Experiment 1: sim $\sim$ earlier $\times$ condition + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.591	[0.541, 0.641]
condition	-0.004	[-0.014, 0.006]
earlier	0.063	[0.051, 0.075]
earlier:condition	-0.008	[-0.011, -0.006]

Table A41: Experiment 2: 6 same describer: sim $\sim$ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.660	[0.619, 0.702]
earlier	0.043	[0.037, 0.05]

Table A42: Experiment 2: 6 full feedback: sim $\sim$ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.605	[0.569, 0.643]
earlier	0.015	[0.006, 0.024]

Table A43: Experiment 2: 6 thin: sim $\sim$ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.533	[0.49, 0.578]
earlier	0.010	[0, 0.019]

Table A44: Experiment 3: sim $\sim$ earlier $\times$ channel $\times$ gameSize + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.714	[0.682, 0.746]
channelthin	-0.124	[-0.159, -0.088]
channelthin:gameSize6	0.000	[-0.051, 0.049]
earlier	0.046	[0.041, 0.052]
earlier:channelthin	-0.010	[-0.018, -0.002]
earlier:channelthin:gameSize6	-0.018	[-0.029, -0.007]
earlier:gameSize6	-0.003	[-0.011, 0.004]
gameSize6	-0.034	[-0.069, 0.003]

Divergence from first

We also looked at how similar an utterance was to the first block utterance for the same image. This is not very informative because first round utterances tend to be pretty noisy with lots of hedges and filler words.

Table A45: Experiment 1: sim $\sim$ later $\times$ condition + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.647	[0.591, 0.705]
condition	-0.010	[-0.022, 0.003]
later	-0.030	[-0.041, -0.019]
later:condition	0.001	[-0.002, 0.004]

Table A46: Experiment 2: 6 same describer: sim $\sim$ later + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.680	[0.628, 0.728]
later	-0.042	[-0.049, -0.035]

Table A47: Experiment 2: 6 full feedback: sim $\sim$ later + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.644	[0.584, 0.706]
later	-0.044	[-0.052, -0.037]

Table A48: Experiment 2: 6 thin: sim $\sim$ later + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.537	[0.49, 0.584]
later	-0.014	[-0.023, -0.004]

Table A49: Experiment 3: sim ~ later × channel × gameSize + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.721	[0.681, 0.76]
channelthin	-0.076	[-0.123, -0.026]
channelthin:gameSize6	-0.062	[-0.127, 0.001]
gameSize6	-0.017	[-0.062, 0.03]
later	-0.034	[-0.039, -0.028]
later:channelthin	0.011	[0.003, 0.019]
later:channelthin:gameSize6	0.021	[0.01, 0.032]
later:gameSize6	-0.011	[-0.019, -0.004]

Exploratory Mega-analytic models

As an exploratory measure, we combined data across all experiments and re-ran the core set of models (same structure and priors as for the individual models). We binarized game thickness: all games that were not thin were counted as thick. The intercept condition is a 2-player game, in the thick condition, for the first block. Thinner means the game was in the thin condition; larger is per additional player, and block is per later block. Group size was coded based on assigned condition, not based on actual number of players at the time.

Table A50: Mega-analytic on accuracy: correct.num $\sim$ block $\times$ thinner $\times$ larger + (1|gameId)

Term	Est.	95% CrI
Intercept	1.83	[1.6, 2.07]
block	0.46	[0.4, 0.52]
block:larger	-0.07	[-0.09, -0.05]
block:thinner	-0.12	[-0.21, -0.02]
block:thinner:larger	0.01	[-0.02, 0.04]
larger	-0.07	[-0.15, 0]
thinner	-0.50	[-0.89, -0.08]
thinner:larger	-0.02	[-0.14, 0.11]

Table A51: Mega-analytic of reduction: words $\sim$ block $\times$ thinner $\times$ larger + (block|gameId)

Term	Est.	95% CrI
Intercept	17.38	[15.59, 19.21]
block	-2.80	[-3.24, -2.36]
block:larger	-0.36	[-0.51, -0.2]
block:thinner	0.79	[0.04, 1.52]
block:thinner:larger	0.26	[0, 0.51]
larger	2.12	[1.5, 2.75]
thinner	-1.59	[-4.61, 1.55]
thinner:larger	-0.90	[-1.92, 0.11]

Table A52: Mega-analytic on divergence between groups: sim $\sim$ block $\times$ thinner $\times$ larger

Term	Est.	95% CrI
Intercept	0.428	[0.426, 0.431]
block	-0.026	[-0.026, -0.025]
block:larger	-0.002	[-0.002, -0.002]
block:thinner	0.005	[0.004, 0.007]
block:thinner:larger	0.004	[0.004, 0.005]
larger	0.012	[0.011, 0.013]
thinner	-0.003	[-0.007, 0.001]
thinner:larger	-0.007	[-0.008, -0.006]

Table A53: Mega-analytic on convergence to last: sim $\sim$ earlier $\times$ larger $\times$ thinner

Term	Est.	95% CrI
Intercept	0.546	[0.534, 0.558]
earlier	0.072	[0.067, 0.077]
earlier:larger	0.000	[-0.002, 0.002]
earlier:larger:thinner	-0.007	[-0.01, -0.004]
earlier:thinner	-0.016	[-0.025, -0.008]
larger	-0.016	[-0.021, -0.012]
larger:thinner	0.009	[0.002, 0.016]
thinner	0.001	[-0.021, 0.021]

Sensitivity analysis models

These are the same specification as the primary models, just on the subset of the data from games that were full and complete.

Accuracy

Table A54: Experiment 1 logistic model of matcher accuracy: correct.num $\sim$ block $\times$ numPlayers + (1|gameId)

Term	Est.	95% CrI
Intercept	2.15	[1.5, 2.78]
block	0.43	[0.29, 0.57]
block:numPlayers	-0.02	[-0.05, 0.01]
numPlayers	-0.08	[-0.22, 0.07]

Table A55: Experiment 2: 6 same describer logistic model of matcher accuracy: correct.num $\sim$ block + (1|gameId)

Term	Est.	95% CrI
Intercept	1.81	[1.38, 2.27]
block	0.45	[0.37, 0.52]

Table A56: Experiment 2: 6 full feedback logistic model of matcher accuracy: correct.num $\sim$ block + (1|gameId)

Term	Est.	95% CrI
Intercept	1.72	[1.41, 2.05]
block	0.46	[0.39, 0.54]

Table A57: Experiment 2: 6 thin logistic model of matcher accuracy: correct.num $\sim$ block + (1|gameId)

Term	Est.	95% CrI
Intercept	0.86	[0.43, 1.29]
block	0.26	[0.21, 0.31]

Table A58: Experiment 3 logistic model of matcher accuracy: correct.num $\sim$ block $\times$ gameSize $\times$ channel + (1|gameId)

Term	Est.	95% CrI
Intercept	1.67	[1.4, 1.96]
block	0.41	[0.32, 0.5]
block:channelthin	-0.06	[-0.18, 0.06]
block:gameSize6	-0.02	[-0.12, 0.09]
block:gameSize6:channelthin	-0.14	[-0.27, 0]
channelthin	-0.36	[-0.77, 0.05]
gameSize6	-0.43	[-0.9, 0.03]
gameSize6:channelthin	0.42	[-0.21, 1.04]

Reduction

Table A59: Experiment 1: words $\sim$ block $\times$ numPlayers + (block|tangram) + (1|playerId) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	17.22	[11.42, 22.74]
block	-3.35	[-4.64, -2.05]
block:numPlayers	-0.09	[-0.39, 0.21]
numPlayers	1.55	[0.43, 2.69]

Table A60: Experiment 2: 6 same describer: words $\sim$ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	29.45	[23.78, 35.14]
block	-5.23	[-6.36, -4.05]

Table A61: Experiment 2: 6 full feedback: words ∼ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	25.17	[20.16, 30.14]
block	-4.40	[-5.5, -3.28]

Table A62: Experiment 2: 6 thin: words ∼ block + (block|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	20.35	[16.21, 24.45]
block	-1.70	[-3.03, -0.38]

Table A63: Experiment 3: words ∼ block × channel × gameSize + (block × channel × gameSize|tangram) + (1|tangram:gameId) + (block|gameId)

Term	Est.	95% CrI
Intercept	14.93	[12.54, 17.43]
block	-2.21	[-2.77, -1.67]
block:channelthin	0.26	[-0.48, 1.05]
block:channelthin:gameSize6	0.07	[-1.28, 1.45]
block:gameSize6	-0.35	[-1.32, 0.64]
channelthin	0.52	[-2.53, 3.52]
channelthin:gameSize6	0.58	[-4.8, 6.19]
gameSize6	3.99	[0.09, 8.14]

Convergence within games

Table A64: Experiment 1: sim $\sim$ earlier $\times$ condition + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.517	[0.458, 0.575]
condition	-0.008	[-0.02, 0.005]
earlier	0.090	[0.077, 0.103]
earlier:condition	-0.008	[-0.011, -0.005]

Table A65: Experiment 2: 6 same describer: sim $\sim$ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.501	[0.443, 0.556]
earlier	0.087	[0.079, 0.095]

Table A66: Experiment 2: 6 full feedback: sim $\sim$ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.438	[0.388, 0.486]
earlier	0.062	[0.052, 0.072]

Table A67: Experiment 2: 6 thin: sim $\sim$ earlier + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.499	[0.448, 0.55]
earlier	0.022	[0.011, 0.032]

Table A68: Experiment 3: sim $\sim$ earlier $\times$ channel $\times$ gameSize + (1|tangram) + (1|gameId)

Term	Est.	95% CrI
Intercept	0.583	[0.549, 0.618]
channelthin	-0.037	[-0.084, 0.008]
channelthin:gameSize6	0.010	[-0.069, 0.09]
earlier	0.080	[0.075, 0.086]
earlier:channelthin	-0.023	[-0.032, -0.015]
earlier:channelthin:gameSize6	-0.031	[-0.046, -0.015]
earlier:gameSize6	0.006	[-0.005, 0.017]
gameSize6	-0.058	[-0.116, -0.003]

Divergence across games

Table A69: Experiment 1: sim $\sim$ block $\times$ condition + (1|tangram)

Term	Est.	95% CrI
Intercept	0.468	[0.426, 0.51]
block	-0.034	[-0.037, -0.03]
block:condition	0.001	[0, 0.002]
condition	0.002	[0, 0.005]

Table A70: Experiment 2: 6 same describer: sim $\sim$ block + (1|tangram)

Term	Est.	95% CrI
Intercept	0.469	[0.429, 0.516]
block	-0.038	[-0.041, -0.036]

Table A71: Experiment 2: 6 full feedback: sim $\sim$ block + (1|tangram)

Term	Est.	95% CrI
Intercept	0.509	[0.461, 0.554]
block	-0.040	[-0.043, -0.037]

Table A72: Experiment 2: 6 thin: sim $\sim$ block + (1|tangram)

Term	Est.	95% CrI
Intercept	0.441	[0.401, 0.478]
block	-0.007	[-0.011, -0.004]

Table A73: Experiment 3: sim $\sim$ block $\times$ channel $\times$ gameSize + (1|tangram)

Term	Est.	95% CrI
Intercept	0.416	[0.368, 0.458]
block	-0.025	[-0.026, -0.024]
block:channelthin	0.005	[0.003, 0.006]
block:channelthin:gameSize6	0.023	[0.019, 0.027]
block:gameSize6	-0.007	[-0.01, -0.004]
channelthin	0.011	[0.006, 0.015]
channelthin:gameSize6	-0.064	[-0.075, -0.052]
gameSize6	0.057	[0.048, 0.066]

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